

Share Buyback Valuation & Simulation

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Abstract

This paper debunks the prevalent belief that share buybacks and dividends are equivalent methods of returning capital to shareholders. In reality, a share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders.

Whether a share buyback causes a gain or loss to the remaining shareholders depends on the price paid relative to the present value of the subsequent dividends and share-prices.

Formulas are presented for properly measuring the effect of a share buyback on intrinsic shareholder value. Simulations are used to give a complete overview of possible outcomes.

Formulas are also provided for measuring the actual gain/loss of historical share buybacks based on the subsequent dividends and share-prices. These formulas are used on thousands of the largest U.S. stocks since the late 1990s, which shows that the actual effect of the many trillions of dollars spent on share buybacks has probably just been an even split between gains and losses for the shareholders.

Several case studies of individual companies show the actual effect of their historical share buybacks on the shareholder value. Simulations also demonstrate how financial managers could have assessed the effect of share buybacks at different points in time.

This paper should be understandable by anyone who invests a little effort. It mostly uses simple math and basic financial concepts like Present Value, and half the pages are plots.

“When the world is sliding into a decline, to excel is easy.” – Hagakure

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AI Disclosure

Artificial Intelligence (AI) has not been used for the main research and writing of this paper, but the Google Gemini 3.1 Pro and “thinking” chat-bot was used after the paper was nearly finished:

Section 11 is a conversation with AI, which explains the basics of share buyback valuation. When another AI session reviewed the paper later on, it amusingly said that this was the best layman’s explanation of share buybacks that it had ever seen. So you may want to read section 11 first.

AI was then used to carefully go through and review each section of the paper in steps. First, AI gained a high-level understanding of the claims and formulas and whether they were logically coherent. Then, AI rigorously checked the derivation of every single formula, where it found an error in Eq.(5) for calculating the excise tax of a share buyback. AI also found a few large rounding errors in the numerical examples. AI also made a few suggestions for clarifications of the text, and checked the spelling and grammar (although sometimes I refused to follow its suggestions). AI also checked that all references to formulas and figures were correct. AI also carefully analyzed the case studies and automatically made some test calculations to verify all the plots looked correct.

This was a very thorough review, far better than most human journal reviews and it only cost \$1.68. It is truly stunning what AI is capable of today compared to just two years ago. It will be interesting to see where this ends. Soon, AI will probably be able to innovate entirely on its own initiative.

Preface

This is the story about how this paper is the culmination of 15 years of trials and tribulations.

I am originally a computer scientist. Investing was a complete mystery to me until I stumbled over a book about some old geezer named Warren Buffett, whom I had never heard of before. His investment philosophy made sense to me, namely that there is a difference between the market price of a stock and its intrinsic value to long-term investors. If you can estimate the intrinsic value and it is very different from the share-price, then you can do well at investing. That made sense to me, but I later found out how difficult it is to do in practice.

After I finished my formal education, I read the books by Warren Buffett's teacher Ben Graham, who was obviously a genius, but I learned through hard experience that his valuation methods were not really suitable for modern times, because Graham's methods were developed during a time when industry was very different. I also studied finance and accounting textbooks by myself, which is why I sometimes reinvented concepts and gave them new names when I was unaware that a concept already existed under another name, and there were no AI chat-bots to explain it back then.

I also made some software to help discover and analyze stocks. I came across a company that had good profits for years, but it spent most or all of the profits on share buybacks instead of dividends, and then suddenly the company went bankrupt. So I wondered what the intrinsic value was for the shareholders who held onto their shares for that whole time, and who had forgone dividends to increase their ownership of the company through share buybacks, but ended up with nothing. They would obviously have been much better off with dividends instead of those share buybacks.

There were only a few books on share buybacks. As I recall, I went to a business school library in Berlin and another library in Copenhagen to read those few books. One book was just a compilation of academic papers on the subject. All the theories were based on the so-called Efficient Market Hypothesis and the Modigliani-Miller theorem. They always assumed that the share-price was exactly equal to the intrinsic value, so there was no difference between share buybacks and dividends; which I knew was false from the company that went bankrupt after making buybacks.

Share buybacks were becoming the main way for corporations to "return cash to shareholders" (a phrase I deplore because it is so misleading), and I realized how important it was and that it was my chance of doing something very important in life. So I started working on the formulas for valuing share buybacks. I thought I could write a short paper in maybe two months. Although the core ideas and formulas are fairly simple, the rabbit hole turned out to be much deeper than I expected.

I used axiomatic proof techniques that I had learned as a student of mathematics and computer science. This is similar to reasoning from "first principles" in physics. You start by only assuming what is obviously true, i.e. a share buyback reduces the company's cash holdings and the number of shares outstanding. You must NOT assume that the intrinsic value is exactly equal to the share-price, which is what all the finance professors assumed, but that is not obviously true in the real world, so you would need to make a big leap of faith to assume that, and that is pseudo-science.

Once you make the distinction between the intrinsic value and share-price, it then becomes a matter of modeling and studying the value transformation caused by a share buyback in different scenarios.

Altogether I probably spent about two years researching and writing what eventually became a 200-page treatise on share buyback valuation [Pedersen 2012]. I knew the treatise was a very important contribution to finance, because it debunked many of the core hypotheses of academic finance, and it also provided valuation formulas that were derived using proper axiomatic reasoning, and which could be used in real life to improve the capital allocation for the majority of corporate profits.

I was naturally excited about my important discovery, so I told a family relative who was a CFO. He exploded in anger, yelling and screaming at me completely out of control. I later learned this is known as a narcissistic rage, because his whole identity was based on him being a finance expert and he saw me as inferior, so when I excitedly told him something very important about finance that he did not know, he felt it was a mortal attack on his ego, so he exploded in a narcissistic rage.

I also wrote to any professor and researcher who was even remotely interested in share buybacks. I think I suggested they use a “sledge hammer and chainsaw” to see if they could make my theory fall apart. All of the main researchers in the field just ignored me, except for one who immediately dismissed the entire treatise after a quick glance, and he was also very condescending. This is even though it is their paid job to engage in discussions with people whose theories oppose their own beliefs, for the sake of scientific progress. Nobody ever refuted my work. They just ignored it.

I also wrote a shorter paper with the main parts of the treatise, which I tried to get published in nearly ten of the leading finance and business journals. I sent letters directly to the editors asking if it would be possible to publish the paper as “free access” if it was accepted, because I had paid for all the research myself and I wanted it to benefit as many people as possible. The editors themselves responded very quickly that they were not interested in publishing the paper at all, but they refused to give me any reason why.

I also contacted financial consultancy agencies because they could use my work to help their clients make better share buyback decisions. Nobody ever responded. I later realized that they might be held legally liable if they suddenly admitted that they had advised their clients to buyback shares for trillions of dollars, if many of those share buybacks had likely destroyed shareholder value.

Since the first edition of my treatise was released almost 14 years ago, I am only aware of a single professor who used my treatise to teach his students how to properly value share buybacks. But he never spread the knowledge beyond his own classroom and sadly he passed away a few years ago.

I continued work on other projects in computer science and investment finance e.g. developing new and vastly superior algorithms for portfolio optimization. But I got tired of writing research papers with highly original and important contributions that nobody cared about.

So I started developing the [SimSim.Run](#) web-site, originally just to make it easy for people to value share buybacks using Monte Carlo simulation with probability distributions from historical data. During the development of the web-site I also extended the formulas from the original treatise, and that is what has now been turned into this paper, which is a major extension of the original treatise.

This is surely one of the most important and useful finance papers ever written. It could change the world if people would just read it and use it. Hopefully it won't be ignored for another 15 years.

I would like to [thank everyone](#) who helped throughout this ordeal.

1 Introduction

A share buyback is when a company repurchases some of its outstanding shares to increase the remaining shareholders' ownership of the company. In USA the majority of corporate profits are being used for share buybacks instead of dividend payouts.

Finance professors used to say that a share buyback signaled the company's management had inside information that the stock was undervalued. But there were countless examples of share buybacks done at peak share-prices just before the company and its stock started to decline. In some cases the companies even went bankrupt just a few years after they had made massive share buybacks.

Finance professors then changed their story so that share buybacks were now a perfect substitute for dividend payouts, possibly even with a tax advantage depending on the tax-rates for dividends and capital gains. This is known as the "substitution hypothesis" which is commonly believed to be true, and is probably the reason that the majority of profits are used for buybacks instead of dividends.

The "proof" of this claim usually takes one of two forms: Either the share-price is assumed to be exactly equal to the so-called intrinsic value to shareholders; or the valuation ratio such as P/E or P/B is assumed to remain the exact same in the future. Neither of these is a valid assumption in reality, but finance professors then utter the phrase "*all else being equal*" as some kind of magic incantation that they believe turns absurd claims into solid science.

Anyone else can see the absurdity of these claims. For example, a company that buys back shares at price \$100 after which the share-price drops to \$50, could obviously have bought back twice as many shares if they had just waited. So timing matters greatly for the efficacy of share buybacks. In the extreme case where a company goes bankrupt after it has made large share buybacks, the shareholders would obviously have been infinitely better off with dividend payouts instead of share buybacks. So share buybacks are clearly not a perfect substitute for dividend payouts.

The reality is that a share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders. Whether this results in a gain or loss for the remaining shareholders depends on how long they hold their shares and what the future dividends and share-prices will be.

Sections 2 and 3 show how to properly value share buybacks using the future earnings and share-prices. As the future is uncertain, we use simulations with probability distributions for the input values, so we can see the entire range of possible outcomes and their probabilities.

Section 4 derives formulas for valuing historical share buybacks and issuances based on subsequent dividends and share-prices. This method is vastly superior to how academia only considers the correlation between share buybacks and subsequent abnormal stock-returns, as these formulas let us see the actual effect of a historical share buyback or issuance on the intrinsic shareholder value.

Section 6 uses these valuation formulas on historical data for thousands of the largest U.S. stocks since the late 1990s. This shows that the actual effect of the many trillions of dollars spent on share buybacks has probably just been an even split between gains and losses for the shareholders.

Sections 7 to 10 are case studies for individual companies that show the actual effect of their historical share buybacks. Simulations are also used to demonstrate how the financial managers of those companies could have assessed the effect of share buybacks at different points in time.

1.1 Other Resources

This paper is a major extension to the original treatise on share buyback valuation [Pedersen 2012]. Section 2 of that treatise contains an extensive literature review which still seems current for today even though it was written almost 14 years ago, because academia is still stuck in their old dogma.

Another recent paper considers the valuation and simulation of share issuances [Pedersen 2024].

The [SimSim.Run](#) web-site makes it easy to run the simulations. [Video tutorials](#) are also available.

1.2 Advice

For financial managers trying to decide whether to make a share buyback:

- A share buyback destroys shareholder value when the price paid is above the present value of the subsequent dividends and share-prices. It is similar to a regular investment in the stock.
- Don't buyback shares if you don't really understand their effect. You cannot destroy shareholder value if you just make a dividend payout instead.
- Don't buyback shares to try and boost Earnings Per Share (EPS) or offset dilution from stock options. It still destroys shareholder value if the share-price is higher than the intrinsic value.
- You can usually only estimate the intrinsic shareholder value without knowing the exact value. So it is essential to make simulations to get an overview of the possible outcomes, and then have a margin of safety to lower the risk of destroying shareholder value from a share buyback.
- Try different inputs for your simulations to better understand how share buybacks would affect the shareholder value in different scenarios for the future of your company. If you need to make wildly optimistic assumptions about the future, in order for the share buyback to result in a gain to shareholders, then you should probably refrain from making the share buyback. Conversely, if your assumptions about the future are very conservative and the simulations still show a significant gain in shareholder value, then you may want to make a large share buyback.
- Avoid making share buybacks when the stock seems to be over-valued because it will likely destroy shareholder value. It would be much better to make a special dividend payout, or retain the cash to improve the financial strength of the company, so you can take full advantage the next time the share-price declines and then buyback a much greater number of shares.
- Once you properly understand the effect of share buybacks, you can use them more strategically to increase shareholder value whenever a good opportunity arises, without having to grow the company's sales and earnings, cut expenses, overwork your employees, etc. All you have to do is buyback shares when the price is significantly below the intrinsic value.
- Inform your shareholders about your share buyback plans for full transparency: How do you value share buybacks to assess whether they will likely cause a gain or loss for shareholders? Do you fund the share buybacks using excess cash, profits or borrowings? You should regularly inform the shareholders about recent share buybacks and your future plans. The shareholders are your business partners so they should be properly informed about these very important issues.

2 Basic Formulas

To properly understand the effect of a share buyback on long-term shareholder value, you should always keep in mind that a share buyback has two direct consequences (these are the two axioms):

1. A share buyback reduces the amount of cash the company has. This is a cost.
2. A share buyback reduces the number of shares outstanding. This is a benefit.

A share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders. The question is whether this benefits or harms the remaining shareholders. The answer is that it depends on the share-price relative to the so-called intrinsic value to shareholders.

To measure the gain/loss effect of a share buyback, we first need to define the intrinsic value v (lower-case) of a company to its shareholders. This is the total amount and not per-share. It can be defined in different ways. The simplest is to consider the intrinsic value to be the company's excess cash that could either be used for dividends or share buybacks now, plus the so-called present value of all future earnings for eternity that could also be paid out as dividends:

$$v = \text{Excess Cash} + \sum_{t=1}^{\infty} \frac{\text{Earnings}_t}{(1+d)^t} \quad (1)$$

Let TaxRateDividend be the tax-rate for dividends, and let Shares be the number of shares outstanding prior to the buyback. Then the per-share, after-tax intrinsic value is denoted V and is:

$$V = v \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}} \quad (2)$$

A share buyback reduces the intrinsic value by the buyback amount so the intrinsic value becomes:

$$w = v - \text{Buyback} \quad (3)$$

The share-count after the buyback is denoted Shares^W (W is a super-script, not an exponent), which can be rewritten as follows when using the definition $\text{MarketCap} = \text{Shares} \cdot \text{SharePrice}$:

$$\text{Shares}^W = \text{Shares} - \frac{\text{Buyback}}{\text{SharePrice}} = \text{Shares} \cdot \left(1 - \frac{\text{Buyback}}{\text{MarketCap}}\right) \quad (4)$$

If there is an excise tax on share buybacks then the formula becomes:

$$\text{Shares}^W = \text{Shares} \cdot \left(1 - \frac{\text{Buyback} / (1 + \text{TaxRateBuyback})}{\text{MarketCap}}\right) \quad (5)$$

After the share buyback, the per-share, after-tax intrinsic value is denoted W and is defined as:

$$W = (v - \text{Buyback}) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^W} \quad (6)$$

2.1 Alternative Definitions of Intrinsic Value

The intrinsic value v can be defined in different ways and it will result in the same formulas for valuing a share buyback, if two requirements are satisfied: The intrinsic value v is decreased with the buyback amount as in Eq.(3), and the number of shares is decreased as in Eqs.(4) and (5), so the combined effect is that a share buyback transforms the per-share intrinsic value from V to W as in Eq.(6). Section 3 also considers share-price changes, so the buyback amount is subtracted from only the excess cash and earnings part of the intrinsic value, and the formula for W is therefore different.

2.2 Equilibrium

The so-called equilibrium is where the per-share intrinsic value is the same with and without a share buyback, that is, $W = V$. If the buyback tax is zero then the equilibrium is where the market-cap is exactly equal to the intrinsic value, that is, $MarketCap = v$. When also having a buyback tax, the market-cap must be proportionately smaller than the intrinsic value. The equilibrium is written as an inequality so it is easy to see when the buyback causes a gain or loss to the per-share intrinsic value:

$$W > V \Leftrightarrow MarketCap < \frac{v}{1 + TaxRateBuyback} \quad (7)$$

2.3 Relative Value

The relative value of a share buyback versus a dividend payout for the same amount, is the ratio between the value WITH the share buyback W from Eq.(6), and the value WITHOUT the share buyback V from Eq.(2). The formula can be reduced to this:

$$\frac{W}{V} = \frac{1 - \frac{Buyback}{v}}{1 - \frac{Buyback / (1 + TaxRateBuyback)}{MarketCap}} \quad (8)$$

Note that the overall numerator is how much of the intrinsic value remains after the share buyback, and the overall denominator is how much of the market-cap remains after the share buyback.

Also note that both the number of shares and the dividend tax-rate cancel out, so if we are considering the value to “eternal” shareholders who will never sell their shares, then the dividend tax-rate is irrelevant when valuing a share buyback. Further below we extend the formulas to also consider capital gains, so the tax-rates for both capital gains and dividends become relevant.

2.4 Return on Intrinsic Value (ROIV)

Subtracting 1 from the relative value makes it easier to see when the share buyback results in a gain (+) or loss (-) relative to a dividend payout. We call this the Return on Intrinsic Value (ROIV):

$$ROIV = \frac{W}{V} - 1 \quad (9)$$

Note that ROIV is ill-defined when the intrinsic value is zero or negative. And because the formula is non-linear, the losses are greater than gains for similar degrees of mispricing, which has profound implications for many aspects of finance, as explained in [Pedersen 2012].

2.5 Return on Buyback (ROB)

The ROIV ratio in Eq.(9) tells us how a share buyback impacts the per-share intrinsic value to the remaining shareholders. But it is also useful to consider the gain/loss of shareholder value relative to the amount that was spent on the share buyback. This makes it easy to compare the gain/loss to other investment returns. This is called the Return on Buyback (ROB) ratio which is defined as:

$$ROB = \frac{W - V}{(Buyback / Shares) \cdot (1 - TaxRateDividend)} \quad (10)$$

Unlike the ROIV ratio, the ROB ratio can also be calculated for intrinsic values v that are zero or negative, because the ROB ratio does not divide by the intrinsic value v .

The numerator $W - V$ is the net effect of the share buyback, as it is the difference between the intrinsic value of the company with W and without V the share buyback.

The first part of the denominator ($Buyback / Shares$) is the amount used for the share buyback on a per-share basis, because W and V are also on a per-share basis. The second part of the denominator adjusts for the dividend-tax which cancels out with the dividend-tax in the numerator. The ROB ratio compares the net effect of the share buyback to the buyback amount, as if the buyback amount had been paid out as a dividend instead. This is done to make a fair comparison to the numerator.

Using the simple definitions of W and V from Eqs.(6) and (2) above, the dividend tax-rate cancels out so the ROB formula can be reduced to the following:

$$ROB = \frac{v}{Buyback} \cdot ROIV \quad (11)$$

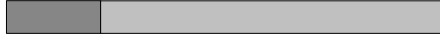
Further below we also consider capital gains in the calculation of W and V , so the dividend tax-rate no longer cancels out in the ROB formula, and the original ROB formula in Eq.(10) must be used.

It is important you understand the difference between the ROIV and ROB ratios: ROIV measures the change in per-share intrinsic value to the remaining shareholders, while ROB measures that change relative to the buyback amount. This will hopefully be made clear from the examples below.

The ROB ratio was not used in [Pedersen 2012] and was first used on the [SimSim.Run](#) web-site.

2.6 Visual Examples

This may be easier to understand with a visual example. The figure below demonstrates the effect of a share buyback when the intrinsic value v is 95% of the *MarketCap* and the share buyback is 20% of the *MarketCap*. If the share-price remains unchanged after the buyback then the mispricing is magnified slightly as the intrinsic value decreases to become 93.75% of *MarketCap*, down from 95% prior to the share buyback. This means the per-share intrinsic value has decreased about -1.3% as a result of the share buyback, which is what the ROIV ratio in Eq.(9) calculates. If instead v is 105% of *MarketCap* then the per-share intrinsic value would have increased by about +1.2% as a result of the share buyback (note that the ROIV ratio is not symmetric). The equilibrium is when $MarketCap = v$ so the share buyback has equal effect on the *MarketCap* and intrinsic value.

Before Share Buyback	
<i>MarketCap</i>	
$v = 95 \% \cdot MarketCap$	
Share Buyback is The Portion in Dark Grey	
$Buyback = 20 \% \cdot MarketCap$	
$Buyback \simeq 21 \% \cdot v$	
After Share Buyback	
<i>MarketCap</i>	
$w = 93.75 \% \cdot MarketCap$	

The figure below demonstrates a much greater mispricing where the intrinsic value v is only 25% of the *MarketCap* and the share buyback is again 20% of the *MarketCap*. If the share-price remains unchanged after the buyback then the mispricing is greatly magnified as the intrinsic value becomes only 6.25% of *MarketCap*, down from 25% prior to the share buyback. So the per-share intrinsic value has decreased -75% as a result of the share buyback, which is the ROIV ratio from Eq.(9).

Before Share Buyback	
<i>MarketCap</i>	
$v = 25 \% \cdot \text{MarketCap}$	
Share Buyback is The Portion in Dark Grey	
$\text{Buyback} = 20 \% \cdot \text{MarketCap}$	
$\text{Buyback} = 80 \% \cdot v$	
After Share Buyback	
<i>MarketCap</i>	
$w = 6.25 \% \cdot \text{MarketCap}$	

2.7 Numeric Examples

Let us now do the math for an example where the stock is slightly over-valued. Say the *MarketCap* of all the company's shares is \$10b, and the company has \$1b in excess cash they want to either use for a share buyback or dividend payout. If the present value of all future earnings is \$7b then the total intrinsic value is $v = \$8b$ because it also includes the excess cash. The gain/loss of per-share intrinsic value from the share buyback versus dividend payout is the ROIV calculated using Eq.(9):

$$ROIV = \frac{W}{V} - 1 = \frac{1 - \frac{\text{Buyback}}{v}}{1 - \frac{\text{Buyback}}{\text{MarketCap}}} - 1 = \frac{1 - \frac{\$1b}{\$8b}}{1 - \frac{\$1b}{\$10b}} - 1 \simeq -2.8 \% \quad (12)$$

So the share buyback would incur a loss to the remaining shareholders of about -2.8% relative to a dividend payout. This may seem like a minor loss, but remember that this is a loss of intrinsic value for all remaining shares, so it may actually be a substantial loss on the buyback amount.

To better understand the magnitude of the loss we may calculate the ROB ratio, which measures the net effect of the share buyback on the per-share intrinsic value, relative to the amount that was used on the buyback. Because this example uses the simple definitions of W and V , we can use the reduced formula for the ROB ratio from Eq.(11) as follows:

$$ROB = \frac{v}{Buyback} \cdot ROIV \simeq \frac{\$8b}{\$1b} \cdot (-2.8\%) \simeq -22.4\% \quad (13)$$

The share buyback can be considered a type of investment that the company is making in itself on behalf of its remaining shareholders, and the ROB ratio measures the return on that investment. In this example it would result in a loss of about -22% on the invested amount. No financial manager would make such a stupid investment, if they knew the math!

If financial managers consistently buyback shares when they are overvalued, then the value destruction is compounded through the years and will become massive over time.

Let us consider another example where the stock is under-valued by 50%, so the market-cap is still \$10b but now the intrinsic value is actually $v = \$20b$, which again contains excess cash of \$1b that we will either use on a share buyback or a dividend. The gain/loss of per-share intrinsic value is the ROIV ratio calculated using Eq.(9):

$$ROIV = \frac{W}{V} - 1 = \frac{1 - \frac{Buyback}{v}}{1 - \frac{Buyback}{MarketCap}} - 1 = \frac{1 - \frac{\$1b}{\$20b}}{1 - \frac{\$1b}{\$10b}} - 1 \simeq 5.6\% \quad (14)$$

So the share buyback would result in a gain to the remaining shareholders of about 5.6% relative to a dividend payout. This may seem like a small gain, but remember that the ROIV ratio is relative to the per-share intrinsic value. So let us calculate the ROB ratio using Eq.(11) to see the valuation impact relative to the amount used for the share buyback:

$$ROB = \frac{v}{Buyback} \cdot ROIV \simeq \frac{\$20b}{\$1b} \cdot 5.6\% \simeq 112\% \quad (15)$$

So the effect of the share buyback would be similar to the company making an investment for \$1b whose present value was about 112% more than that amount. The return is more than twice the invested amount, simply because the repurchased shares were under-valued by 50%. Every financial manager would make such an investment, if they knew the math!

These examples show that share buybacks may have a significant impact on long-term shareholder value when the share-price does not exactly equal the intrinsic value, and the impact can be very large in case of severe over- and under-valuation of the stock.

You should try and experiment with different inputs in the formulas above, to see how a share buyback would impact the long-term shareholder value.

2.8 Simulation

We typically don't know the exact intrinsic value of a company, because it is based on estimates of the future earnings. So we should repeat the calculations above with many different assumptions. You could enter the valuation formulas in a computer spreadsheet and run some simulations, but it is much easier to use the [SimSim.Run](#) web-site that was made specifically for running this kind of simulation (click the link to try it). We use the same input assumptions as above, except for the intrinsic value which is now a normal-distributed (bell-shaped) random variable with mean \$10b and standard deviation \$1b.

Figure 1 shows the simulated ROIV ratio from Eq.(9), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. Note the ROIV distribution is skewed so the losses are greater than the gains, even though the intrinsic value v is normal-distributed. The asymmetry of the ROIV ratio is because Eq.(9) is non-linear with respect to the intrinsic value v .

Figure 2 again shows the simulated ROIV ratio from Eq.(9), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the current share-price changes, while the other variables remain the same.

Figure 3 shows the simulated ROB ratio from Eq.(10), which also measures the gain/loss of per-share intrinsic value for the company's original shareholders, but relative to the buyback amount. Note the ROB distribution is symmetric and bell-shaped like the normal-distributed intrinsic value v (but with sampling noise), which is because Eq.(10) is linear with respect to the intrinsic value v .

Figure 4 again shows the simulated ROB ratio from Eq.(10), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis. This makes it easy to see how the ROB ratio would change when the current share-price changes, while the other variables remain the same.

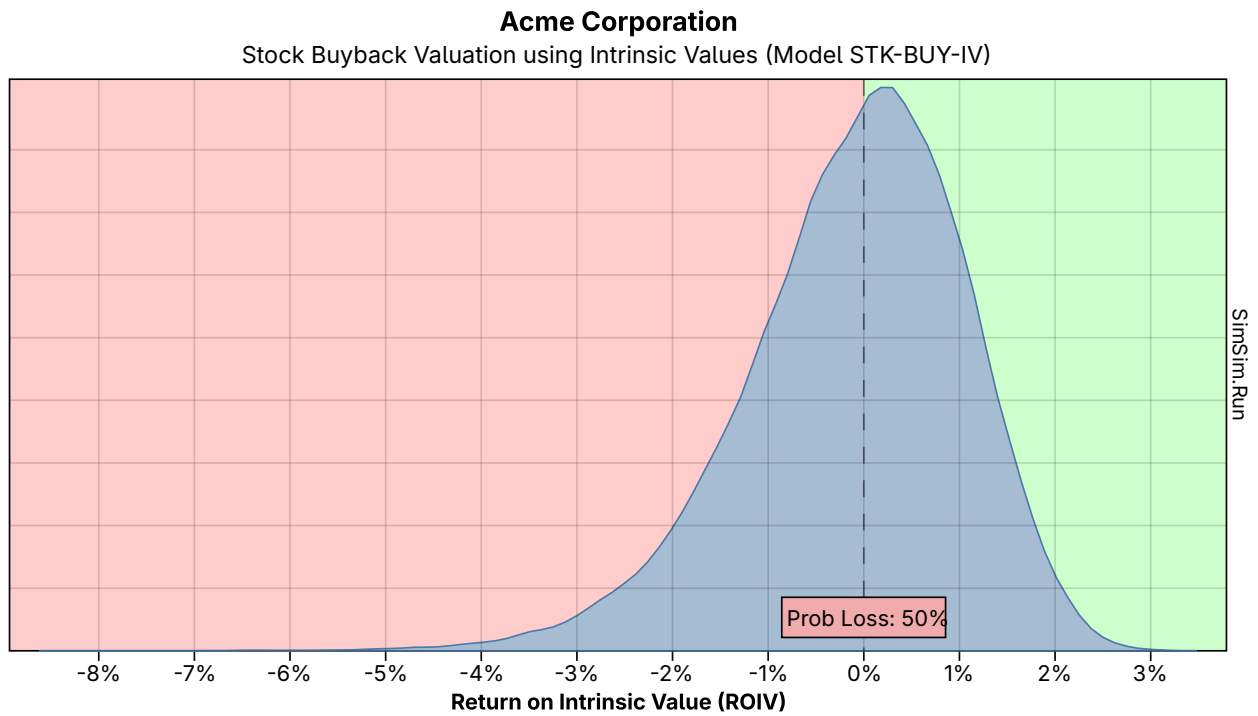


Figure 1: Simulation of $ROIV$ in Eq.(9) using a random variable for the intrinsic value v .

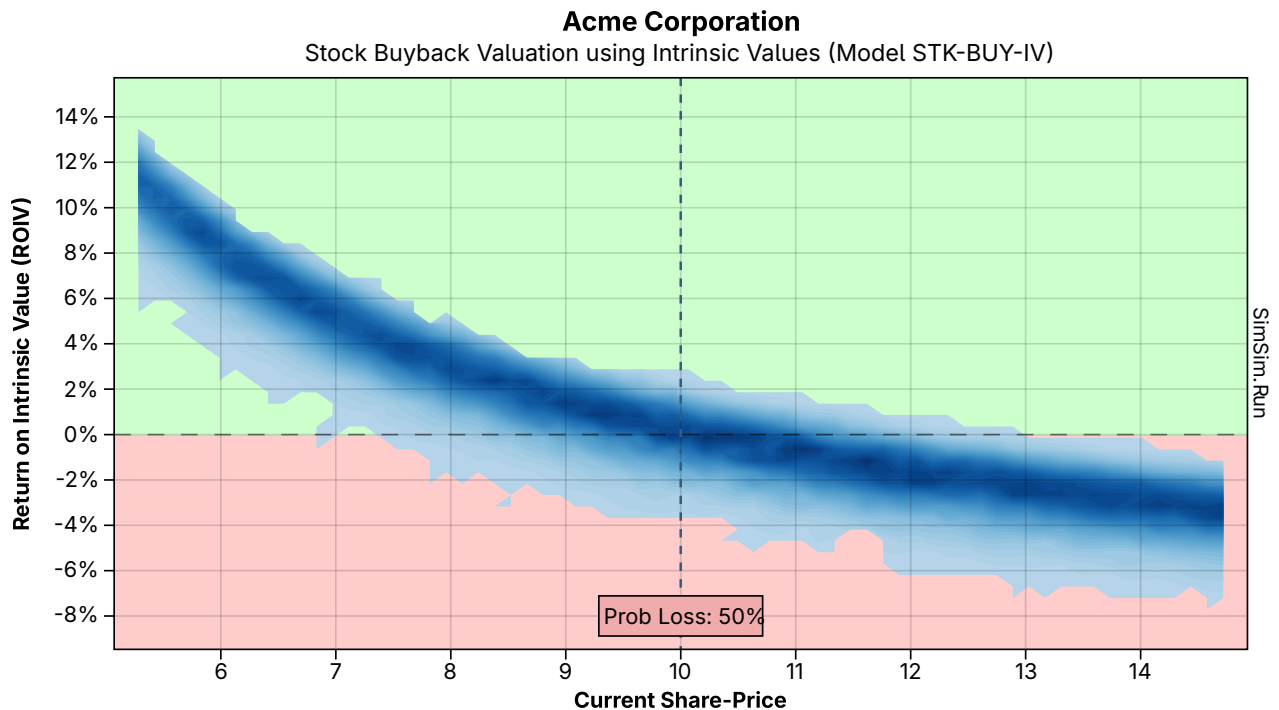


Figure 2: Simulation of $ROIV$ in Eq.(9) using a random variable for the intrinsic value v , and varying the current share-price (or MarketCap) on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.
The distribution in Figure 1 is the vertical dashed blue line here.

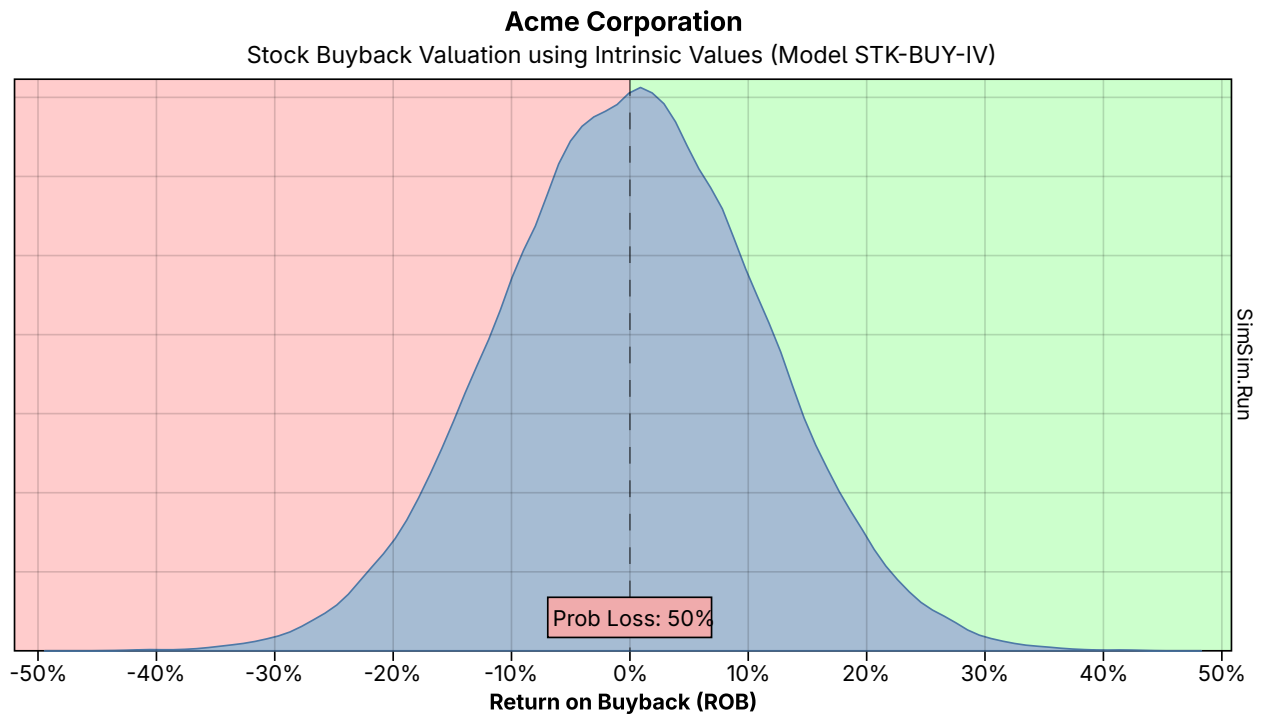


Figure 3: Simulation of **ROB** in Eq.(10) using a random variable for the intrinsic value v .

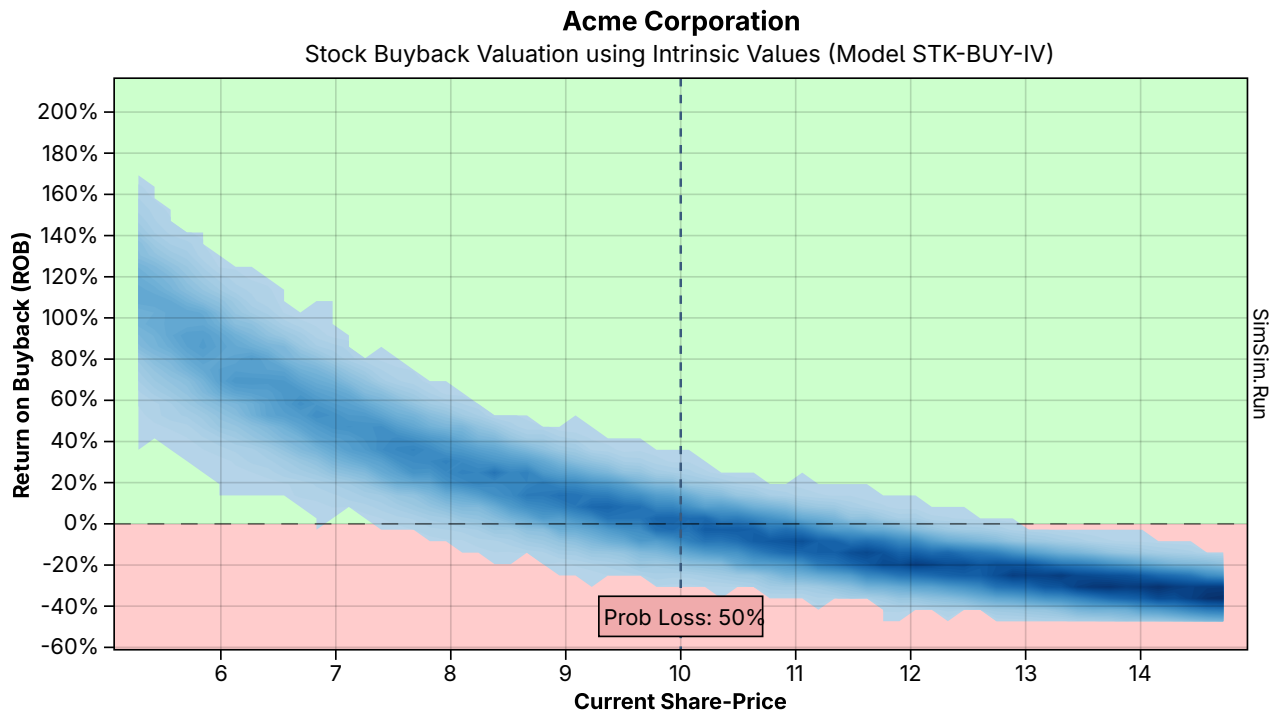


Figure 4: Simulation of **ROB** in Eq.(10) using a random variable for the intrinsic value v , and varying the current share-price (or MarketCap) on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.
The distribution in Figure 3 is the vertical dashed blue line here.

3 Capital Gains

In [Pedersen 2012] it was argued that share buybacks should always be made for the sake of the remaining shareholders, which ultimately means that the valuation should only consider the so-called “eternal” shareholders who will never sell their shares, so there is no need to consider future share-prices when valuing share buybacks.

But one of the main arguments in favor of share buybacks over dividends is that if capital gains taxes are lower than dividend taxes, then share buybacks are claimed to be a more tax-efficient way of “returning” capital to shareholders. The “proof” of this claim typically assumes that the valuation ratio remains constant in the future, but that is obviously not a valid assumption in the real world.

We can extend the formulas for share buyback valuation to also include capital gains/losses from future changes in share-prices. The intrinsic value for shareholders is now calculated from the excess cash, the present value of the earnings until the future year N which are assumed to be paid out as dividends, and the present value of $SharePrice_N$ for the future year N .

We use the same future $MarketCap_N$ in year N for both scenarios without and with a share buyback. We typically estimate it from the product of the future earnings (or sales) and the P/E (or P/S) ratio, which are usually much easier to estimate than trying to estimate the future $MarketCap_N$ directly:

$$MarketCap_N = Earnings_N \cdot P/E_N \quad (16)$$

The share-price in future year N is different for the two scenarios without and with a share buyback, because $MarketCap_N$ is divided by $Shares$ without, and $Shares^W$ with a share buyback from Eq.(5):

$$\begin{aligned} SharePrice_N &= MarketCap_N / Shares \quad \text{WITHOUT Share Buyback} \\ SharePrice_N^W &= MarketCap_N / Shares^W \quad \text{WITH Share Buyback} \end{aligned} \quad (17)$$

The capital gains depend on each investor's purchase price for the shares, so they may be different for all investors and we therefore have to make a reasonable compromise and calculate the capital gains between the current $SharePrice$ and $SharePrice_N$ for future year N (and for $SharePrice_N^W$):

$$Capital Gains_N = SharePrice_N - SharePrice \quad (18)$$

The tax on only positive capital gains can be calculated as follows (and similarly for $SharePrice_N^W$):

$$TaxCapGains_N = TaxRateCapGains \cdot \max(Capital Gains_N, 0) \quad (19)$$

The after-tax, present value of $SharePrice_N$ is then as follows (and similarly for $SharePrice_N^W$):

$$PV SharePrice_N = \frac{SharePrice_N - TaxCapGains_N}{(1+d)^N} \quad (20)$$

We can now calculate the after-tax, per-share intrinsic value WITHOUT the share buyback:

$$V_N = \left(\text{Excess Cash} + \sum_{t=1}^N \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}} + \text{PV SharePrice}_N \quad (21)$$

And the after-tax, per-share intrinsic value WITH the share buyback:

$$W_N = \left(\text{Excess Cash} - \text{Buyback} + \sum_{t=1}^N \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^W} + \text{PV SharePrice}_N^W \quad (22)$$

3.1 Equilibrium

The equilibrium is where a share buyback does not change the after-tax, per-share intrinsic value. It is written as an inequality $W_N > V_N$ so it is easy to see when the share buyback causes a gain or loss. Using the definitions from Eqs.(21) and (22) above, the equilibrium can be reduced to:¹

$$W_N > V_N \Leftrightarrow$$

$$\text{MarketCap} < \frac{\text{Excess Cash} + \sum_{t=1}^N \frac{\text{Earnings}_t}{(1+d)^t} + \frac{\text{MarketCap}_N}{(1+d)^N} \cdot \frac{(1 - \text{TaxRateCapGains})}{(1 - \text{TaxRateDividend})}}{1 + \text{TaxRateBuyback}} \quad (23)$$

Note that the tax-rates are just some of the factors that determine whether a share buyback causes a gain or loss to the remaining shareholders. The main factor is still the stock's overall mispricing.

3.2 Return Ratios & Relative Value

In practice, we calculate V_N and W_N using Eqs.(21) and (22) and use the results in Eqs.(9) and (10) to calculate the ROIV and ROB ratios for assessing the gain or loss caused by the share buyback.

It is also interesting to study the formula for the relative value of the share buyback W_N/V_N , which can be reduced to the following when using the definitions of V_N and W_N from Eqs.(21) and (22):

$$\frac{W_N}{V_N} = \frac{1 - \frac{\text{Buyback}}{V_N \cdot \text{Shares}} \cdot \left((1 - \text{TaxRateDividend}) + \frac{\text{TaxRateCapGains}}{(1 + \text{TaxRateBuyback}) \cdot (1+d)^N} \right)}{1 - \frac{\text{Buyback} / (1 + \text{TaxRateBuyback})}{\text{MarketCap}}} \quad (24)$$

Note that the overall numerator is how much of the intrinsic value remains after the share buyback, and the overall denominator is how much of the market-cap remains after the share buyback. This is similar to Eq.(8) for an “eternal” shareholder where there are no capital gains, except in Eq.(24) the overall numerator is also adjusted for all the different tax-rates in a complex manner.

1 Eqs.(23) and (24) are the only formulas in this paper that are somewhat challenging to derive because there are so many variables. We assume there are capital gains both WITH and WITHOUT a share buyback. If both cases have losses instead then the capital gains tax-rate can be set to zero. The formulas are more complicated when the future share-price WITH a buyback has a capital gain, and the future share-price WITHOUT a buyback has a capital loss.

3.3 Example

Now consider an example that uses the formulas above with the following assumptions:

$$\begin{aligned}
 \text{Shares} &= 1 \text{ b} & \text{SharePrice} &= \$ 10 & \text{MarketCap} &= \text{Shares} \cdot \text{SharePrice} = \$ 10 \text{ b} \\
 \text{Buyback} &= \text{Excess Cash} = \$ 1 \text{ b} \\
 \text{Earnings}_t &= \$ 500 \text{ m for all future years } t \\
 \text{MarketCap}_N &= P/E_N \cdot \text{Earnings}_N = 20 \cdot \$ 500 \text{ m} = \$ 10 \text{ b} \\
 d &= 10 \% & \text{TaxRateDividends} &= 30 \% & \text{TaxRateCapGains} &= 10 \%
 \end{aligned} \tag{25}$$

The number of shares after the share buyback is calculated using Eq.(4), assuming zero excise tax:

$$\text{Shares}^W = \text{Shares} \cdot \left(1 - \frac{\text{Buyback}}{\text{MarketCap}} \right) = 1 \text{ b} \cdot \left(1 - \frac{\$ 1 \text{ b}}{\$ 10 \text{ b}} \right) = 0.9 \text{ b} \tag{26}$$

The future share-price in year N without and with a share buyback is calculated using Eq.(17):

$$\begin{aligned}
 \text{SharePrice}_N &= \text{MarketCap}_N / \text{Shares} = \$ 10 \text{ b} / 1 \text{ b} = \$ 10 \\
 \text{SharePrice}_N^W &= \text{MarketCap}_N / \text{Shares}^W = \$ 10 \text{ b} / 0.9 \text{ b} \simeq \$ 11.11
 \end{aligned} \tag{27}$$

The capital gains tax in the two scenarios are:

$$\begin{aligned}
 \text{TaxCapGains}_N &= 10 \% \cdot (\text{SharePrice}_N - \text{SharePrice}) = 10 \% \cdot (\$ 10 - \$ 10) = \$ 0 \\
 \text{TaxCapGains}_N^W &= 10 \% \cdot (\text{SharePrice}_N^W - \text{SharePrice}) = 10 \% \cdot (\$ 11.11 - \$ 10) \simeq \$ 0.11
 \end{aligned} \tag{28}$$

The present value of the future share-prices net of capital gains tax in the two scenarios are:

$$\begin{aligned}
 PV \text{ SharePrice}_N &= (\text{SharePrice}_N - \text{TaxCapGains}_N) / 1.1^N = (\$ 10 - \$ 0) / 1.1^N \\
 PV \text{ SharePrice}_N^W &= (\text{SharePrice}_N^W - \text{TaxCapGains}_N^W) / 1.1^N = (\$ 11.11 - \$ 0.11) / 1.1^N
 \end{aligned} \tag{29}$$

For example, if we use $N = 5$ years for the investment duration then the present values are:

$$\begin{aligned}
 PV \text{ SharePrice}_5 &= (\$ 10 - \$ 0) / 1.1^5 \simeq \$ 6.21 \\
 PV \text{ SharePrice}_5^W &= (\$ 11.11 - \$ 0.11) / 1.1^5 \simeq \$ 6.83
 \end{aligned} \tag{30}$$

The after-tax, per-share intrinsic value WITHOUT the share buyback is calculated using Eq.(21):

$$\begin{aligned}
 V_5 &= \left(\text{Excess Cash} + \sum_{t=1}^5 \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}} + \text{PV SharePrice}_5 \\
 &= \left(\$1\text{ b} + \sum_{t=1}^5 \frac{\$500\text{ m}}{1.1^t} \right) \cdot \frac{1 - 30\%}{1\text{ b}} + \$6.21 \simeq \$8.236
 \end{aligned} \tag{31}$$

The after-tax, per-share intrinsic value WITH the share buyback is calculated using Eq.(22):

$$\begin{aligned}
 W_5 &= \left(\text{Excess Cash} - \text{Buyback} + \sum_{t=1}^5 \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^w} + \text{PV SharePrice}_5^w \\
 &= \left(\$1\text{ b} - \$1\text{ b} + \sum_{t=1}^5 \frac{\$500\text{ m}}{1.1^t} \right) \cdot \frac{1 - 30\%}{0.9\text{ b}} + \$6.83 \simeq \$8.304
 \end{aligned} \tag{32}$$

The gain/loss of per-share intrinsic value from the share buyback versus a dividend payout is the ROIV ratio calculated using Eq.(9) with the values W_5 and V_5 above for $N = 5$ years:

$$\text{ROIV}_5 = \frac{W_5}{V_5} - 1 \simeq \frac{\$8.304}{\$8.236} - 1 \simeq 0.8\% \tag{33}$$

So the share buyback would increase the per-share intrinsic value to the remaining shareholders by about +0.8% relative to a dividend payout.

We may also calculate the ROB ratio, which measures the net effect of the share buyback relative to the amount that was used on the buyback. Because the tax-rates are different for dividends and capital gains, we must use the original formula for the ROB ratio in Eq.(10) as follows:

$$\begin{aligned}
 \text{ROB}_5 &= \frac{W_5 - V_5}{(\text{Buyback} / \text{Shares}) \cdot (1 - \text{TaxRateDividend})} \\
 &\simeq \frac{\$8.304 - \$8.236}{(\$1\text{ b} / 1\text{ b}) \cdot (1 - 30\%)} \simeq 9.7\%
 \end{aligned} \tag{34}$$

A share buyback is an investment the company is making in itself on behalf of the remaining shareholders, and the ROB ratio measures the return on that investment. In this example it would be a gain of about +9.7% on the invested amount.

When using these formulas by hand, you should be careful with rounding errors in the intermediate calculations, which can significantly affect the final result. This is not a problem on a computer.

You should try and experiment with different inputs in the formulas above, to see how a share buyback would impact the intrinsic shareholder value. For example, if you just set the investment duration to $N = 10$ years instead of $N = 5$ as above, then the share buyback actually causes a loss in shareholder value. Why is that? Try and ponder this question before continuing to the next section.

3.4 Simulation

We obviously don't know the exact future earnings and share-prices, so we should repeat the calculations above with many different assumptions to assess the possible effect of a share buyback. You could enter the formulas and run simulations in a computer spreadsheet, but it is much easier to use the [SimSim.Run](#) web-site that was specifically made for this purpose (click the link to try it).

We use the same assumptions as Eq.(25) in the example above, except that some of the variables are now normal-distributed (bell-shaped) random variables with mean μ and standard deviation σ :

$$\begin{aligned} \text{Earnings}_t &\sim N(\mu = \$500\text{ m}, \sigma = \$100\text{ m}) \text{ for all future years } t \\ P/E_N &\sim N(\mu = 20, \sigma = 3) \\ d &\sim N(\mu = 10\%, \sigma = 1\%) \end{aligned} \tag{35}$$

Figure 5 shows the simulated ROIV ratio from Eq.(9), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. This is a violin-plot that shows the ROIV distributions vertically for all investment periods N between 1 and 10 years. The ROIV ratios are calculated from V_N and W_N in Eqs.(21) and (22) and thousands of random values from Eq.(35).

Figure 6 again shows the simulated ROIV ratio from Eq.(9), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis, and all investment periods N from 1 to 10 years are also combined. This makes it easy to see how the ROIV ratio would change for all investment periods overall when the current share-price changes, while all other variables remain the same.

Figure 7 shows the simulated ROB ratio from Eq.(10), which also measures the gain/loss of per-share intrinsic value for the company's original shareholders, but relative to the buyback amount.

Figure 8 again shows the simulated ROB ratio from Eq.(10), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis, and all investment periods N from 1 to 10 years are also combined. This makes it easy to see how the ROB ratio would change for all investment periods overall when the current share-price changes, while all other variables remain the same.

Note in Figure 5 and Figure 7 that the ROIV and ROB distributions are drifting downwards for longer investment periods N , even though the distributions in Eq.(35) are the same for all years. If you try running this simulation yourself, you will see that the drift gets worse for larger discount-rates. It is also possible for the distributions to drift upwards instead of downwards. It all depends on the choice of discount-rate, earnings, valuation ratios, etc.

In this example, the distributions are drifting downwards because the future earnings and P/E ratios don't grow in successive years. The present value of the share-price decreases from one more year of discounting, and adding one more year of discounted earnings is not sufficient to make up for that decrease, so the total present value drifts downwards.

What this means in practice is that you should always use probability distributions when valuing share buybacks, and consider many investment periods, as there is rarely just a single intrinsic value of a stock. Even if the future earnings, share-prices, and discount-rates are all held constant, the present value of a stock may vary simply from changing the investment period.

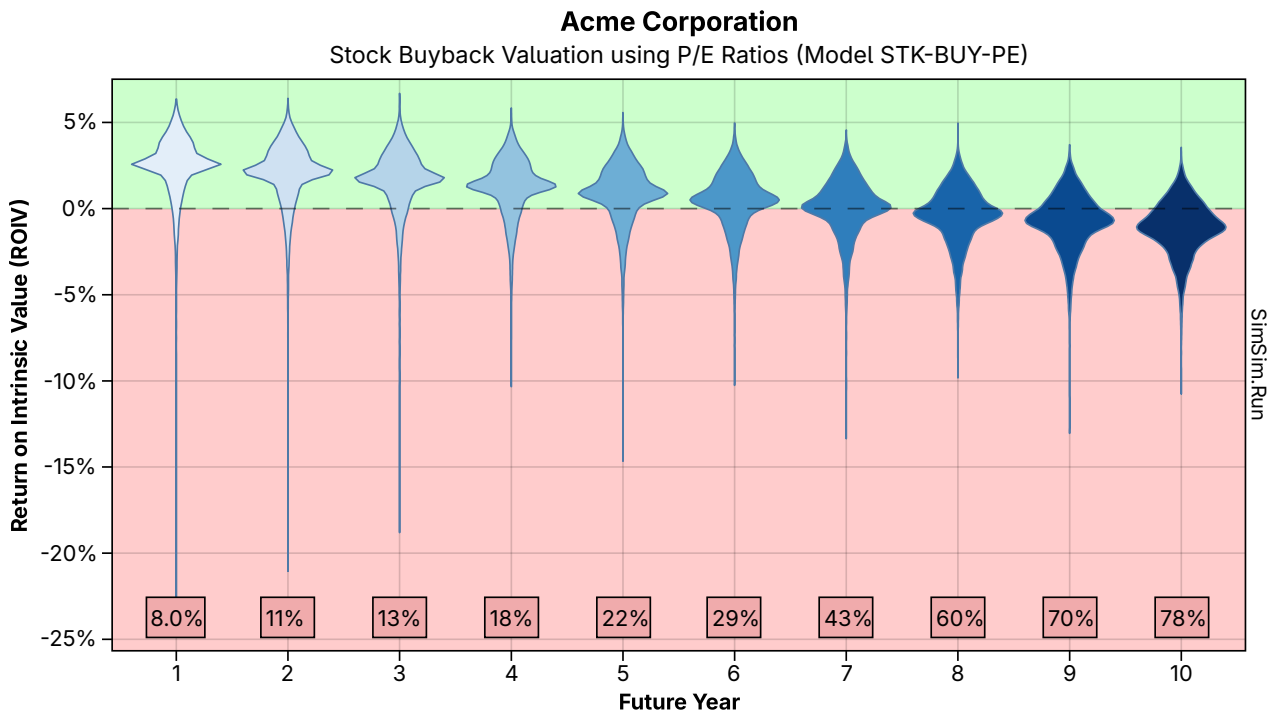


Figure 5: Simulation of **ROIV** in Eq.(9) using random variables for V_N and W_N in Eqs.(21) and (22).

This violin-plot shows the ROIV distributions for investment periods between 1-10 years.

Red boxes show the probability of loss for each investment period when making a share buyback.

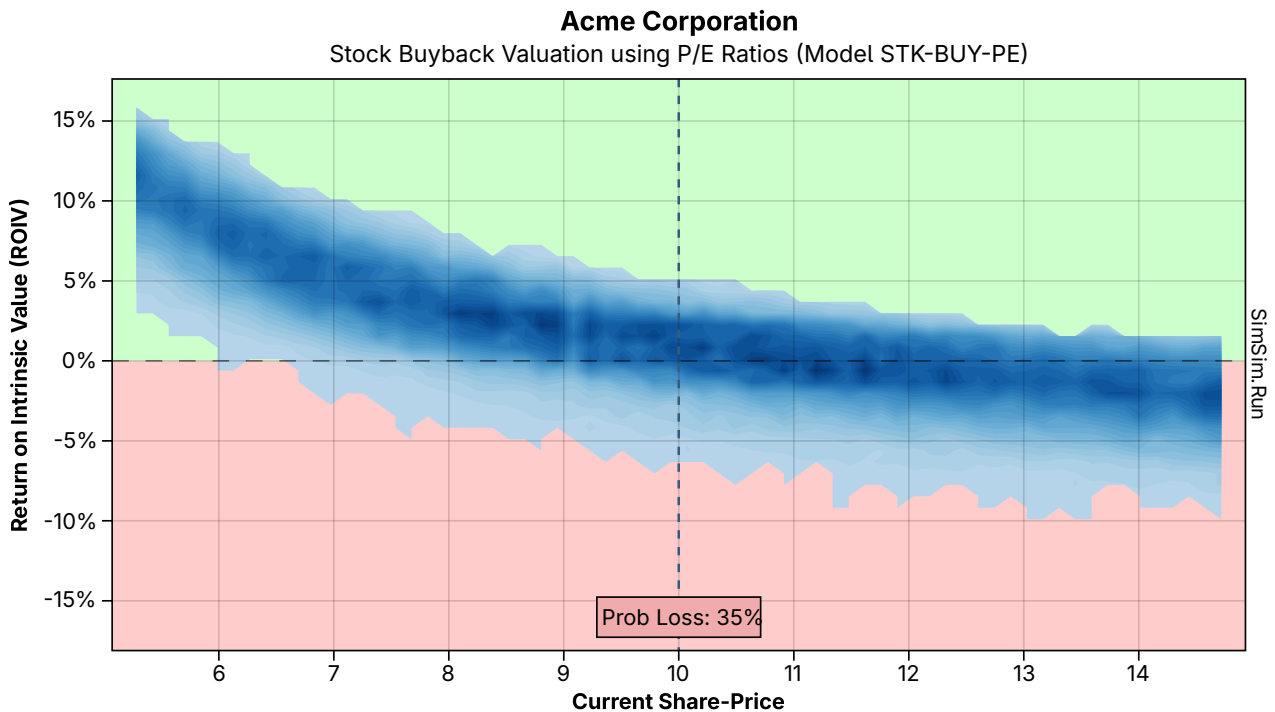


Figure 6: Simulation of **ROIV** in Eq.(9) using random variables for V_N and W_N in Eqs.(21) and (22),

and varying the current share-price (or MarketCap) on the x-axis.

This shows the ROIV ratio for all investment periods between 1-10 years combined.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The red box shows the overall probability of loss for all investment periods when making a share buyback.

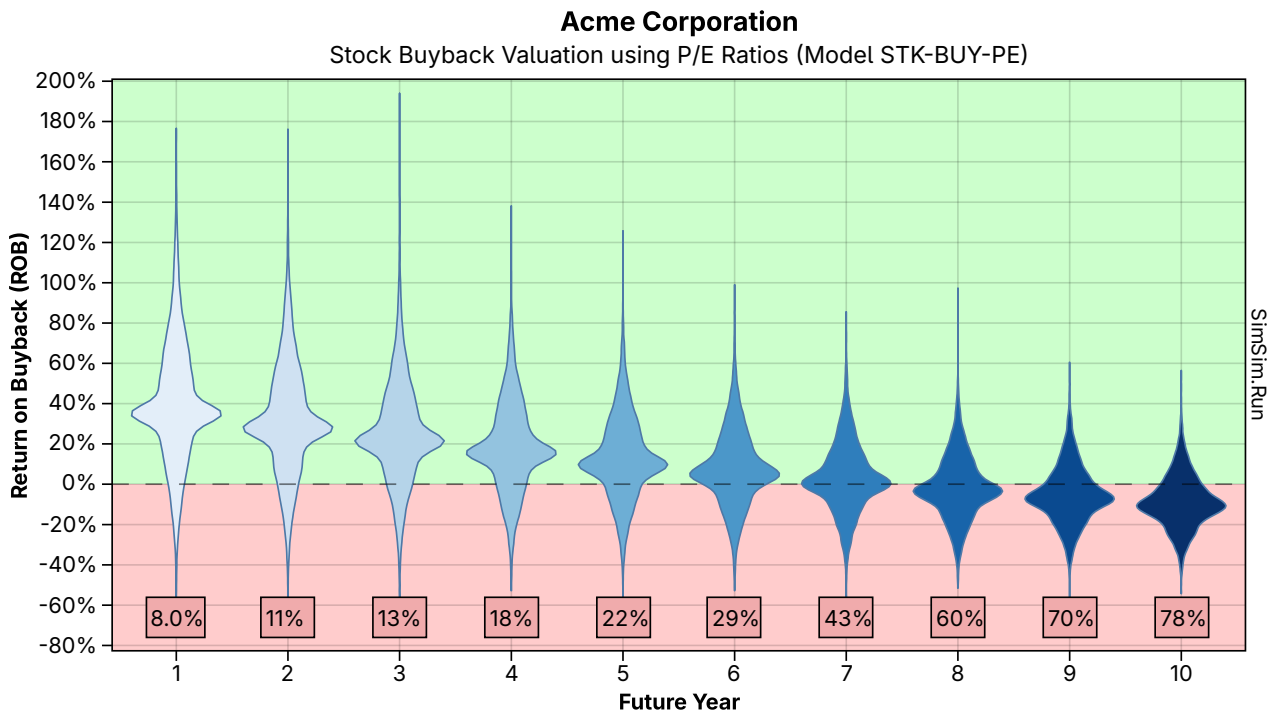


Figure 7: Simulation of **ROB** in Eq.(10) using random variables for V_N and W_N in Eqs.(21) and (22).
This violin-plot shows the ROB distributions for investment periods between 1-10 years.
Red boxes show the probability of loss for each investment period when making a share buyback.

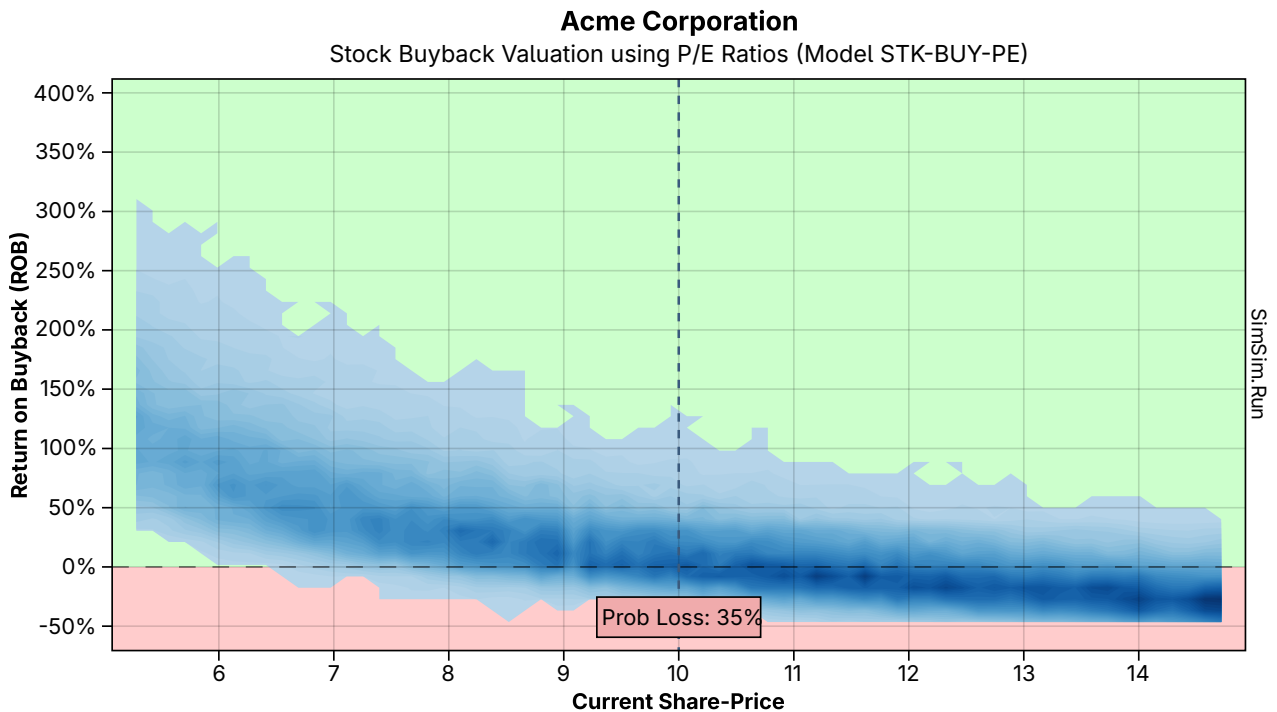


Figure 8: Simulation of **ROB** in Eq.(10) using random variables for V_N and W_N in Eqs.(21) and (22),
and varying the current share-price (or MarketCap) on the x-axis.
This shows the ROB ratio for all investment periods between 1-10 years combined.
This is a 2D distribution where darker blues indicate more simulated values in a region.
The red box shows the overall probability of loss for all investment periods when making a share buyback.

4 Historical Data

In the previous sections 2 and 3, we derived formulas for valuing share buybacks using estimates of the company's future earnings that could be paid out as dividends.

In this section, we present valuation formulas using historical data for dividends and share-prices to assess the actual gain/loss of past share buybacks and issuances for the shareholders at that time.

You can think of the previous sections as valuing share buybacks looking forward into an uncertain future, while this section is valuing past share buybacks and issuances using actual historical data.

4.1 Intrinsic Value

In the previous sections, we defined the per-share intrinsic value *WITHOUT* the share buyback, which was denoted V , and from this we subtracted the buyback amount, and adjusted the number of shares to get the intrinsic value *WITH* the share buyback, which was denoted W .

Here we do the opposite; we start by defining the intrinsic value *WITH* the historical share buyback or issuance W , and from this we estimate the value *WITHOUT* the share buyback or issuance V .

We also want to assess the gain/loss of the share buybacks and issuances for different investors who might have held the shares for 1, 2, or 3 years, etc. To do this, we consider the intrinsic value for shareholders between *Start* and *End* times, which are typically quarterly or yearly data-periods.

The per-share intrinsic value *WITH* the historical share buyback or issuance during the *Start* period, for a shareholder who owns the shares between the *Start* and *End* periods, is defined as the present value of the dividends during that time-range and the share-price at the end:

$$W_{[Start, End]} = \sum_{t=1}^N \frac{\text{Dividend Per Share}_{Start+t}}{(1+d)^t} + \frac{\text{SharePrice}_{End}}{(1+d)^N}, \text{ where } N = End - Start \quad (36)$$

A company's financial cash-flow statements should have the share buyback and issuance amounts reported separately so we can value them separately. But in some financial databases like the one we use in the case-studies further below, the amounts are combined into a *Net Buyback* amount that is positive for a share buyback and negative for an issuance. You can still use these formulas if you have a financial database with separate amounts for the share buybacks and issuances, by simply using the same signs + or - for those amounts, as we have defined for *Net Buyback* here.

Let $Shares_{Start}^W$ denote the number of historical shares outstanding at the end of the *Start* period *WITH* the share buyback or issuance that occurred during the *Start* period (e.g. quarter or year).

Let $Shares_{Start}^V$ denote the number of shares *WITHOUT* the share buyback or issuance during the *Start* Period. This can be estimated as the inverse of Eq.(4) to "undo" the net share buyback:

$$Shares_{Start}^V = \frac{Shares_{Start}^W}{1 - \frac{\text{Net Buyback}_{Start}}{\text{MarketCap}_{Start}}} \quad (37)$$

We might also consider using the actual number of shares outstanding in the previous period:

$$Shares_{Start}^V = Shares_{Start-1}^W \quad (38)$$

But using Eq.(38) causes problems because it conflates share buybacks during the *Start* period with the exercise of stock options that may have been granted several years prior at much lower share-prices. Some companies make very large share buybacks just to offset the dilution from exercised stock options. This would greatly distort the valuation of the share buyback or issuance during the *Start* period. So it is best to use Eq.(37) to estimate what the number of shares WITHOUT the share buyback or issuance should have been when ignoring the exercised stock options. This gives a more accurate measure of the gain/loss for the share buyback or issuance during the *Start* period.

If you have access to a more detailed financial database, you could use the actual number of shares bought back instead of estimating the number as we do in Eq.(37).

The per-share intrinsic value WITHOUT the share buyback or issuance during the *Start* period V , is estimated from the intrinsic value WITH the share buyback or issuance W , by adding the amount for the net buyback or issuance during the *Start* period, and adjusting the number of shares:

$$V_{[Start, End]} = \frac{W_{[Start, End]} \cdot Shares_{Start}^W + Net\ Buyback_{Start}}{Shares_{Start}^V} \quad (39)$$

4.2 Equilibrium

The equilibrium is where a share buyback does not change the per-share intrinsic value. It is written as an inequality $W > V$ so it is easy to see when the share buyback causes a gain or loss. Using the definition of V in Eq.(39) and $Shares^V$ in Eq.(37), the equilibrium can be reduced to this:

$$W_{[Start, End]} > V_{[Start, End]} \Leftrightarrow MarketCap_{Start} < V_{[Start, End]} \cdot Shares_{Start}^V \quad (40)$$

Note that this is essentially the same as the basic equilibrium in Eq.(7), except for taxes which are ignored here, and the subscripts that indicate the *Start* and *End* periods.

4.3 Relative Value

The relative value of a share buyback versus a dividend payout is the ratio between the per-share value WITH and WITHOUT the share buyback in Eqs.(36) and (39). It can be reduced to this:

$$\frac{W_{[Start, End]}}{V_{[Start, End]}} = \frac{1 - \frac{Net\ Buyback_{Start}}{V_{[Start, End]} \cdot Shares_{Start}^V}}{1 - \frac{Net\ Buyback_{Start}}{MarketCap_{Start}}} \quad (41)$$

Note that the overall numerator is how much of the intrinsic value remains after the share buyback, and the overall denominator is how much of the market-cap remains after the share buyback. Apart from taxes and the subscripts that indicate the *Start* and *End* periods, this is the same as Eq.(8).

4.4 Return Ratios

The Return on Intrinsic Value (ROIV) measures the gain/loss in per-share intrinsic value resulting from the share buyback or issuance during the *Start* period, for the shareholders who continued to own their shares until the *End* period. It is defined similarly to Eq.(9) but using W from Eq.(36) and V from Eq.(39):

$$ROIV_{[Start, End]} = \frac{W_{[Start, End]}}{V_{[Start, End]}} - 1 \quad (42)$$

The Return on Buyback (ROB) measures the gain/loss in per-share intrinsic value, relative to the amount that was either spent on a share buyback or received in a share issuance, for an investor who owned the shares between the *Start* and *End* periods. The formula for the ROB ratio is similar to Eq.(10), but in order to handle both share buyback (positive) and issuance (negative) in a single formula, we need to take the absolute value of the net buyback amount, so the sign of the ROB ratio is only determined by the difference in the per-share intrinsic value:

$$ROB_{[Start, End]} = \frac{W_{[Start, End]} - V_{[Start, End]}}{|Net\ Buyback_{Start}| / Shares_{Start}^V} \quad (43)$$

We have experimented with different ways of plotting $ROIV_{[Start, End]}$ and $ROB_{[Start, End]}$ using 2D distributions to show the return ratios for all possible *Start* and *End* periods, but those plots were hard to understand. The plot that is easiest to understand only shows the mean and standard deviation for each *Start* period and all subsequent *End* periods. Several examples are shown in the case studies further below.

4.5 Example

Now consider an example with the following historical data for the *Start* period:

$$\begin{aligned} Shares_{Start}^W &= 1\ b & SharePrice_{Start} &= \$100 \\ MarketCap_{Start} &= Shares_{Start}^W \cdot SharePrice_{Start} = \$100\ b \\ Net\ Buyback_{Start} &= \$10\ b \end{aligned} \quad (44)$$

The number of shares WITHOUT the share buyback in the *Start* period is estimated using Eq.(37):

$$Shares_{Start}^V = \frac{Shares_{Start}^W}{1 - \frac{Net\ Buyback_{Start}}{MarketCap_{Start}}} = \frac{1\ b}{1 - \frac{\$10\ b}{\$100\ b}} \simeq 1.111\ b \quad (45)$$

The following table shows the dividends and share-prices for the following 5 years, as well as the results of using the formulas above to calculate the intrinsic values and return ratios for the share buyback in the *Start* period. This uses a discount-rate of $d = 8\%$ in the present value calculations:

Year t	1	2	3	4	5
<i>Dividend Per Share</i> $_t$	\$5	\$5.25	\$5.50	\$5.75	\$6
<i>SharePrice</i> $_t$	\$110	\$50	\$80	\$120	\$150
$W_{[Start, t]}$	\$106.48	\$52.00	\$77.00	\$105.93	\$123.89
$V_{[Start, t]}$	\$104.83	\$55.79	\$78.30	\$104.33	\$120.50
$ROIV_{[Start, t]}$	+1.6%	-6.8%	-1.7%	+1.5%	+2.8%
$ROB_{[Start, t]}$	+18%	-42%	-14%	+18%	+38%

Whether the share buyback in the *Start* period resulted in a gain or loss to the shareholders at that time depends on how long they held on to their shares, as shown in the table's last two rows.

For example, for the shareholders who sold their shares after 1 year, the share buyback during the *Start* period caused a gain in the per-share intrinsic value of $ROIV_{[Start, 1]} \simeq +1.6\%$, and when compared to the amount used for the share buyback the gain was $ROB_{[Start, 1]} \simeq +18\%$.

But for the shareholders who sold their shares after 2 years when the share-price crashed to half of what it was during the *Start* period, the share buyback caused a loss in the per-share intrinsic value of $ROIV_{[Start, 2]} \simeq -6.8\%$, and when compared to the amount used for the share buyback the loss was $ROB_{[Start, 2]} \simeq -42\%$. This is because the intrinsic value would have been significantly greater if the amount used for the share buyback in the *Start* period had instead been paid out as a dividend.

For all holding periods between 1 and 5 years, the mean ROIV was about -0.5% and the standard deviation was about 3.5%. The mean ROB was about +3.4% and the standard deviation was about 28%. Note that the two means have opposite signs, which happens sometimes because of the non-linearity of the ROIV formula, which makes losses have greater magnitude than gains. This is an issue that was analyzed in great detail in [Pedersen 2012] because it has important implications.

Let us demonstrate how the values in the table above are calculated. Consider an investment period of 3 years and a discount-rate of $d = 8\%$. We calculate the per-share intrinsic value with and without the share buyback in the *Start* period, so we can compare those two scenarios.

The per-share intrinsic value WITH the share buyback is calculated using Eq.(36):

$$\begin{aligned} W_{[Start, 3]} &= \sum_{t=1}^3 \frac{Dividend\ Per\ Share_{Start+t}}{(1+d)^t} + \frac{SharePrice_3}{(1+d)^3} \\ &= \frac{\$5}{1.08^1} + \frac{\$5.25}{1.08^2} + \frac{\$5.50}{1.08^3} + \frac{\$80}{1.08^3} \simeq \$77.00 \end{aligned} \quad (46)$$

The per-share intrinsic value WITHOUT the share buyback is estimated using Eq.(39):

$$\begin{aligned} V_{[Start, 3]} &= \frac{W_{[Start, 3]} \cdot Shares_{Start}^W + Net\ Buyback_{Start}}{Shares_{Start}^V} \\ &\simeq \frac{\$77.00 \cdot 1\ b + \$10\ b}{1.111\ b} \simeq \$78.30 \end{aligned} \quad (47)$$

The ROIV ratio is calculated using Eq.(42):

$$ROIV_{[Start, 3]} = \frac{W_{[Start, 3]}}{V_{[Start, 3]}} - 1 = \frac{\$77.00}{\$78.30} - 1 \simeq -1.7\% \quad (48)$$

The ROB ratio is calculated using Eq.(43):

$$ROB_{[Start, 3]} = \frac{W_{[Start, 3]} - V_{[Start, 3]}}{|Net\ Buyback_{Start}| / Shares_{Start}^V} \simeq \frac{\$77.00 - \$78.30}{\$10\ b / 1.111\ b} \simeq -14\% \quad (49)$$

In practice, we let a computer calculate the ROIV and ROB ratios like this for all *Start* and *End* periods, and then it calculates the mean and standard deviation for each *Start* period and all the subsequent *End* periods. This results in a time-series of means and standard deviations which can then be plotted, as demonstrated in the case studies further below.

4.6 Caveats

- Taxes for dividends and capital gains are ignored here because they are different between investors and the taxes may also change over time. But Eqs.(36) and (39) could be extended similarly to section 3 to include taxes for dividends and capital gains.
- Funding of share buybacks is ignored because it is too difficult to try and determine whether a share buyback was funded from excess cash, operating cash-flows, or debt issuance.
- Historical share issuance valuation can be quite misleading, because a share issuance is normally used to fund growth, or pay down debt to avoid bankruptcy. So the intrinsic value without a share issuance usually cannot be calculated accurately by merely adjusting the intrinsic value with the share issuance amount and adjusting the number of shares, as we do here, because the future earnings might be very different without the share issuance. So the valuation formulas above only give a very rough estimate of the gain/loss resulting from a historical share issuance.
- Share issuance to fund an acquisition or merger with another company does not appear on the cash-flow statement of the acquiring company because it is not a cash transaction. It is disclosed in the notes of the financial statements. This data is not available in the financial database we use here, so this kind of share issuance is not shown in the plots further below.
- A spin-off is when a parent company divests a subsidiary into a separate company, and gives the shares in the new company to the shareholders of the parent company. This should be included when valuing historical share buybacks and issuances for the parent company, by considering the market-value of the new shares as a special dividend payout in Eq.(36). This data is available in the financial database we use here, so it is included in our calculations.
- A split-off is when a parent company divests a subsidiary into a separate company, and the shareholders of the parent company have the choice of getting shares in the new company in exchange for some of their shares in the parent company, which are then cancelled / retired. This is a form of share buyback where the parent company pays for its own shares with a subsidiary company instead of cash. This is a non-cash transaction that is disclosed in the notes of the financial statements of the parent company. This data is not available in the financial database we use here, so split-offs and related buybacks are not shown in the plots further below. The subsequent dividends and share-prices for the parent company already reflect the decreased share count and divestiture of the subsidiary, so further adjustments for the split-off are not needed in Eq.(36). A future research project could study the valuation of split-offs to assess whether a long-term shareholder should choose the parent or subsidiary company, when the market-value of their shares may not equal their intrinsic values.

5 Other Issues

5.1 Borrowing

Section 4.9 of [Pedersen 2012] studied how share buybacks funded with borrowed money impacts the intrinsic value. It was suggested increasing the discount-rate in the present value calculations to account for the increased risk of the company defaulting on its debt. Alternatively a revaluation term could be subtracted from the intrinsic value to account for the increased risk of default. Both methods resulted in valuation formulas that were overly sensitive to even small adjustments, so it was concluded that more research was needed.

A much better solution was found when developing the simulation models on the SimSim.Run website. Instead of only having the debt issuance in the value WITH a share buyback W , we now also have the debt issuance in the value WITHOUT a share buyback V . You can also increase the discount-rate if you want to account for the increased risk of the company defaulting on its debt.

This creates an even comparison between the dividend and share buyback scenarios, because the money is borrowed in both scenarios, and it is either used for a share buyback, or paid out as a dividend immediately, so the company would change its capital structure equally in both scenarios.

The formulas for the intrinsic value in Eqs.(1) and (21) have the borrowed amount added, and the interest payments subtracted from the future earnings, similar to this basic formula:

$$v = \text{Excess Cash} + \text{Borrow} + \sum_{t=1}^{\infty} \frac{\text{Earnings}_t - \text{Borrow} \cdot \text{Interest Rate}}{(1+d)^t} \quad (50)$$

The interest-rate should be given as the after-tax rate. For example, if the interest-rate is 5% and the corporate tax-rate is 20%, then the after-tax interest rate is only $5\% \cdot (1 - 20\%) = 4\%$

The interest payments should also affect the future share-prices if they are simulated from e.g. the future earnings and P/E ratios, because the interest payments are subtracted from the earnings.

Section 4.9 of [Pedersen 2012] had some closing remarks that are worth repeating here:

A share buyback for borrowed money should only be made when the market-cap is significantly below the intrinsic value to long-term shareholders. If the market-cap is near the intrinsic value then the recapitalization has no potential for gain but a large risk of loss of shareholder value.

Conversely, a share issuance to repay debt should only be made when the market-cap is significantly above the intrinsic value to existing shareholders, otherwise the dilution will decrease the value to existing shareholders. But a share issuance to repay debt is usually only done under distress when there is no choice. Therefore companies should avoid getting into unsustainable levels of debt, as it may have to be repaid from a share issuance when the share-price is low and hence causing a loss to existing shareholders.

5.2 Earnings Yield

The future earnings yield of a company is the estimated earnings for the next year divided by the current market-cap (or equivalently the earnings per share divided by the current share-price).

The earnings yield can be compared to e.g. low-risk bond yields, to quickly assess whether the stock is cheap or expensive. This is such a useful valuation tool that it is featured in several places on the [SimSim.Run](#) web-site, but it is also limited as it only considers a single year into the future.

For example, using data from the case study in section 7 further below, the company's historical sales and net profit margin data can be selected as shown in Figure 16 and Figure 17, and the selected data is then multiplied with each other to simulate the earnings shown in Figure 18, which is then divided by the market-cap at the time to simulate the earnings yield shown in Figure 9 here.

U.S. government bond yields with 10-year maturity were about 4.3% in 2025. The company had no debt so there was no risk of default. But the company's future is still much more uncertain than low-risk government bonds, so a significant risk premium would be appropriate. A reasonable hurdle rate would probably be 9%. If we compare that to the simulated earnings yield in Figure 9, we see that well over half of the distribution is above the hurdle rate, but a significant part is below. Therefore, it is quite possible that the actual earnings yield could be less than the required return, so a share buyback would result in a loss for the remaining shareholders.

But this is a very limited way of assessing whether a share buyback may result in a gain or loss to the remaining shareholders, in part because it only considers earnings for a single future year, and also because it cannot quantify the percentage gain/loss as done using the ROIV and ROB ratios.

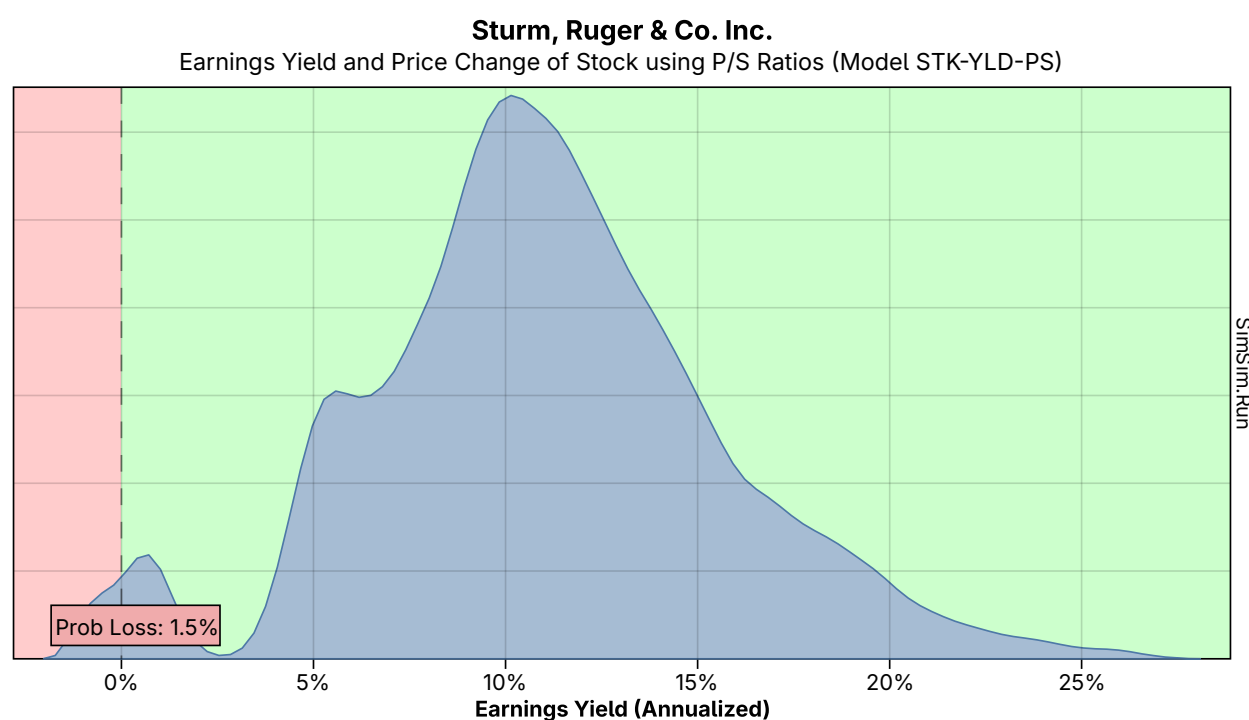


Figure 9: **Sturm, Ruger & Co.** Simulated earnings yield using the simulated earnings in Figure 18 divided by the company's market-cap at the time. See section 7 for the full case study.

5.3 Internal Rate of Return (IRR)

The Internal Rate of Return (IRR) is the discount-rate that makes the present value equal to the purchase price.² The IRR measures the annualized rate of return on an investment, so it can easily be compared to e.g. government bond yields to assess whether a stock is cheap or expensive. The IRR also supports multiple years of earnings, as opposed to the earnings yield which only considers a single year. Unfortunately there is no formula for finding the IRR, so an algorithm has to search for the IRR, which sometimes fails for difficult input values. In some cases the IRR is not even well-defined mathematically, or there may be several different values that satisfy the equation.

For a company's intrinsic value in Eq.(1) the IRR is the discount-rate that satisfies this equation:

$$MarketCap = Excess\ Cash + \sum_{t=1}^{\infty} \frac{Earnings_t}{(1 + IRR)^t} \quad (51)$$

When using this definition of intrinsic value, ignoring dividend tax, and assuming the IRR is well-defined and unique for the input values, then the IRR is the same before and after a share buyback. This is proven by considering the equation after the share buyback, where the buyback amount must be subtracted from both sides of the equation, and let us denote IRR^W as being WITH the buyback:

$$MarketCap - Buyback = Excess\ Cash - Buyback + \sum_{t=1}^{\infty} \frac{Earnings_t}{(1 + IRR^W)^t} \quad (52)$$

The *Buyback* amount cancels out on both sides of this equation, and because the original IRR was assumed to be well-defined and unique we must have that $IRR = IRR^W$.

The IRR is generally not the exact same before and after a share buyback when also considering dividend taxes in the formulas above, and when considering changes in share-prices in Eq.(22).

You can use the IRR to assess whether a share buyback may cause a gain or loss for the remaining shareholders, by comparing the stock's IRR to a reasonable hurdle rate, but you should keep in mind that the IRR is not always well-defined mathematically, and it also cannot quantify the percentage gain/loss as done using the ROIV and ROB ratios.

The [SimSim.Run](#) web-site has several simulation models for the IRR, and you can easily copy the data between the IRR and share buyback models to see how they behave with different input data.

² This was called "Value Yield" in [Pedersen 2012] because I had reinvented the concept without knowing it already existed under the somewhat misleading name "Internal Rate of Return".

5.4 Alternative Investment

Section 4.7 of [Pedersen 2012] derived valuation formulas for comparing a share buyback relative to an alternative investment that the company could make for the same money (e.g. build a new factory or acquire another company). It is easier and probably better to just compare the probability distributions for the returns on the share buyback and alternative investment, to assess which would be the most profitable for the company's shareholders. We can use one of these methods:

- The IRR of the company's stock can be compared to the IRR of the alternative investment. We don't need to select appropriate discount-rates. We don't even need to use specialized simulation models for share buybacks, as the normal IRR simulation models are sufficient. But the problem is that the IRR is not always well-defined mathematically.
- The ROB ratio of the share buyback can be compared to the NPV ratio of the alternative investment. We need to select appropriate discount-rates and use a specialized simulation model to get the ROB ratio and compare it to the NPV ratio of the alternative investment.

It is very important that you do not mix these two valuation methods! You must not compare IRR and NPV ratios, because it would grossly distort the comparison. This would be like using metric units for the company's stock, and imperial units for the alternative investment.

Figure 10 shows an example with simulated ROB ratios for different share-prices for the buyback, compared to the NPV ratio of an alternative investment. It is not always clear which choice is best.

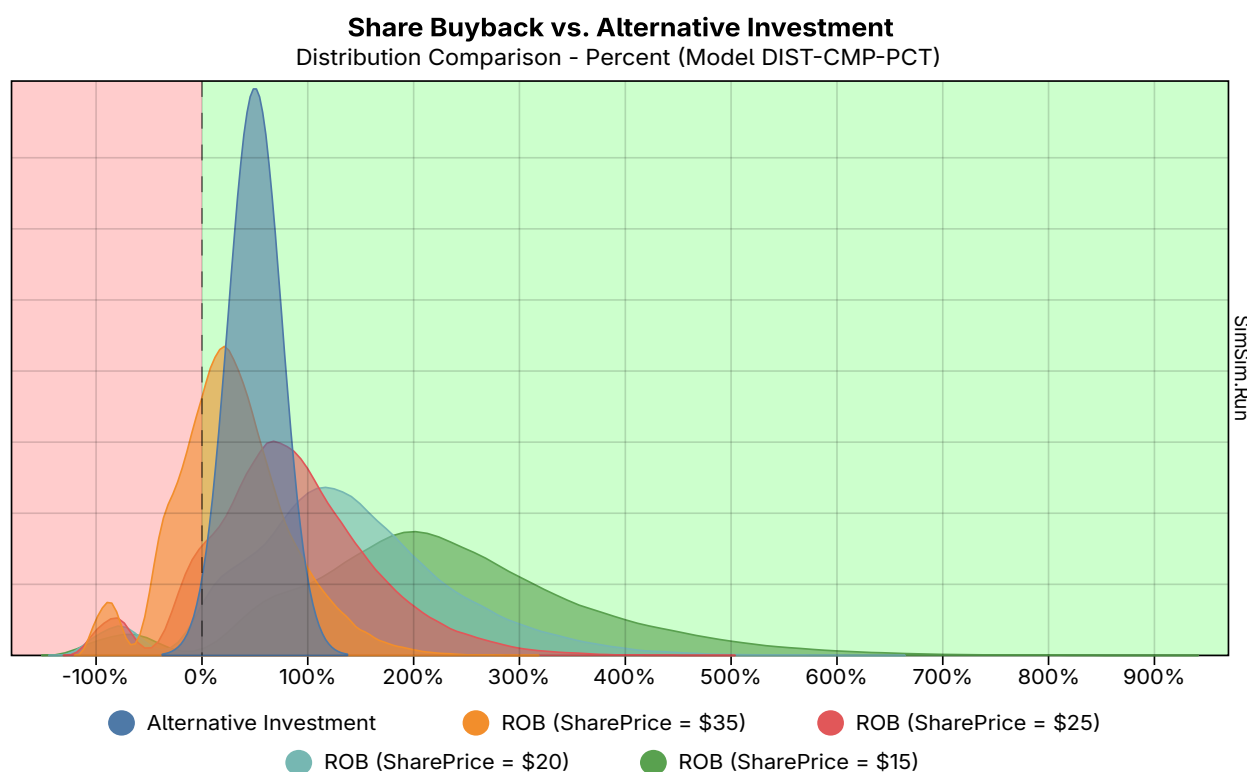


Figure 10: **Sturm, Ruger & Co.** Comparing a share buyback and alternative investment.

ROB ratio is simulated as in section 7.3 with different share-prices for the buyback.

NPV Ratio of the alternative investment is normal-distributed with mean 50% and std.dev. 25%.

6 Case Studies

Let us now consider some real-world case studies using the valuation formulas from the previous sections. The plots in these case studies are taken from the [SimSim.Run](#) web-site, which currently uses a low-cost financial database that aggregates and standardizes the raw financial data for U.S. stocks from the SEC EDGAR database with minimal human intervention. This database has fewer details and is sometimes missing data, and it may contain errors that were also present in the original corporate filings; but it costs a tiny fraction of the human-curated financial databases.

It is not always the people with the most expensive tools who do the best work. Although we use a low-cost financial database, our plots are infinitely more useful than what academia has produced for decades using very expensive financial databases, because they have basically just studied the correlation between share buybacks and abnormal stock-returns, whereas we study the actual change in long-term intrinsic shareholder value that was caused by the historical share buybacks.

The discount-rate is 8% for plots that use the present value of historical dividends and share-prices.

6.1 U.S. Stocks Overall

We first consider the 2600 largest U.S. stocks in aggregate. The number 2600 is a result of various selection criteria for the stocks. These stocks represent nearly the entire U.S. stock-market in terms of market-cap. We then calculate the historical ROIV and ROB ratios using Eqs.(42) and (43) for all these stocks and for all possible *Start* and *End* quarterly periods from the late 1990s to early 2026. This gives a total of about 6.8 million historical data-points with ROIV and ROB ratios. Nearly 3% of these data-points ($\pm 5 \times$ Inter-Quartile Range) were removed because they were extreme outliers and we want to focus the plots on the bulk of the distributions.

The plots show 2D distributions similar to 2D histograms or heat-maps where darker blues indicate more values in a region. Recent areas of the plots are “thinner” because there are naturally fewer data-points available for recent years.

Figure 11 shows the historical ROIV ratio for the 2600 stocks. The clipped range is roughly $\pm 0.8\%$.

Figure 12 shows the historical ROB ratio for the 2600 stocks. The clipped range is roughly $\pm 400\%$.

If share buybacks and issuances were truly value-neutral to shareholders, then these plots would have to show all the historical ROIV and ROB ratios being equal to zero (if the discount-rate was correct). In reality, both plots have their distributions centered around zero, but the ROIV and ROB ratios for different stocks and different *Start* and *End* periods are greatly dispersed, and the historical share buybacks and issuances normally resulted in gains or losses for the shareholders.

Also note that there is a pulsating effect that is especially visible in Figure 12 with the ROB ratios. This suggests there were more share buyback and issuance activity during certain periods. For example, note how the ROB ratios are “thinner” during the years 2008-2010, which was the period of a great financial crisis and stock-market crash.

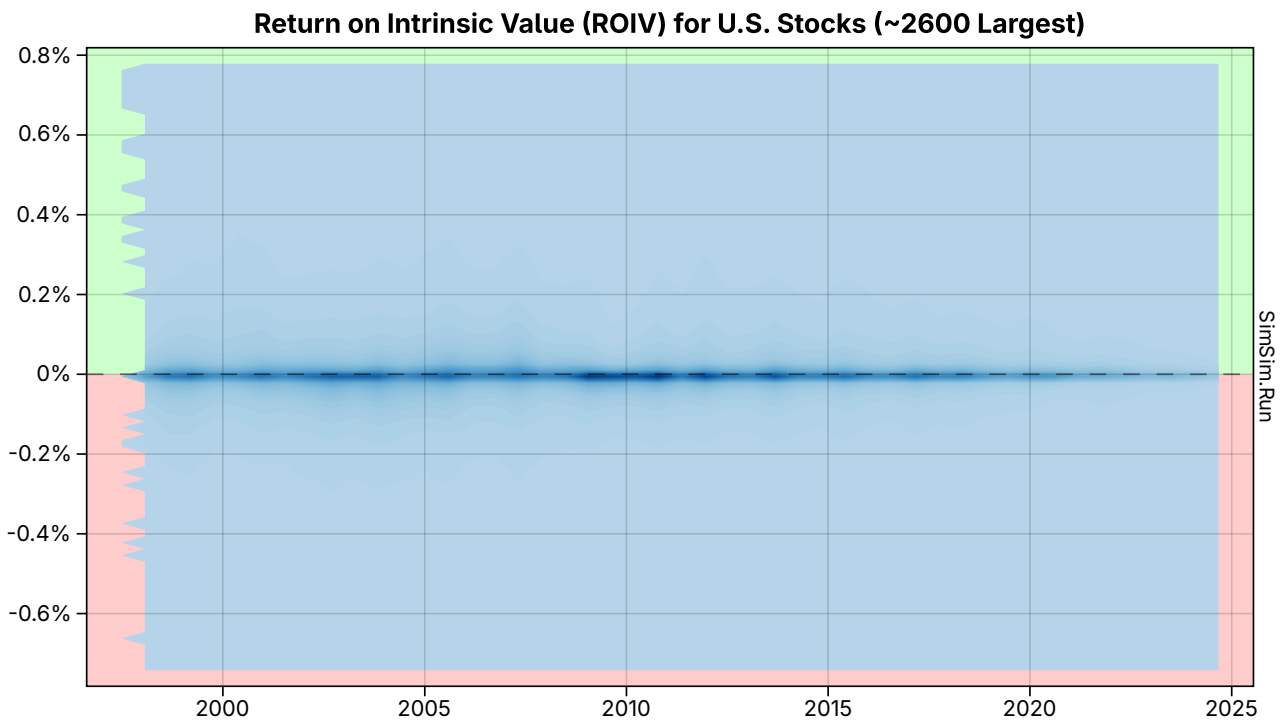


Figure 11: Historical **ROIV** using Eq.(42) for about 2600 of the largest U.S. stocks. This is a 2D distribution where darker blues indicate more values in a region. Clipped at $\pm 5 \times$ Inter-Quartile-Range (IQR).

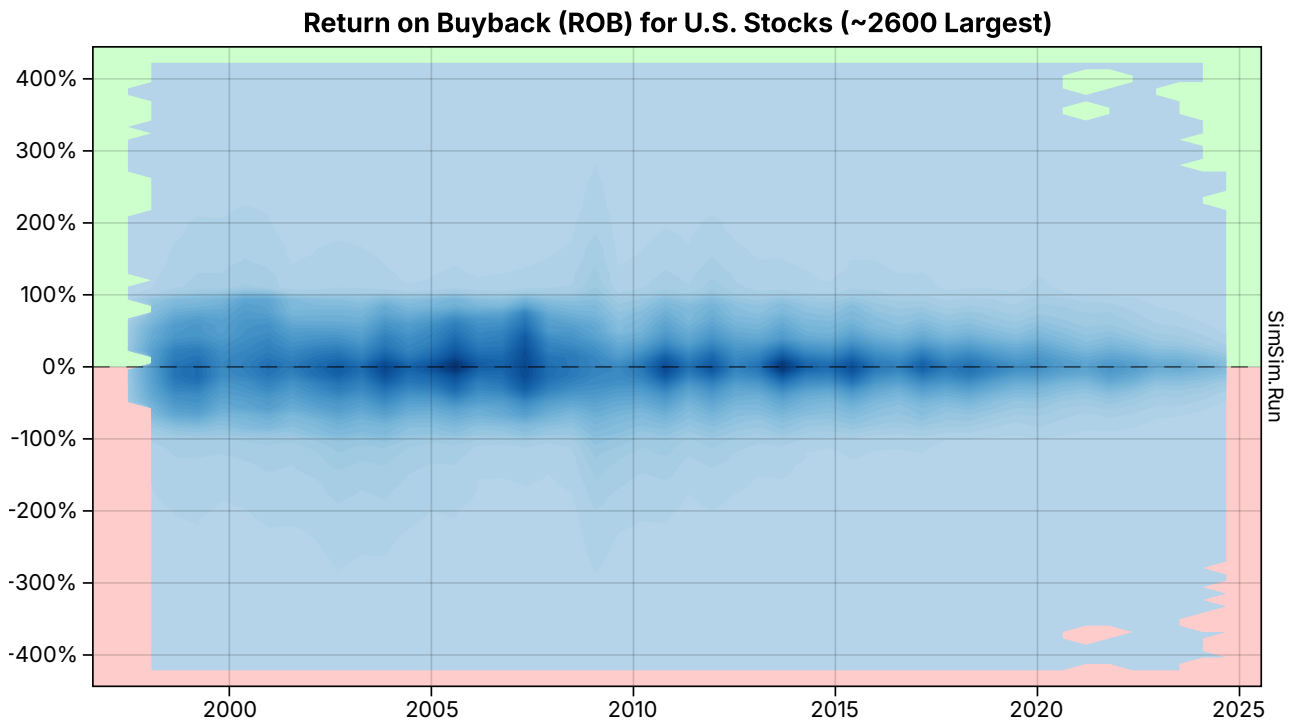


Figure 12: Historical **ROB** using Eq.(43) for about 2600 of the largest U.S. stocks. This is a 2D distribution where darker blues indicate more values in a region. Clipped at $\pm 5 \times$ Inter-Quartile-Range (IQR).

7 Case Study: Sturm, Ruger & Co. (RGR)

This company is a firearms manufacturer founded in 1947 in the USA. This is a very rare example of a company that has been very successful at making share buybacks at the exact right time. The stock-data shown here is taken from the [SimSim.Run](#) web-site (click the link to view the data).

7.1 Historical Share Buyback Valuation

Figure 13 shows the historical valuation of share buybacks and issuances in four sub-plots (on the web-site you can click on the plots to pinpoint information for each data-point):

1. **P/S Ratio** calculated from the average market-cap in the quarter and the TTM sales. This makes it easy to see if the company has had a tendency to buyback or issue shares when the valuation ratio was relatively high or low.
2. **Net Share Buyback (% of Assets)** is positive when the company made a net share buyback and negative for a net share issuance. It is shown relative to the company's total assets so it is easy to see the relative magnitude over time. Share issuances to pay for mergers and acquisitions are typically not shown in this sub-plot.
3. **Return on Intrinsic Value (ROIV)** measures the change in per-share intrinsic value from the net share buyback or issuance. A positive ROIV is a gain for the remaining shareholders and a negative ROIV is a loss. It is calculated using Eq.(42) for all possible *Start* and *End* periods, and the plot shows the mean and two standard deviations for each *Start* period.
4. **Return on Buyback (ROB)** measures the change in per-share intrinsic value relative to the net buyback or issuance amount. A positive ROB is a gain for the remaining shareholders and a negative ROB is a loss. It is calculated using Eq.(43) for all possible *Start* and *End* periods, and the plot shows the mean and two standard deviations for each *Start* period.

Figure 13 shows that the company rarely made share buybacks during the last 30 years, and the few share buybacks they made were generally at low P/S ratios for the stock. Notice a few things:

- In 2006 and 2007, two large share buybacks were made. The ROIV ratios were about 5% and 9% on average, and the ROB ratios were about 170% and 320% on average. So these share buybacks were incredibly good investments for the remaining shareholders.
- In late 2008, a small share buyback was made during a major financial crisis and stock-market crash. The P/S ratio was even lower, but the buyback amount was fairly low so the ROIV ratio was also low. Compared to the buyback amount, the ROB ratio was over 320% on average, so it was still a fantastic investment for the remaining shareholders.
- In late 2014, a large share buyback was made whose average ROIV and ROB ratios were around zero. That is because the net present value of future dividends and share-prices was roughly equal to the share-price at the time of the buyback, as shown in Figure 15.
- In early 2017, a large share buyback was made which had an average ROIV ratio around -1.3% and a ROB ratio around -18%. So far this share buyback has resulted in an average loss to the remaining shareholders, because the share-price fluctuated in the following years,

and in late 2023 the share-price started declining significantly below the 2017 share-prices when the buyback was made. The share-price would have to gain significantly in the next few years, if the buyback in 2017 should have been a gain for long-term shareholders.

- Apart from a small buyback in 2019, the company did not make any buybacks until 2023 when the P/S ratio started becoming low again. This was only a few years ago so there is only a few years of subsequent dividend and share-price data, so we cannot properly value those buybacks yet and we will instead make a simulation of the value further below.

7.2 Market Timing

Figure 14 shows the company's historical dividend payout and net share buyback, both shown as percentages of total assets so it is easy to see the relative magnitude over time. This data is Trailing Twelve Months (TTM) instead of the quarterly data shown in Figure 13, which means that an isolated share buyback in a single quarter will show up in the following three quarters as well.

Figure 15 shows the historical Net Present Value (NPV) ratio, where the present value is calculated using Eq.(36) from subsequent dividends and share-prices, and then divided by the *Start* period's average share-price. These are calculated for all possible *Start* and *End* periods, and the plots show the mean and two standard deviations for each *Start* period and all subsequent *End* periods.

Notice the following from comparing Figure 14 and Figure 15:

- Between 2006 and 2009, dividends were halted and the cash was spent on share buybacks instead (along with cash from divesting unproductive assets, not shown here). This turned out to be a fantastic investment for the remaining shareholders. The NPV ratio was around 60% to 400% on average during this period, because not only was the stock's valuation ratio low, but a few years later the company's sales also grew 4x and its net profit margin expanded to more than 10%. This was partly because the company modernized its manufacturing process and partly due to the political situation in USA at the time.
- After 2009, dividends were reinstated and they were then adjusted to the company's irregular earnings (which are shown on the [SimSim.Run](#) web-site). Share buybacks were only made when the company's management deemed the share-price was sufficiently low.

7.3 Simulation

When valuing historical share buybacks for a *Start* period, we use the subsequent dividends and share-prices to estimate the intrinsic value during the *Start* period. This means the recent share buybacks cannot be valued accurately because there are few or no data-points available for the subsequent dividends and share-prices. This can also be seen from Figure 15 where the NPV ratio narrows around zero in the recent years, because the present value of subsequent dividends and share-prices is nearly the same as the *Start* share-price.

To assess the value of the company's recent share buybacks, we can either just simulate the future earnings and use the valuation formulas from section 2, or we can simulate both the future earnings and share-prices using the formulas from section 3.

We do not have any special insight about the company's future earnings, and in the past the company's sales and profit margins have fluctuated greatly from year to year. That is exactly why we must use probability distributions in these valuations. Figure 16 shows the company's historical sales where we have selected the period from 2015 onwards. Figure 17 shows the company's historical profit margins where we have selected the period from 2010 onwards. Multiplying all the sales and net profit margins from these periods gives the probability distribution for earnings shown in Figure 18, which ranges roughly between USD -5m and +160m with a peak around USD 60m.

We will only simulate the share buyback valuation using future earnings here. We use a model on the [SimSim.Run](#) web-site that supports terminal growth-rates (click the link to try it). You can easily try one of the other models to simulate future share-prices as well, similar to section 3.4.

We use the simulated earnings in Figure 18 and the following assumptions from 2025:

$$\begin{aligned} \text{Shares} &= 17 \text{ m} & \text{SharePrice} &= \$ 35 \\ \text{Buyback} &= \$ 30 \text{ m} & \text{Excess Cash} &= \$ 70 \text{ m} \\ \text{Terminal Growth Rate} &= 0 \% & d &\sim N(\mu = 9 \%, \sigma = 1 \%) \end{aligned} \quad (53)$$

Figure 19 shows the simulated ROIV ratio from Eq.(9), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. The distribution is skewed down because of the non-linearity of Eq.(9), which also creates outliers that have been removed. The clipped range is -10% to +4% with 26% probability of loss. The most likely ROIV ratio is the peak around 1.8%.

Figure 20 again shows the simulated ROIV ratio from Eq.(9), but here the buyback share-price (or equivalently the *MarketCap*) is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the buyback share-price changes, while all other variables remain the same. For example, when the share-price is around \$50 the most likely ROIV ratio would be around 0%.

Figure 21 shows the simulated ROB ratio from Eq.(10), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. The range is roughly -100% to +300% with a peak at 15-20%. The probability of loss is 29%.

Figure 22 again shows the simulated ROB ratio from Eq.(10), but here the buyback share-price (or equivalently the *MarketCap*) is varied on the x-axis. For example, when the share-price is around \$45 the most likely ROB ratio would be around 0%. This is different from the share-price that is most likely value neutral for the ROIV ratio in Figure 20 because of its non-linearity and skewness.

Note that the probability of loss is slightly different between the ROIV and ROB ratios, because it is measured after outliers have been removed, and more outliers were removed for the ROIV ratios.

7.4 Summary

The company's share buybacks in 2006 and 2007 during the turnaround were masterfully done and significantly increased the per-share intrinsic value for the remaining shareholders. The buybacks in 2015 were probably value neutral. The buybacks in 2017 probably caused a small loss unless the share-price increases significantly within the next few years. Simulations for the recent buybacks indicate that they will likely cause a gain in shareholder value, but only time will tell for sure.

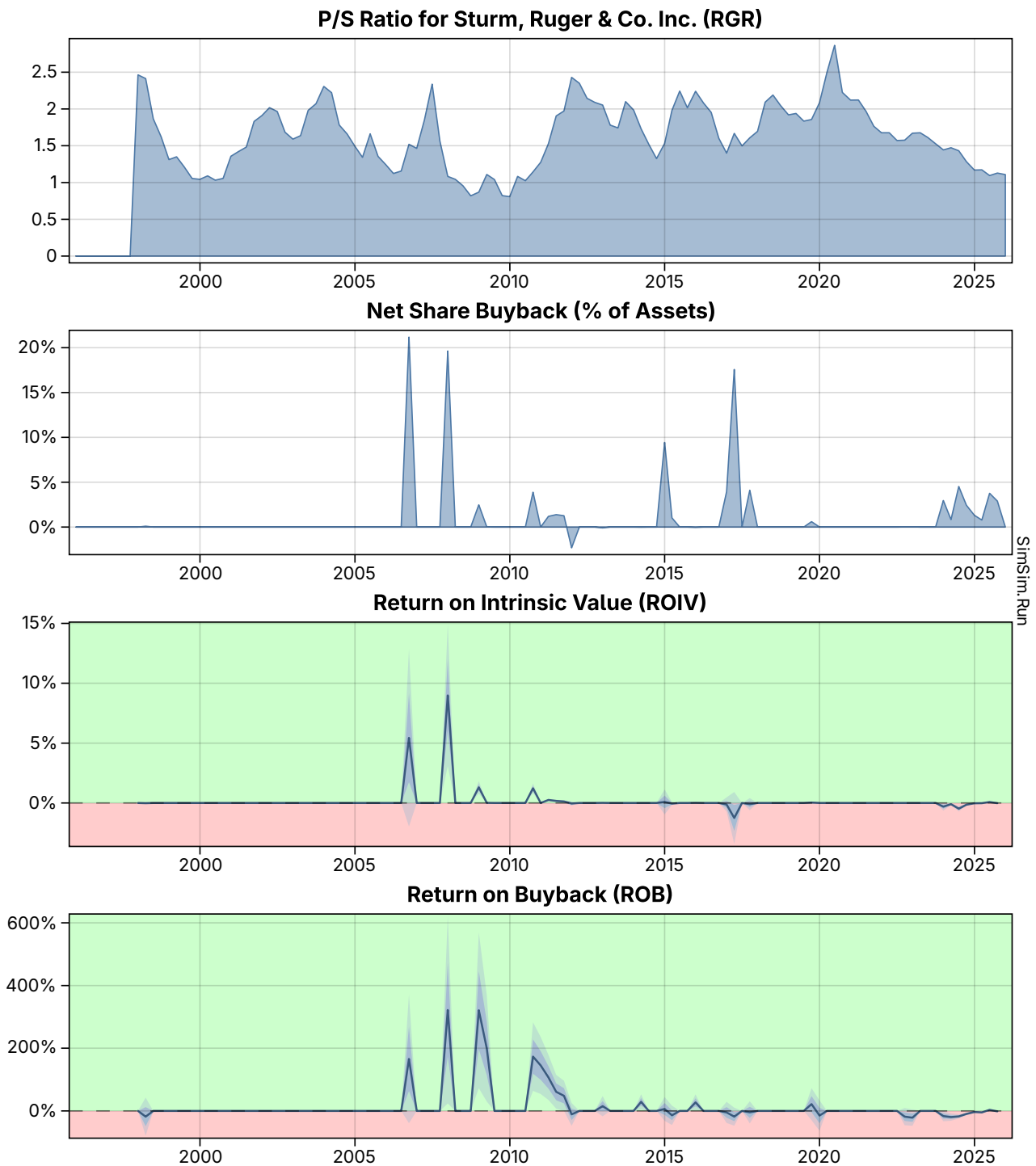


Figure 13: **Sturm, Ruger & Co.** Valuation of historical share buybacks and issuances using ROIV in Eq.(42) and ROB in Eq.(43), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. Also shown is historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)

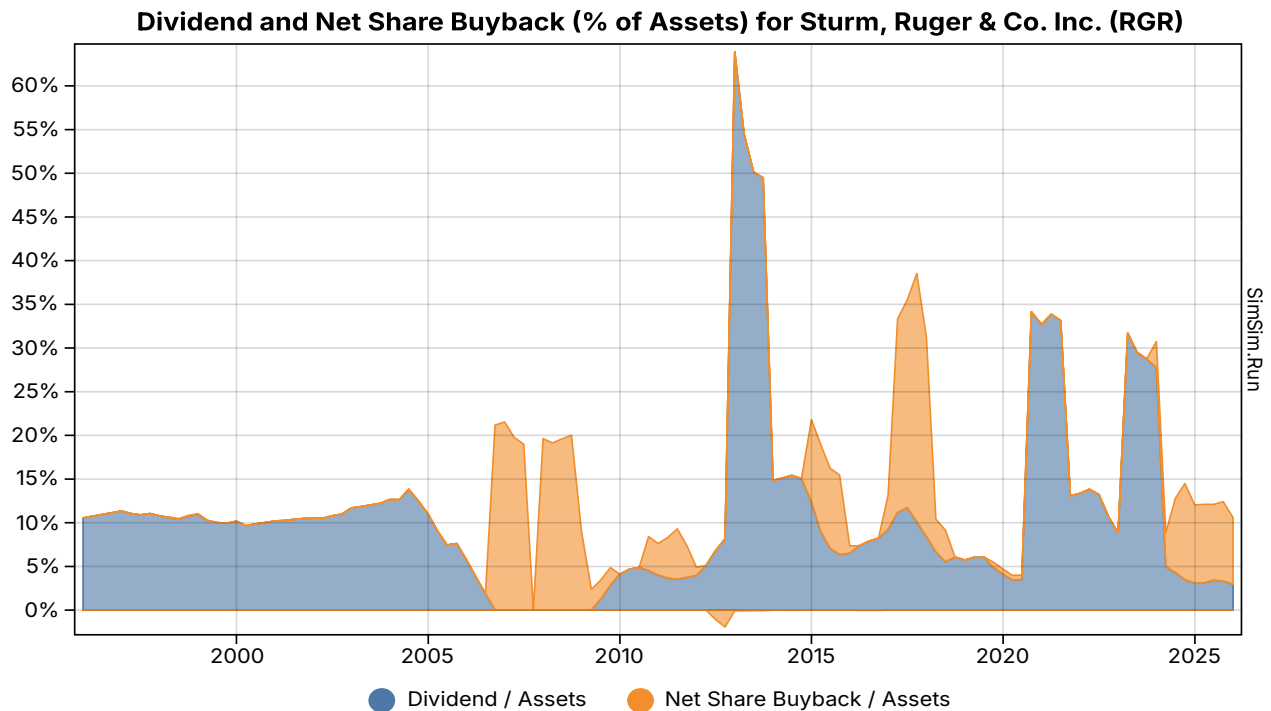


Figure 14: **Sturm, Ruger & Co.** Historical dividend payout and net share buyback as percentages of total assets. (TTM data where each data-point covers the preceding four quarters.)

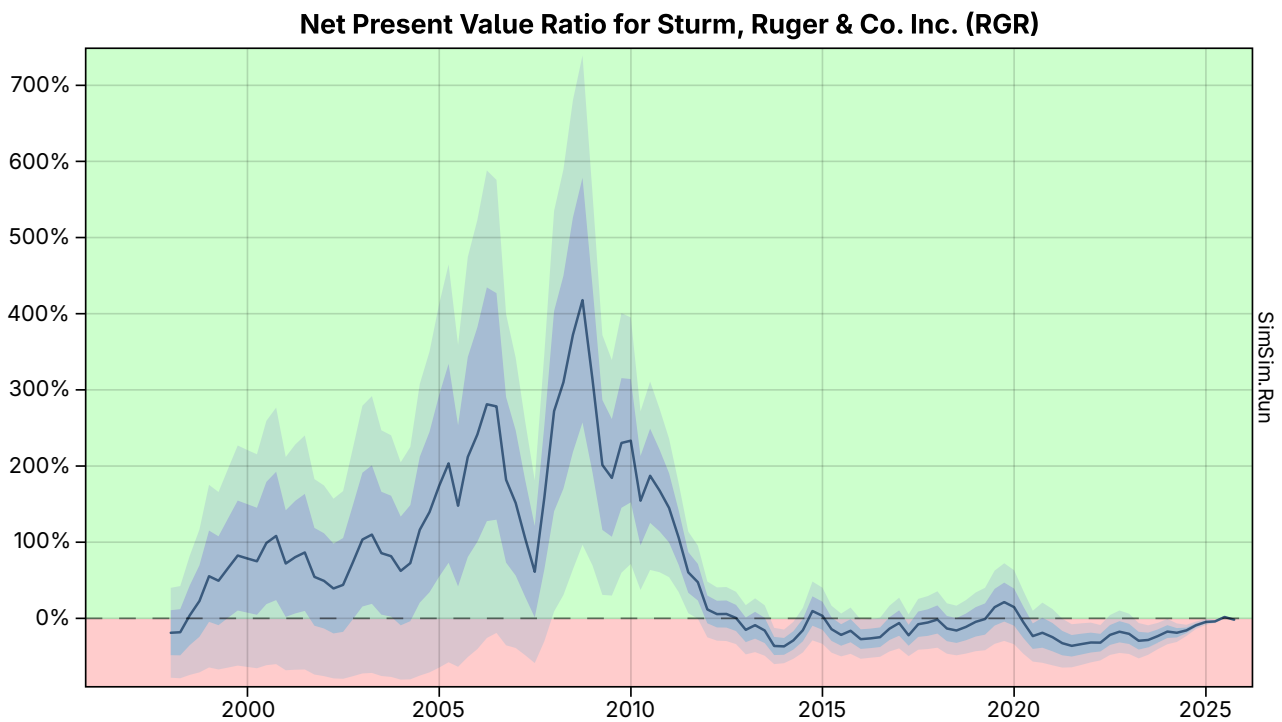


Figure 15: **Sturm, Ruger & Co.** Historical net present value ratio, where the present value is calculated using Eq.(36) from the future dividends and share-prices, and then divided by the Start period's average share-price. These are calculated for all possible Start and End periods, and the plot shows the mean and two standard deviations for each Start period and all subsequent End periods. (Quarterly data.)

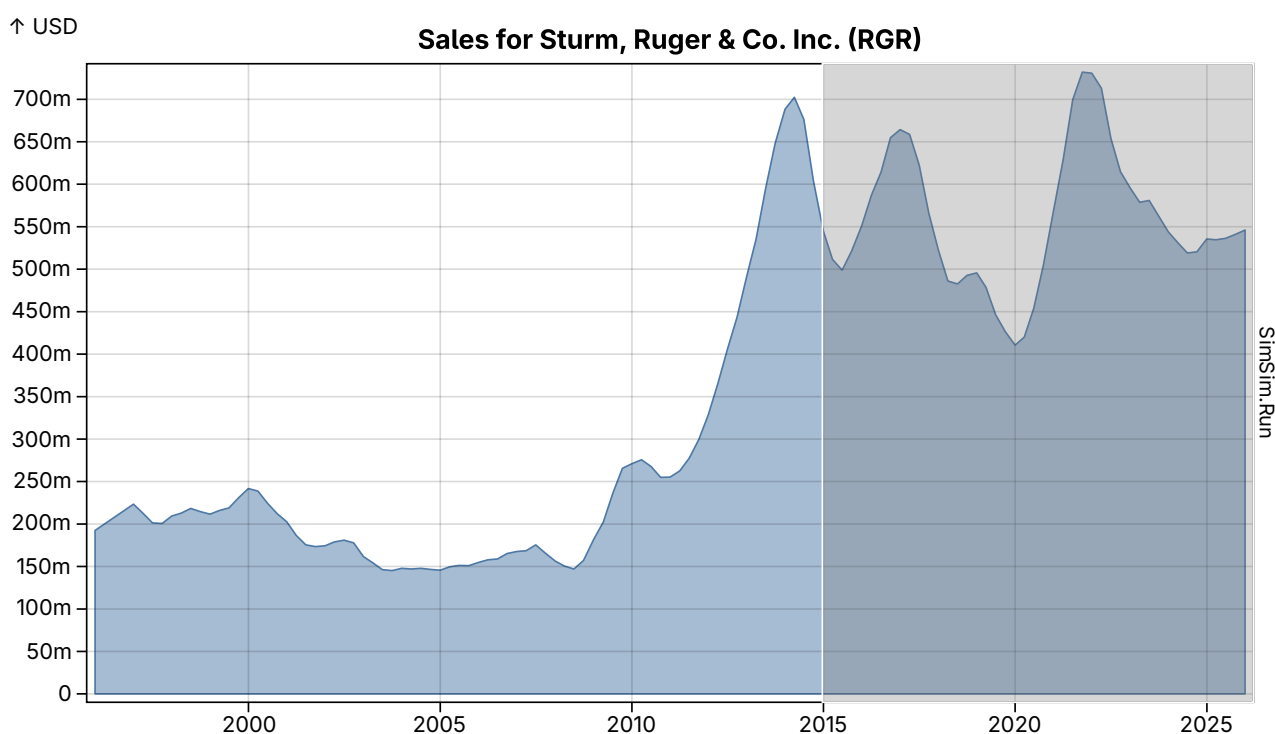


Figure 16: **Sturm, Ruger & Co.** Historical sales for the company. (TTM data.)

The shaded area is the selected data-range that is used to simulate the earnings in Figure 18.

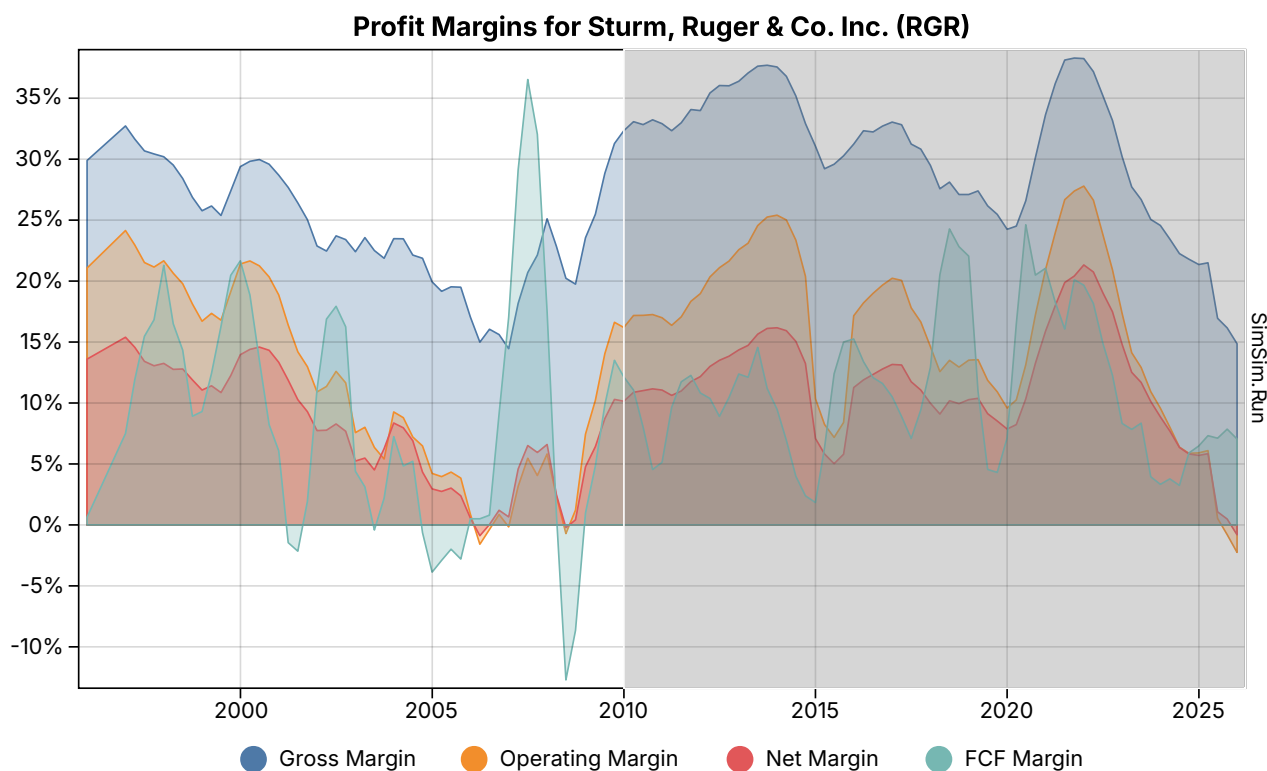


Figure 17: **Sturm, Ruger & Co.** Historical profit margins for the company. (TTM data.)

The shaded area is the selected data-range whose net profit margin is used to simulate the earnings in Figure 18.

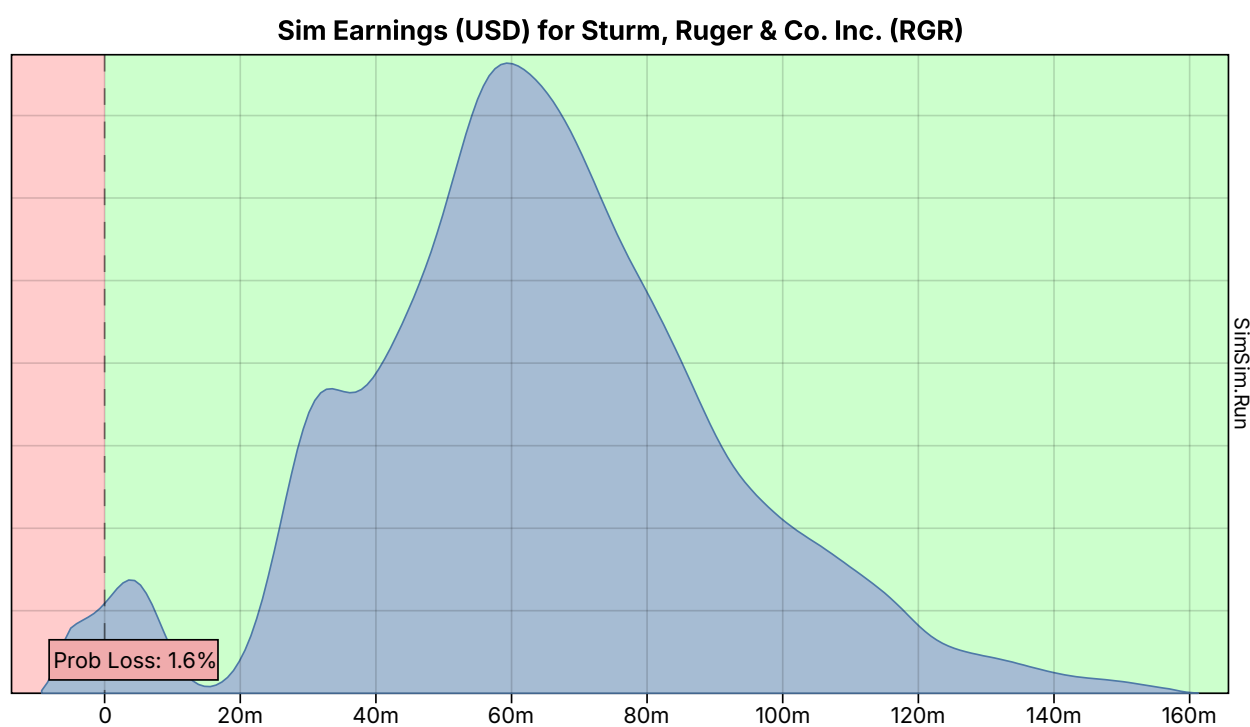


Figure 18: **Sturm, Ruger & Co.** Simulated earnings by multiplying the historical sales that are selected in Figure 16 with the historical net profit margins that are selected in Figure 17.

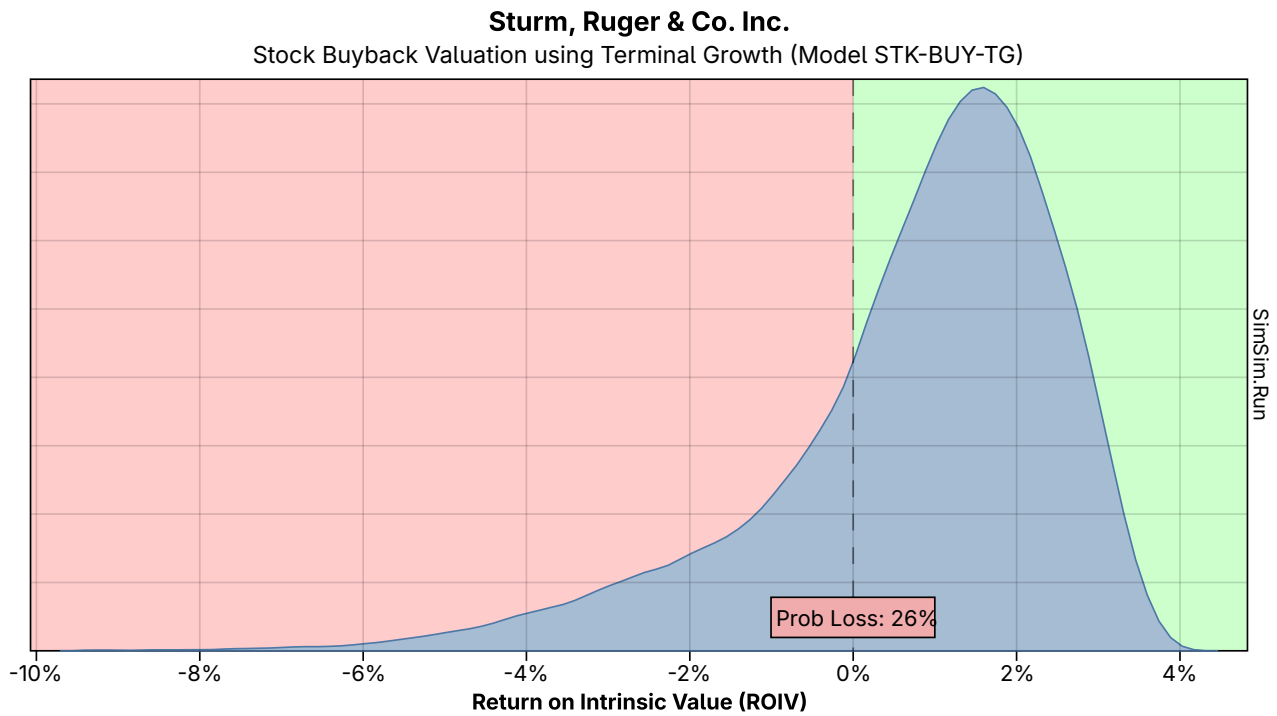


Figure 19: **Sturm, Ruger & Co.** Simulation of **ROIV** in Eq.(9) using the simulated earnings from Figure 18.

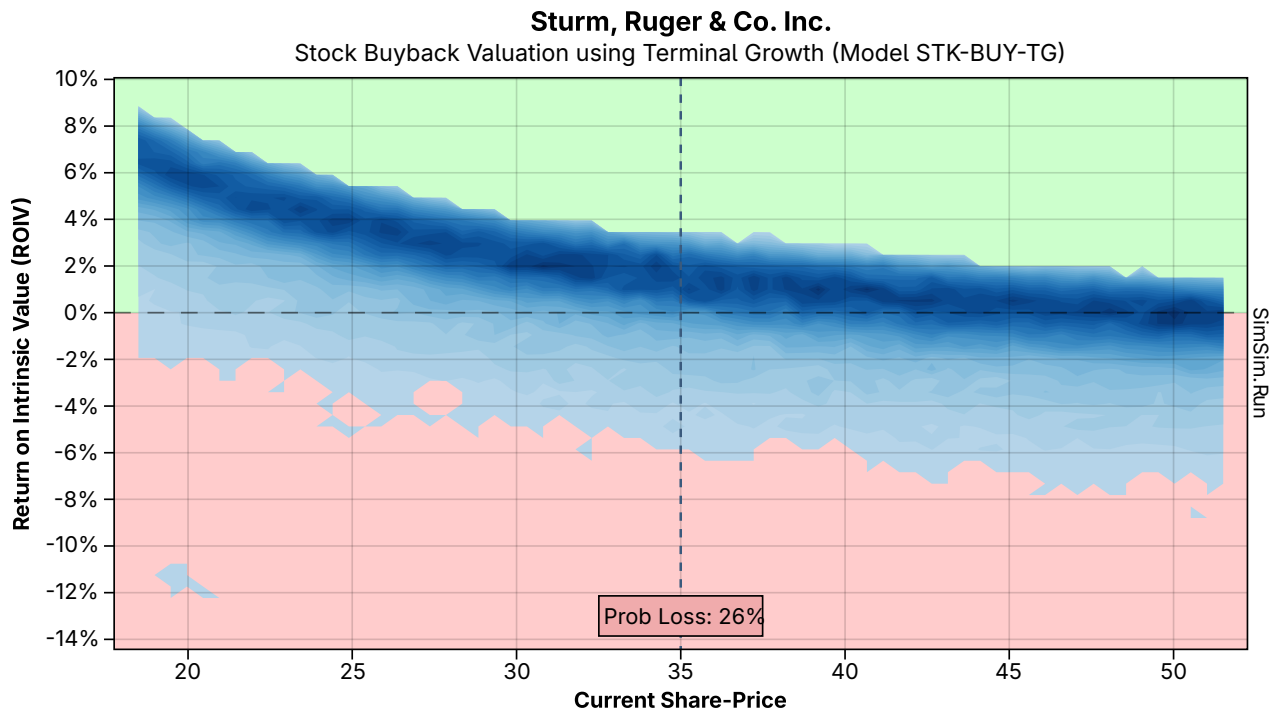


Figure 20: **Sturm, Ruger & Co.** Simulation of **ROIV** in Eq.(9) using the simulated earnings from Figure 18, and varying the buyback share-price (or MarketCap) on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 19 is the vertical dashed blue line here.

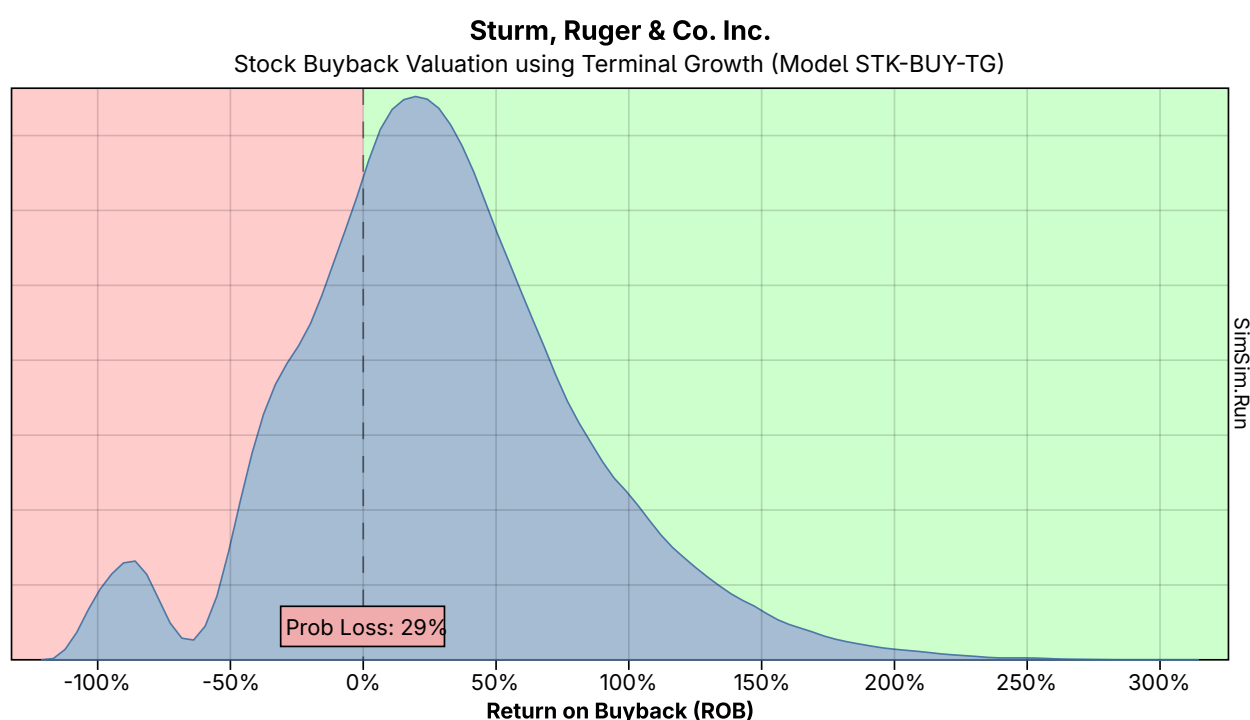


Figure 21: **Sturm, Ruger & Co.** Simulation of **ROB** in Eq.(10) using the simulated earnings from Figure 18.

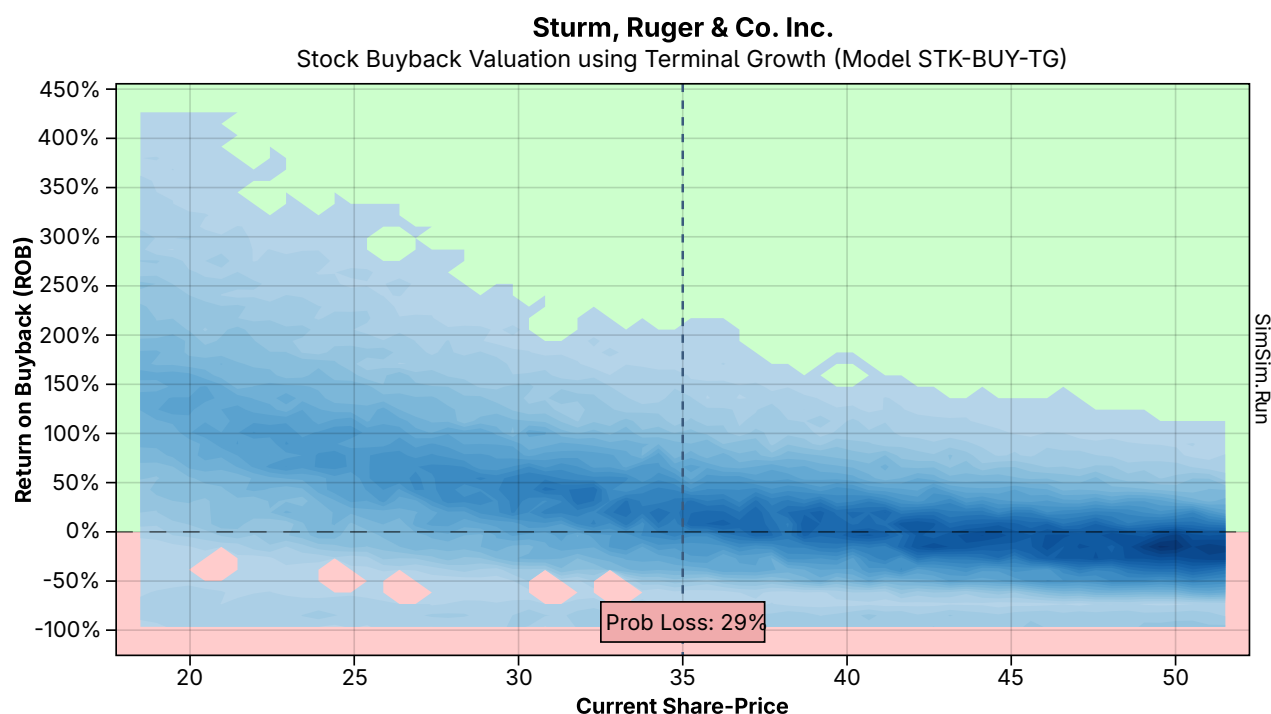


Figure 22: **Sturm, Ruger & Co.** Simulation of **ROB** in Eq.(10) using the simulated earnings from Figure 18, and varying the buyback share-price (or MarketCap) on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 21 is the vertical dashed blue line here.

8 Case Study: Harley Davidson (HOG)

This company is a motorcycle manufacturer founded in 1903 in the USA. This is a typical example of a company that made share buybacks at high prices and halted when the stock-market crashed. The stock-data used here is taken from the [SimSim.Run](#) web-site (click the link to view the data).

8.1 Historical Share Buyback Valuation

Figure 23 shows the historical valuation of share buybacks and issuances in four sub-plots that were explained in section 7.1, so we will just analyze the results here:

- Between 2004 and 2007, the company made a series of large share buybacks when the P/S ratio was significantly higher than in the following decades.
- In 2008, the company stopped its share buybacks as the stock-market crashed in the great financial crisis of the time, and the P/S ratio had dropped 90% from its peak 8 years prior.
- In 2011, the company started making regular share buybacks again until it stopped the buybacks during the Covid pandemic between 2020 and 2022.
- The ROIV ratio shows the gain/loss in per-share intrinsic value from the share buybacks and issuances. The company's historical share buybacks mostly resulted in losses. The worst losses were for the large share buybacks between 2004 and 2007, and again in 2015.
- The ROB ratio shows the gain/loss in per-share intrinsic value, relative to the net buyback or issuance amount. This shows the losses on the share buybacks were often around -50% on the amounts that the company invested in buying its own shares.
- Note how the ROB ratio sometimes zig-zags between gains and losses. It is easier to see why this happens on the web-site where data-points can be pinpointed. It typically happens when the net share buyback amount switches between positive and negative values in successive quarters. For example, if the company makes regular share buybacks, but for some reason they stop in a single quarter and make a small share issuance instead e.g. from employees exercising their stock options. If the share-price was higher than in the following years, then the share issuance resulted in a gain to per-share intrinsic value, while the share buybacks in surrounding quarters resulted in a loss to per-share intrinsic value.
- The present values of future dividends and share-prices that are used in the ROIV and ROB ratios are all made with a discount-rate of 8%. It may be argued that this was too low for this particular company, in which case the present values would have been lower and hence the ROIV and ROB ratios would have shown an even greater loss from the share buybacks.

8.2 Market Timing

Figure 24 shows the company halted its share buybacks between 2009 and 2011 and again between 2020 and 2021, which was the exact opposite of what they should have done, because Figure 25 shows these were the only two periods (apart from the late 1990s) where the present value of future dividends and share-prices was much greater than the share-price, so only share buybacks during

these periods would have increased the per-share intrinsic value to shareholders.

8.3 Simulation A

Let us try and simulate a share buyback during the stock-market crash in late 2008 and early 2009 where the company halted its share buybacks. Assume the company had not made share buybacks in the few years prior and instead accumulated the cash, so the share-count was the same as in 2007. The lowest share-price during the stock-market crash was about \$8 for a short while, but we use \$20 instead because it was below that level for about 8 months or so. These are the assumptions:

$$\begin{aligned} \text{Shares} &= 257 \text{ m} & \text{SharePrice} &= \$ 20 \\ \text{Buyback} &= \text{Excess Cash} = \$ 1.2 \text{ b} \end{aligned} \tag{54}$$

Figure 26 shows the company's historical present value calculated using Eq.(36) with the future dividends and share-prices, and multiplied by the actual share-count at the time to get the total intrinsic value and not per-share. The discount-rate is 8% because it is the same for all companies in the database. The shaded area in Figure 26 is a selection of the present values between 2008-2009. This data is not normal-distributed but the plot only shows the mean and standard deviation for each *Start* period. The selected data has a mean around \$6.75b and a standard deviation around \$2.3b. We add *Excess Cash* = \$ 1.2 b, which we assume was saved from avoiding the share buybacks in the previous years. So we get the following normal-distributed intrinsic value in 2008-2009:

$$v \sim N(\mu = \$ 7.95 \text{ b}, \sigma = \$ 2.3 \text{ b}) \tag{55}$$

We use Eq.(54) and (55) in a simulation model on the [SimSim.Run](#) web-site (click the link to try it).

Figure 27 shows the simulated ROIV ratio from Eq.(9), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. The distribution is skewed down because of the non-linearity of Eq.(9), which also creates outliers that have been removed. The clipped range is -30% to +25% with 10% probability of loss. The most likely ROIV ratio is the peak around 13%.

Figure 28 again shows the simulated ROIV ratio from Eq.(9), but here the buyback share-price is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the buyback share-price changes, while all other variables remain the same. For example, when the share-price is around \$12 the most likely ROIV ratio would be around +40%. The lowest share-price during the 2008-2009 stock-market crash was only around \$8 which is not even shown in this plot.

Figure 29 shows the simulated ROB ratio from Eq.(10), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. The range is roughly -150% to +300% with 11% probability of loss (different from the ROIV ratio due to non-linearity and removal of outliers). The most likely ROB ratio is the peak around +75%.

Figure 30 again shows the simulated ROB ratio from Eq.(10), but here the buyback share-price is varied on the x-axis. For example, when the share-price is around \$29 the ROB ratio would be centered around 0%, which is not the case for the ROIV ratio in Figure 29 due to its skewness.

8.4 Simulation B

Let us try and repeat the simulation but now using data for the following 10 years individually instead of lumping them all together in a single distribution for the intrinsic value, as we did above.

We use a simulation model on the SimSim.Run web-site (click the link to try it) with probability distributions for the P/S ratio and sales for each future year, which are multiplied to simulate the future share-prices for that year. We can use different input values for the future “earnings”:

1. Net Income as reported in the company’s financial statements, adjusted for non-cash items such as goodwill impairments, and subtracting the part of the earnings that is used to grow the future sales. This can be difficult to estimate accurately.
2. Dividends plus net share buyback and stock spin-offs. This is easier to use because it avoids the adjustments needed for the net income, and only considers the cash that was actually paid out to shareholders in some form. But this also includes dividends and share buybacks made for borrowed money e.g. during periods of real earnings losses or re-capitalizations.

Neither of these earnings estimates are perfect. We use the 2nd estimate here, but for this particular company, these two earnings estimates give very similar results in the share buyback simulation.

We use the company’s historical financial data between the years 2009 and 2018 with the same assumptions from Eq.(54) that were used in simulation A. We use a higher discount-rate that is normal-distributed with a mean of 10% and standard deviation of 1%.

The simulation model uses Eqs.(21) and (22) for the per-share intrinsic value without and with the share buyback, which are then used in Eqs.(9) and (10) to calculate the ROIV and ROB ratios.

Figure 33 is a violin-plot of the ROIV ratios for investment periods starting in 2008 / 2009 until each of the 10 years between 2009 and 2018. The first year (2009) had both gains and losses, but the remaining years all had very large gains resulting from the share buyback in 2008 / 2009. The peak was in years 5 and 6, corresponding to years 2013 and 2014 where the share-price peaked.

Figure 34 combines the ROIV ratios for all 10 years and also varies the buyback share-price on the x-axis. This is similar to Figure 28 which only uses a single distribution for the intrinsic value.

Figure 35 is a violin-plot of the ROB ratios for investment periods starting in 2008 / 2009 until each of the 10 years between 2009 and 2018. This has similar features as the ROIV ratios in Figure 33, but with greater magnitudes because the ROB ratio is relative to the net share buyback amount.

Figure 36 combines the ROB ratios for all 10 years and also varies the buyback share-price on the x-axis. This covers roughly the same area as Figure 30 when only using a single distribution for the intrinsic value, but Figure 36 is a bit tighter around the ridge with peak values (the darker blues).

8.5 Summary

This is a typical example of a company that made large share buybacks during good times when the share-price was high, which resulted in losses for the remaining shareholders. The company then halted the share buybacks during stock-market crashes when the share-price was low, where the share buybacks would have caused tremendous gains for the remaining shareholders.

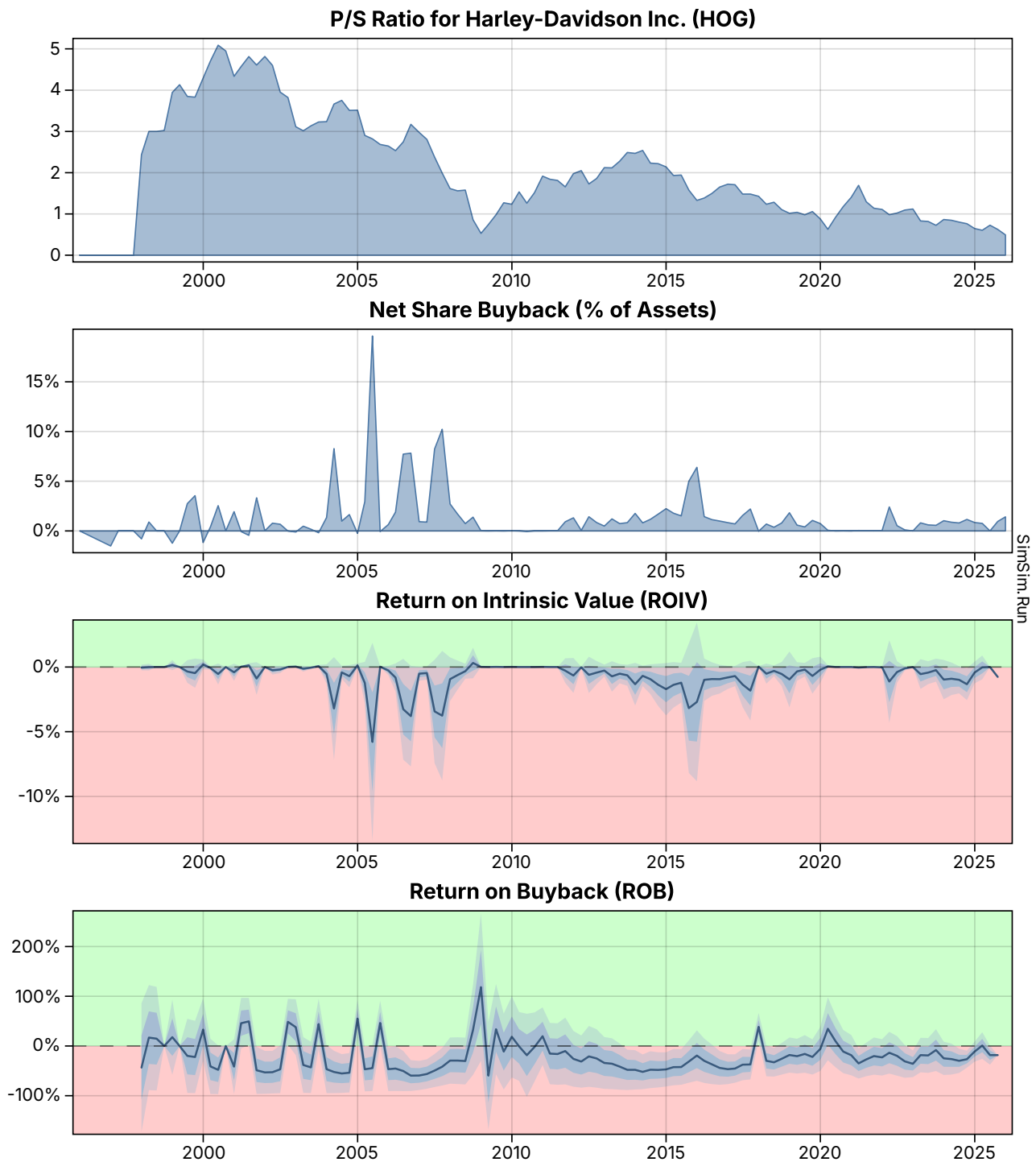


Figure 23: **Harley Davidson**. Valuation of historical share buybacks and issuances using ROIV in Eq.(42) and ROB in Eq.(43), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. Also shown is historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)

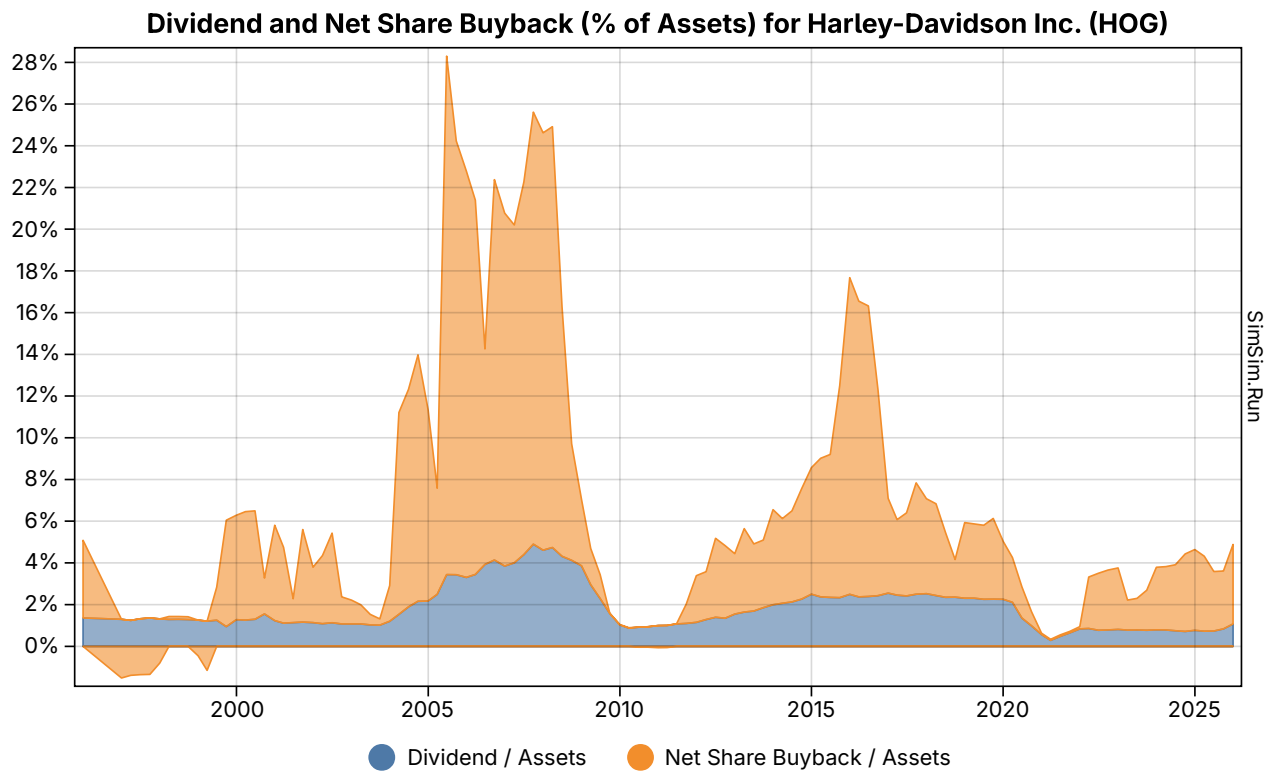


Figure 24: **Harley Davidson.** Historical dividend payout and net share buyback as percentages of total assets. (TTM data where each data-point covers the preceding four quarters.)

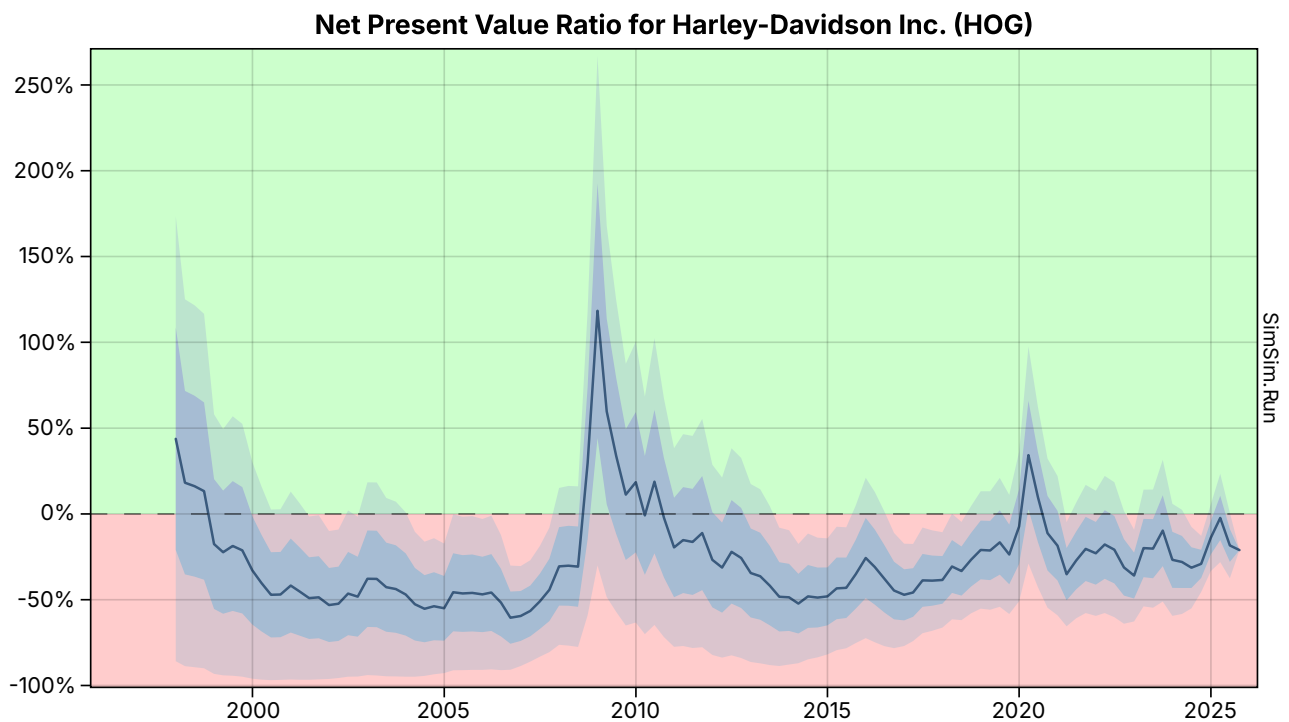


Figure 25: **Harley Davidson.** Historical net present value ratio, using the present value from Figure 26 and then divided by the Start period's average share-price. Plot shows the mean and two standard deviations. (Quarterly data.)

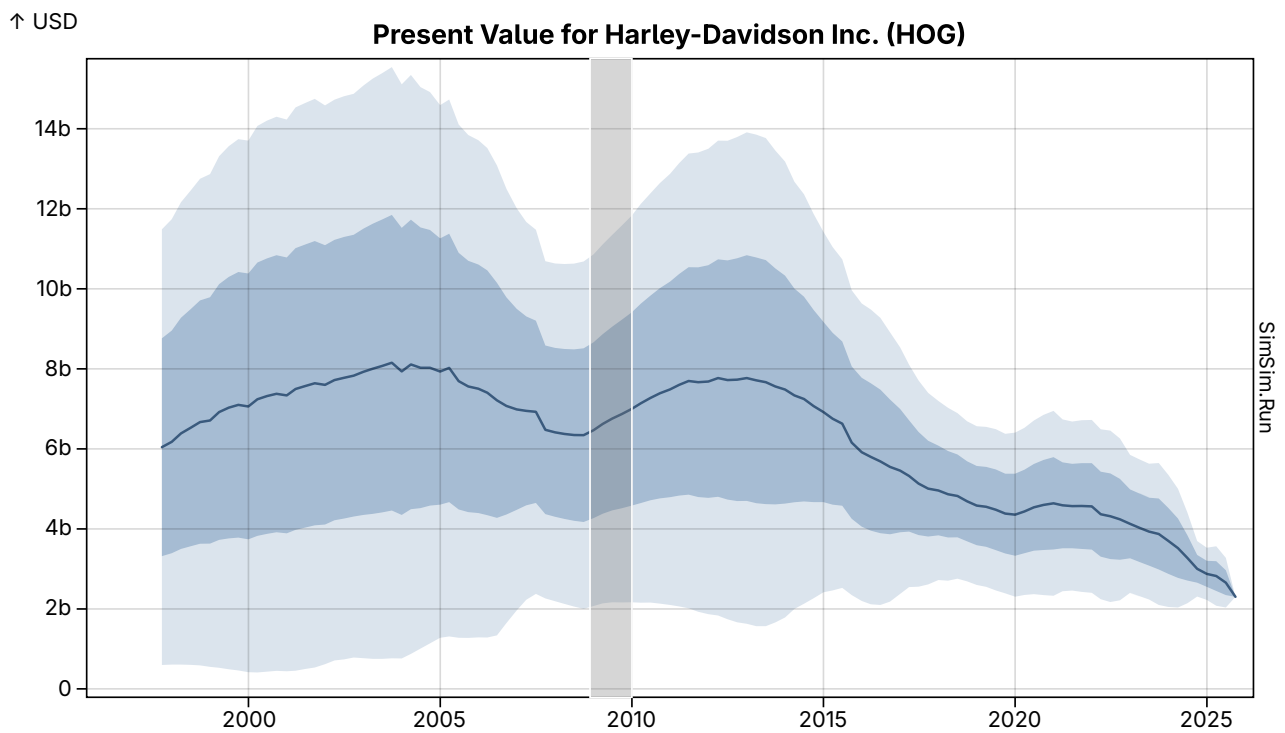


Figure 26: **Harley Davidson.** Historical present value calculated using Eq.(36) with the future dividends and share-prices. Plot shows the mean and two standard deviations for each Start period and all subsequent End periods.

Shaded area is the selected data-range used as the intrinsic value when simulating a share buyback in 2008 / 2009.

(Quarterly data.)

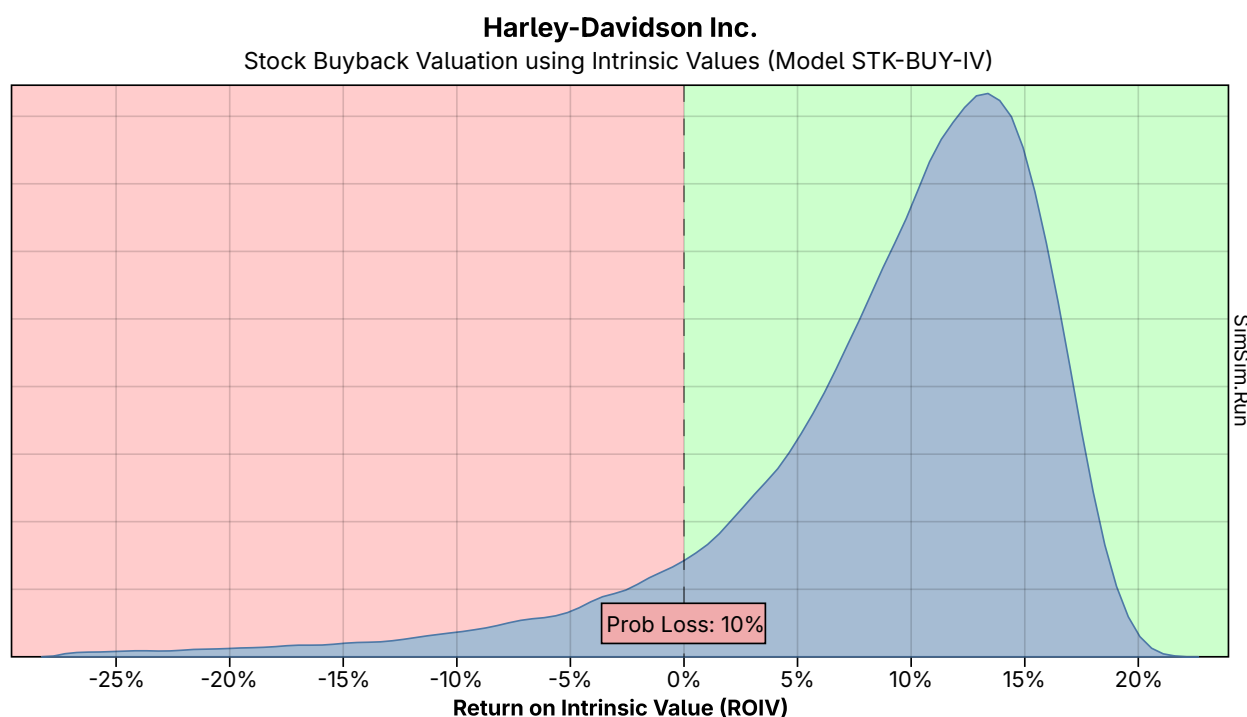


Figure 27: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 26 as the intrinsic value when calculating the **ROIV** ratio using Eq.(9).

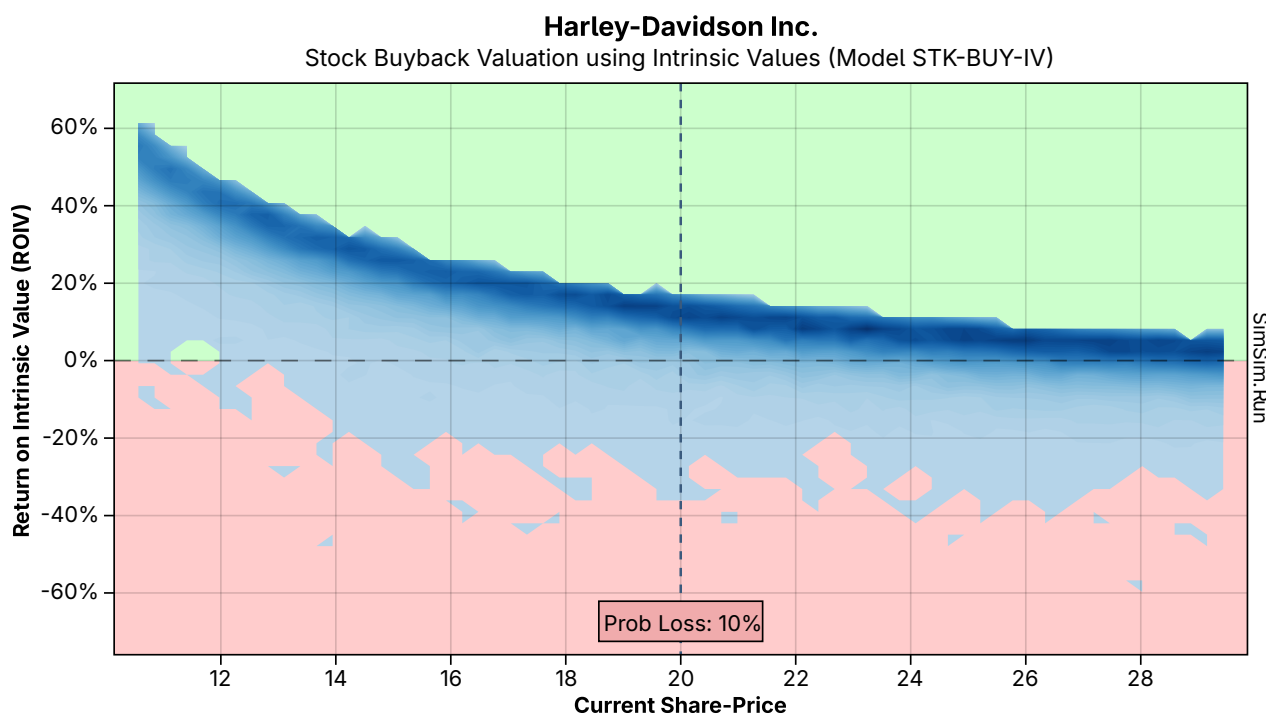


Figure 28: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 26 as the intrinsic value when calculating the **ROIV** ratio using Eq.(9), and varying the buyback share-price on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The vertical dashed line is the distribution shown in Figure 27.

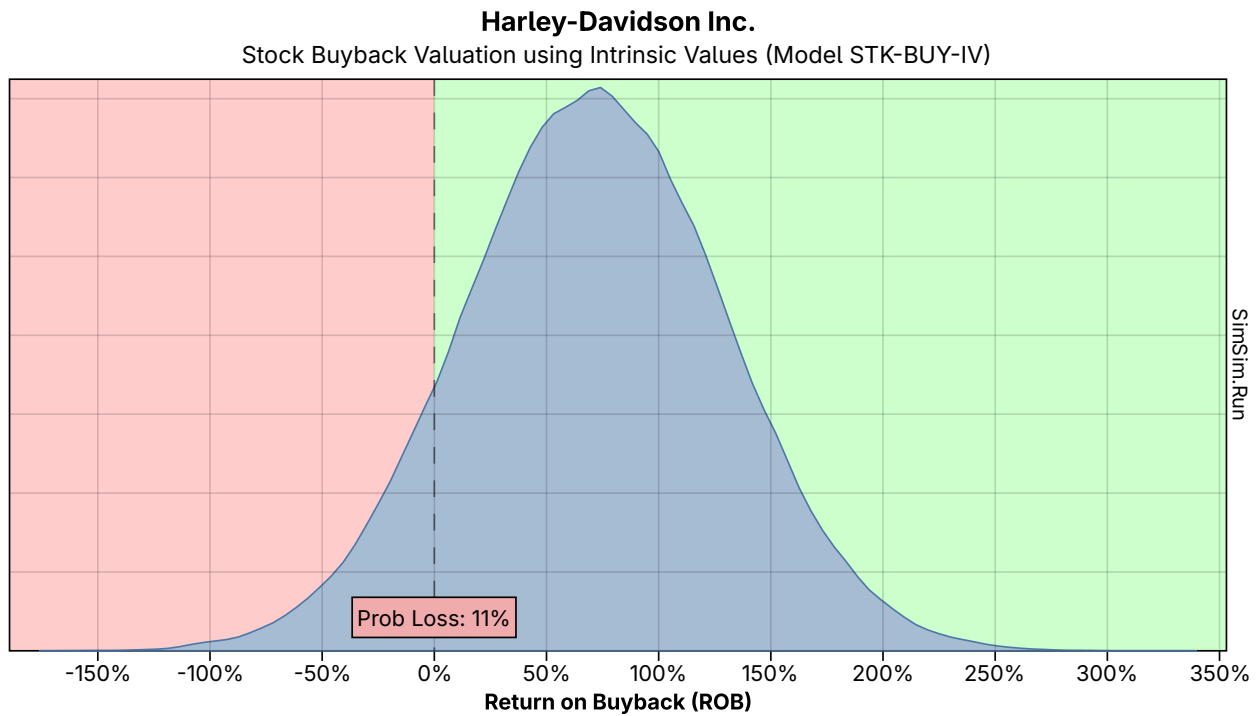


Figure 29: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 26 as the intrinsic value when calculating the **ROB** ratio using Eq.(10).

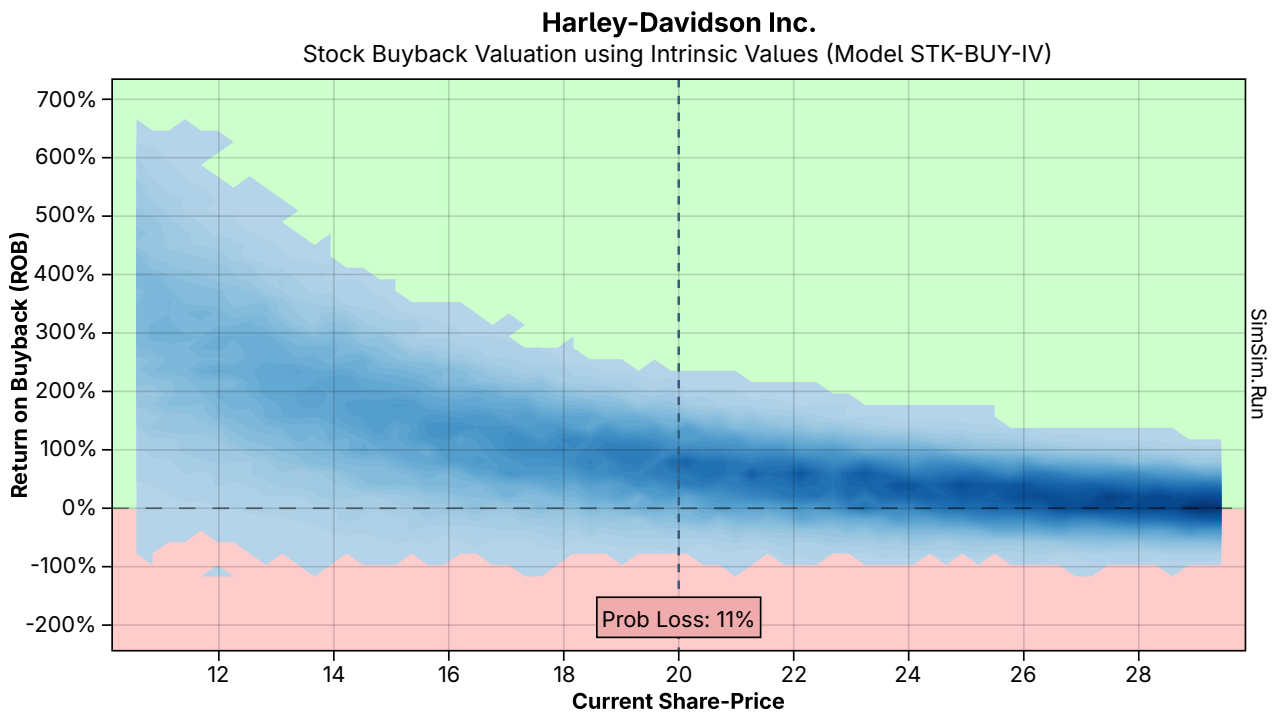


Figure 30: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 26 as the intrinsic value when calculating the **ROB** ratio using Eq.(10), and varying the buyback share-price on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The vertical dashed line is the distribution shown in Figure 29.

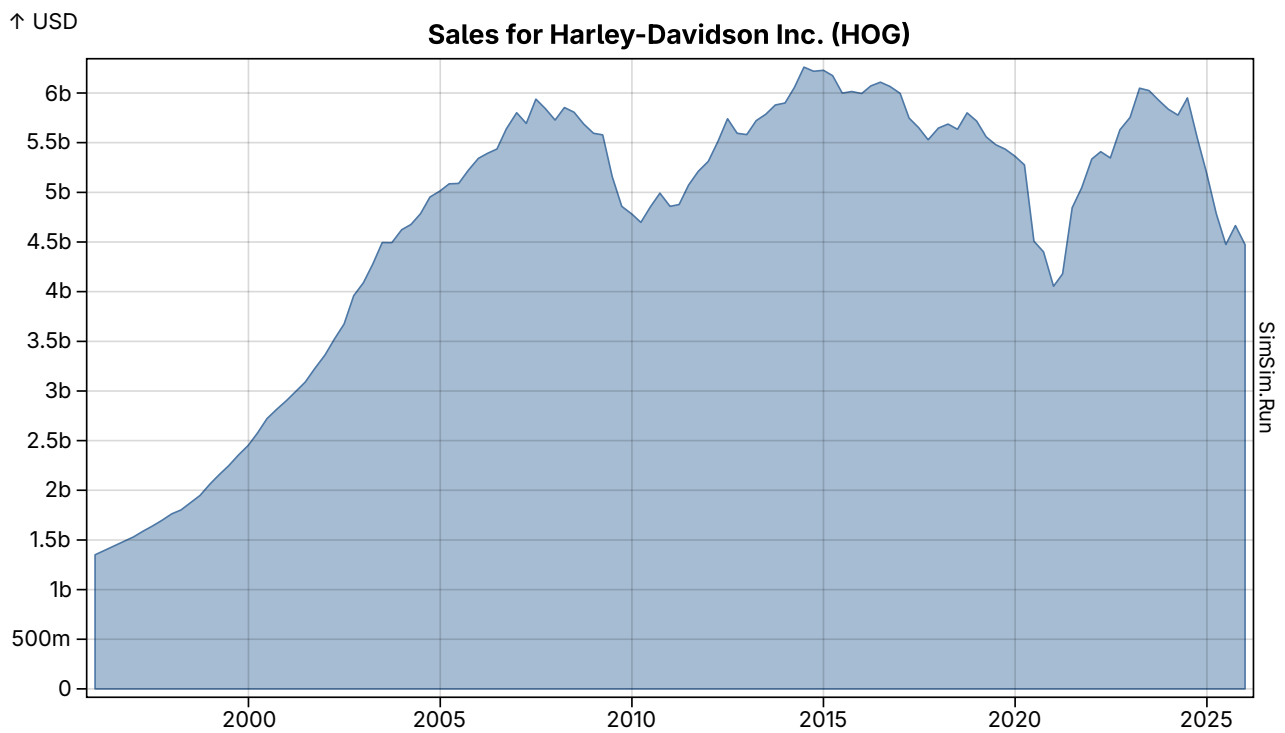


Figure 31: **Harley Davidson.** Historical sales for the company. (TTM data.)

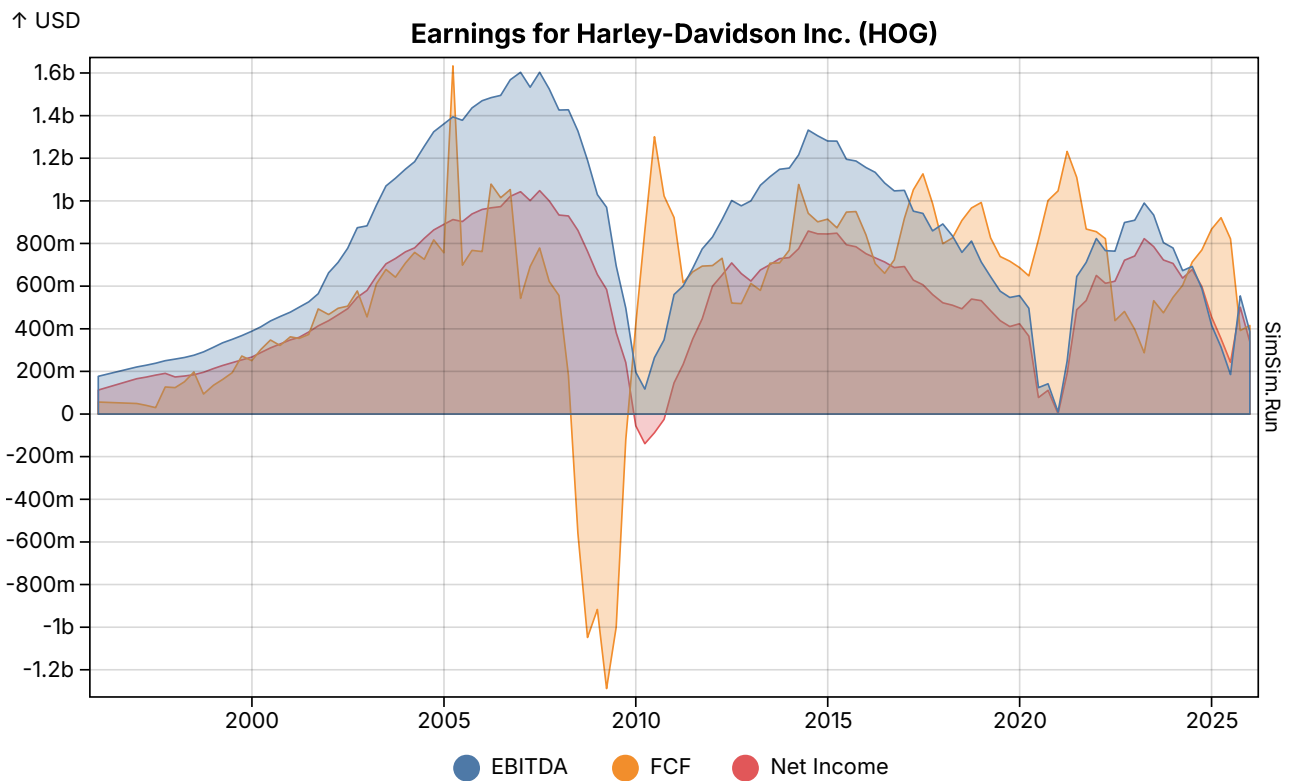


Figure 32: **Harley Davidson.** Historical earnings for the company. (TTM data.)

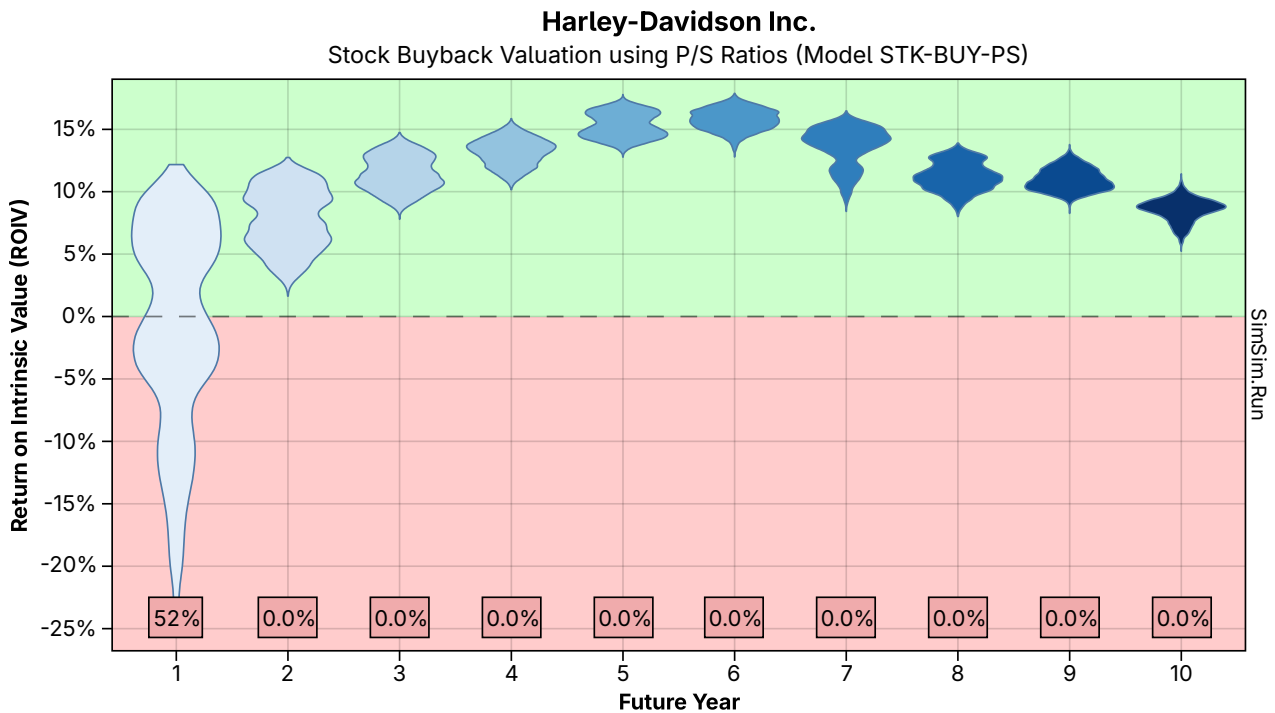


Figure 33: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009.
Violin-plot of **ROIV** distributions for investment periods between 1-10 years covering the years 2009-2018.
Red boxes show the probability of loss for each investment period when making a share buyback.

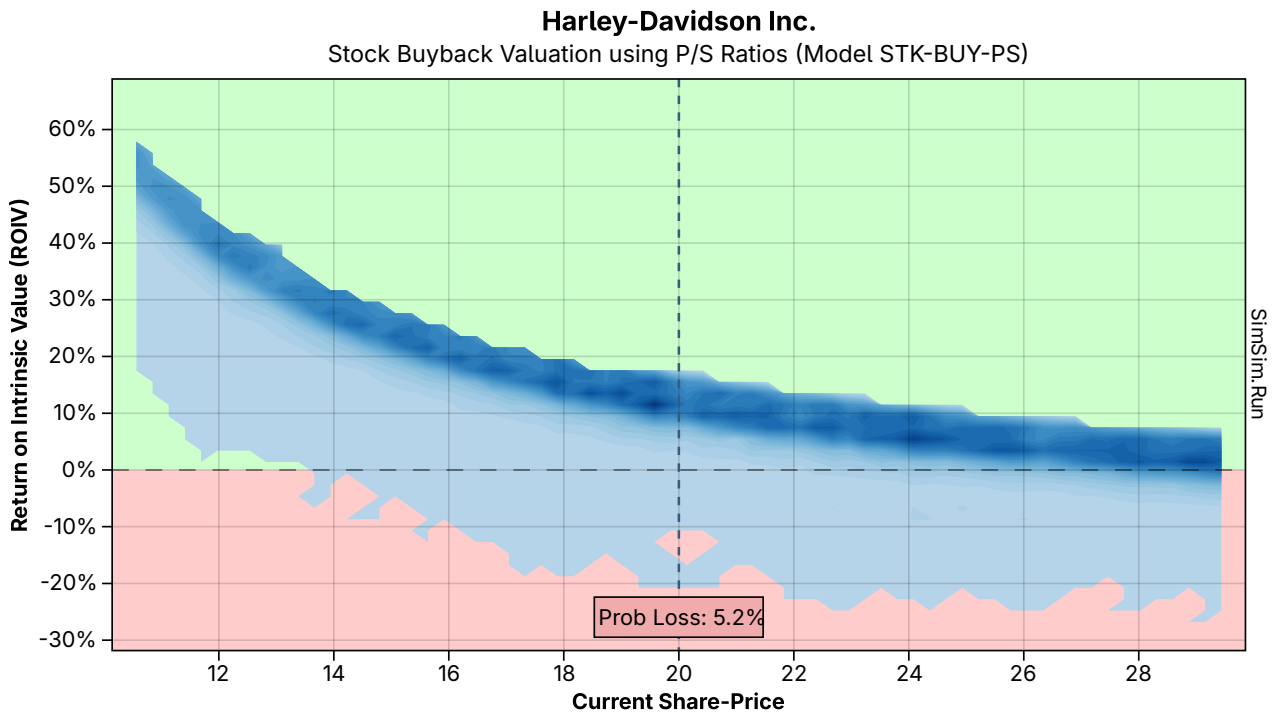


Figure 34: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 when varying the buyback share-price on the x-axis. This shows the **ROIV** ratio for all investment periods between 1-10 years combined. This is a 2D distribution where darker blues indicate more simulated values in a region. The red box at the bottom shows the overall probability of loss for all investment periods when making a share buyback.

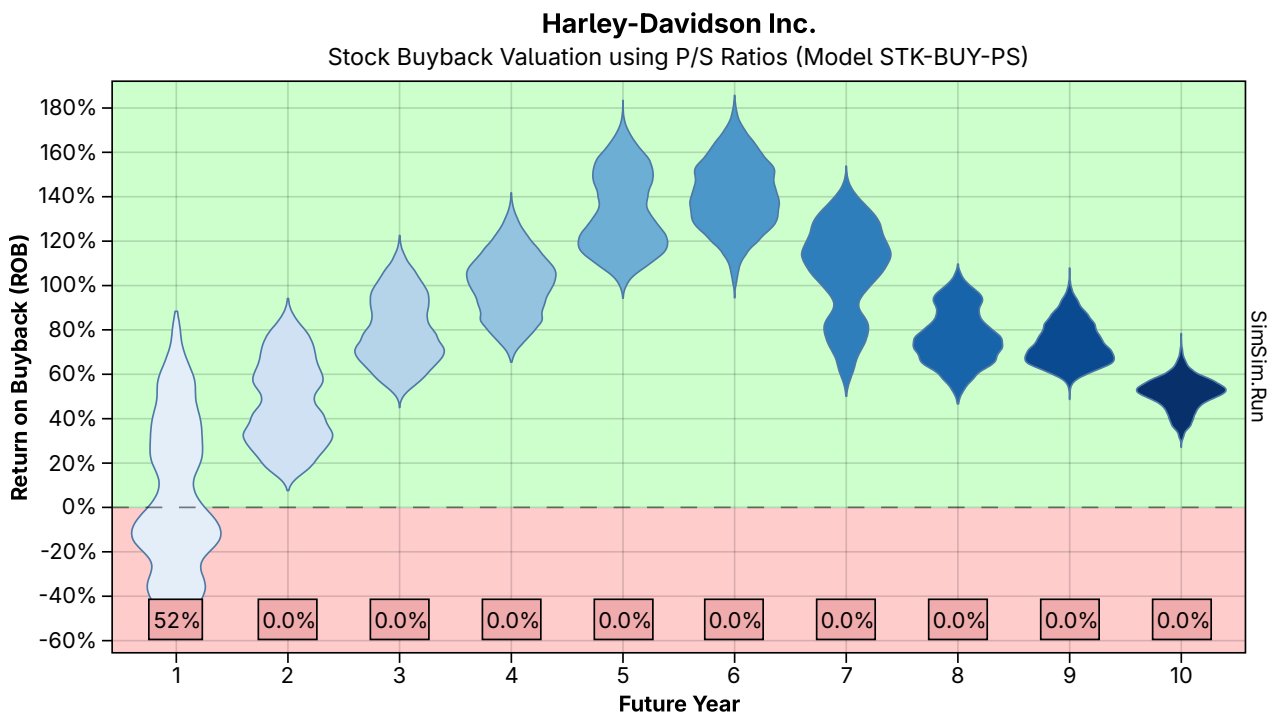


Figure 35: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009.
Violin-plot of **ROB** distributions for investment periods between 1-10 years covering the years 2009-2018.
Red boxes show the probability of loss for each investment period when making a share buyback.

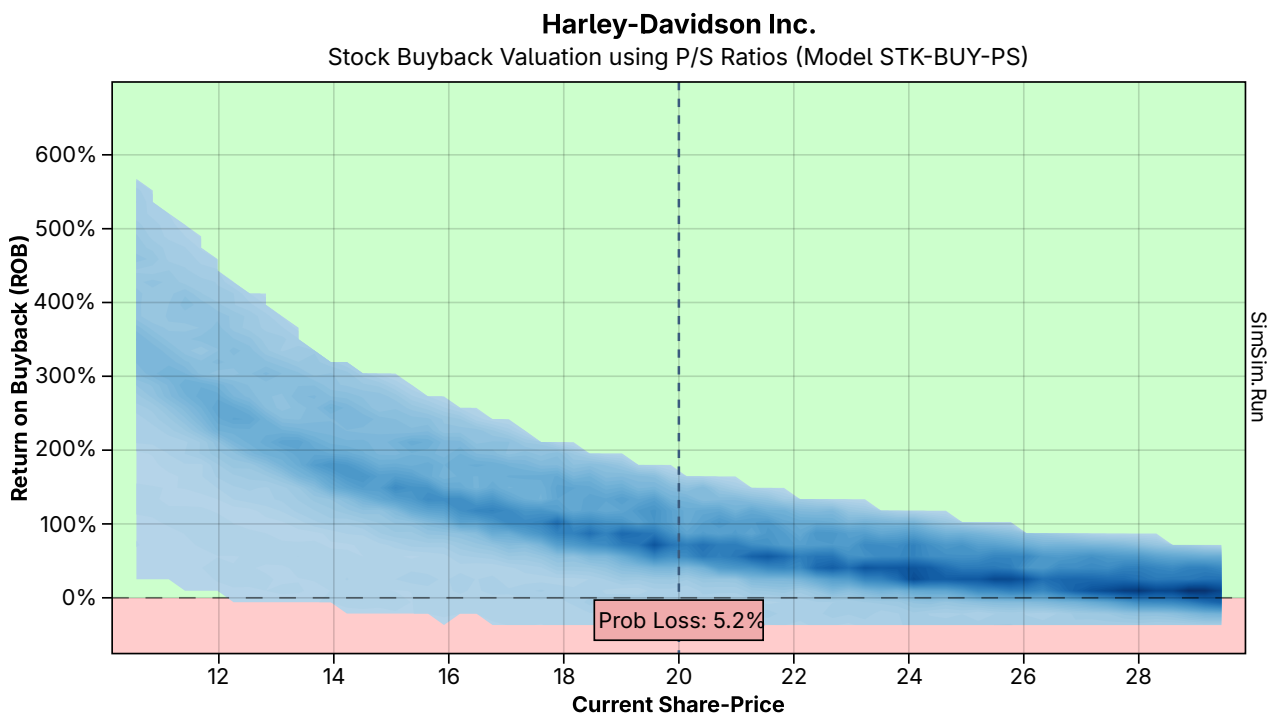


Figure 36: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 when varying the buyback share-price on the x-axis. This shows the **ROB** ratio for all investment periods between 1-10 years combined. This is a 2D distribution where darker blues indicate more simulated values in a region. The red box at the bottom shows the overall probability of loss for all investment periods when making a share buyback.

9 Case Study: AutoZone (AZO)

This company is a car-parts retailer founded in 1979 in the USA. This is an example of a company that only made share buybacks and did not pay any dividends for nearly 30 years since the late 1990s. They have bought back and retired nearly 90% of their shares. The stock-data shown here is taken from the [SimSim.Run](https://simrun.com) web-site (click the link to view the data).

9.1 Historical Share Buyback Valuation

Figure 37 shows the historical valuation of share buybacks and issuances in four sub-plots that were explained in section 7.1, so we will just analyze the results here:

- Since the late 1990s until now, the company has consistently bought back shares, except for very short pauses e.g. during the Covid pandemic in 2020.
- The P/S ratio has gradually risen, especially since 2010. This is partly because the profit margins have increased, but the stock's valuation ratios have generally expanded as well.
- The ROIV ratio shows the gain/loss in per-share intrinsic value from the share buybacks and issuances. The company's historical share buybacks mostly resulted in gains. The greatest gains were prior to 2010 where the stock had lower valuation ratios.
- The ROB ratio shows the gain/loss in per-share intrinsic value relative to the net buyback or issuance amount. Before 2010, the gains from the share buybacks were often 200-300% or more on average, relative to the amounts that the company invested in buying its own shares. After 2010, the stock's valuation ratios gradually increased so the gains from the share buybacks gradually decreased as measured by the ROB ratios.

Figure 38 shows the company's historical cash-flows as a percentage of the total positive cash-flows. On the web-site you can click on the plot to pinpoint information for each data-point. Note that the share buybacks are usually much larger than the net income, and the company has borrowed money to partially fund the share buybacks. Also note that capital expenditures were usually much higher than the depreciation and amortization, showing the company invested a lot in its growth.

Figure 39 shows the company's historical liabilities as percentages of the total assets. Although the company has been consistently profitable, it has bought back shares in part using borrowed money, so the equity became negative in 2008 and has remained negative since. Another plot on the web-site shows the EBITDA interest coverage ratio has been 9-20 during this period, so although the company had negative equity, it was so profitable that it could easily pay interest on its debt.

9.2 Simulation

Let us try and simulate a share buyback as if we were in mid 2007 using only financial data available at that time. Figure 40 shows the company's historical sales where we have selected the year prior to 2007. Figure 41 shows the company's historical profit margins where we have selected the period from 2003 to 2007. Multiplying all the sales and net profit margins from these periods gives the probability distribution for earnings shown in Figure 42, which ranges roughly between USD 515m and 625m with the bulk of the distribution between roughly USD 570m and 625m.

We use a model on the SimSim.Run web-site that supports terminal growth-rates (click the link to try it). We use the simulated earnings in Figure 42 and the following assumptions from 2007:

$$\begin{aligned} \text{Shares} &= 70 \text{ m} & \text{SharePrice} &= \$ 130 \\ \text{Buyback} &= \text{Excess Cash} = \$ 750 \text{ m} & & (56) \\ \text{Terminal Growth Rate} &= 0 \% & d &\sim N(\mu = 10 \%, \sigma = 1 \%) \end{aligned}$$

Figure 43 shows the simulated ROIV ratio from Eq.(9), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. Almost all of these are losses down to about -10% with the most likely ROIV ratio being around -4%.

Figure 44 shows the simulated ROB ratio from Eq.(10), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. Almost all of these are losses down to about -60% with the most likely ROB ratio around -35%.

So the company would have to grow its future earnings in order for the share buyback in 2007 to result in a gain in per-share intrinsic value for the remaining shareholders. In the following 20 years the average annual growth-rate was about 6% for the company's sales, so let us try and use that as the terminal growth-rate for the earnings in the simulation. This company funded its growth from borrowed money and an increase in accounts payable. If the company had used retained earnings to fund the growth, then we would have to subtract that amount from the earnings in the simulation.

Figure 45 shows the simulated ROIV ratio with the 6% terminal growth-rate. The range is roughly between -2% and +8% with the peak around +3.5%.

Figure 46 shows the simulated ROB ratio with the 6% terminal growth-rate. The range is roughly between -20% and +500% with the peak around +45%.

9.3 Summary

The company only made share buybacks and no dividend payouts for almost 30 years. The share buybacks mostly resulted in large gains in per-share intrinsic values for the remaining shareholders. But this required significant and continued sales growth, otherwise many of the buybacks would have resulted in losses for the remaining shareholders, as we demonstrated using simulations.

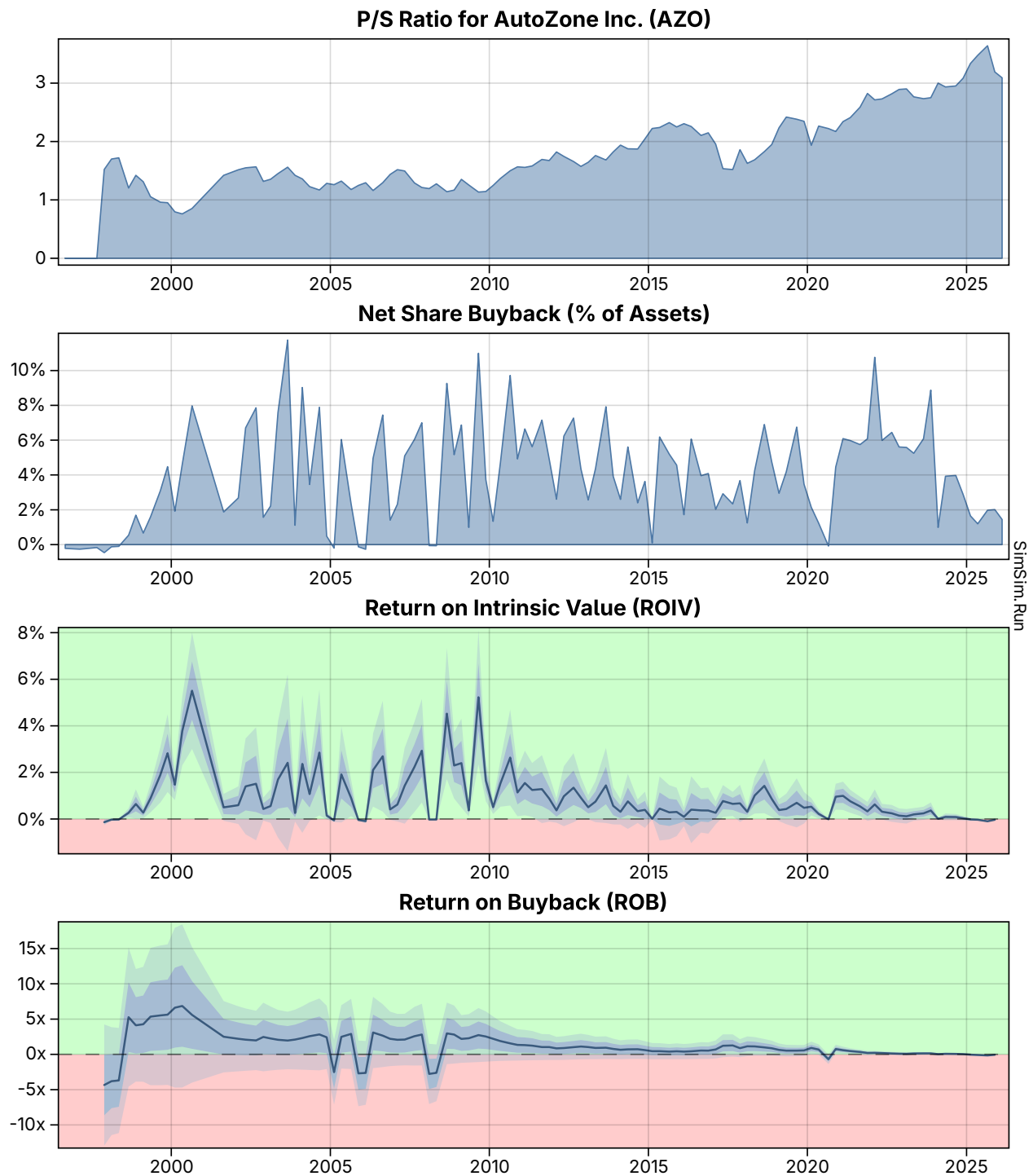


Figure 37: **AutoZone**. Valuation of historical share buybacks and issuances using ROIV in Eq.(42) and ROB in Eq.(43), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. ROB uses x-scales instead of % e.g. 15x = 1,500%. Also shown is historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)

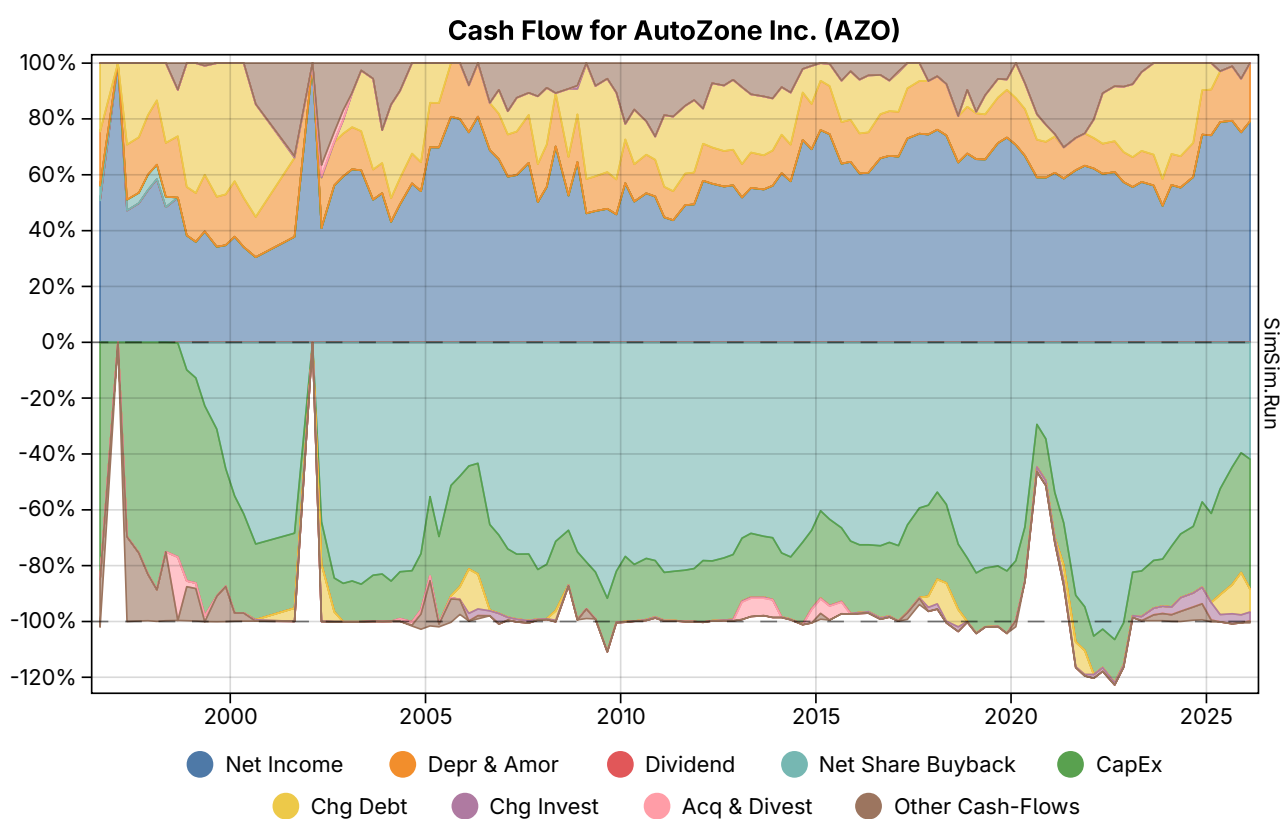


Figure 38: **AutoZone**. Historical cash-flows for the company as percentages of total positive cash-flows. (TTM data.)

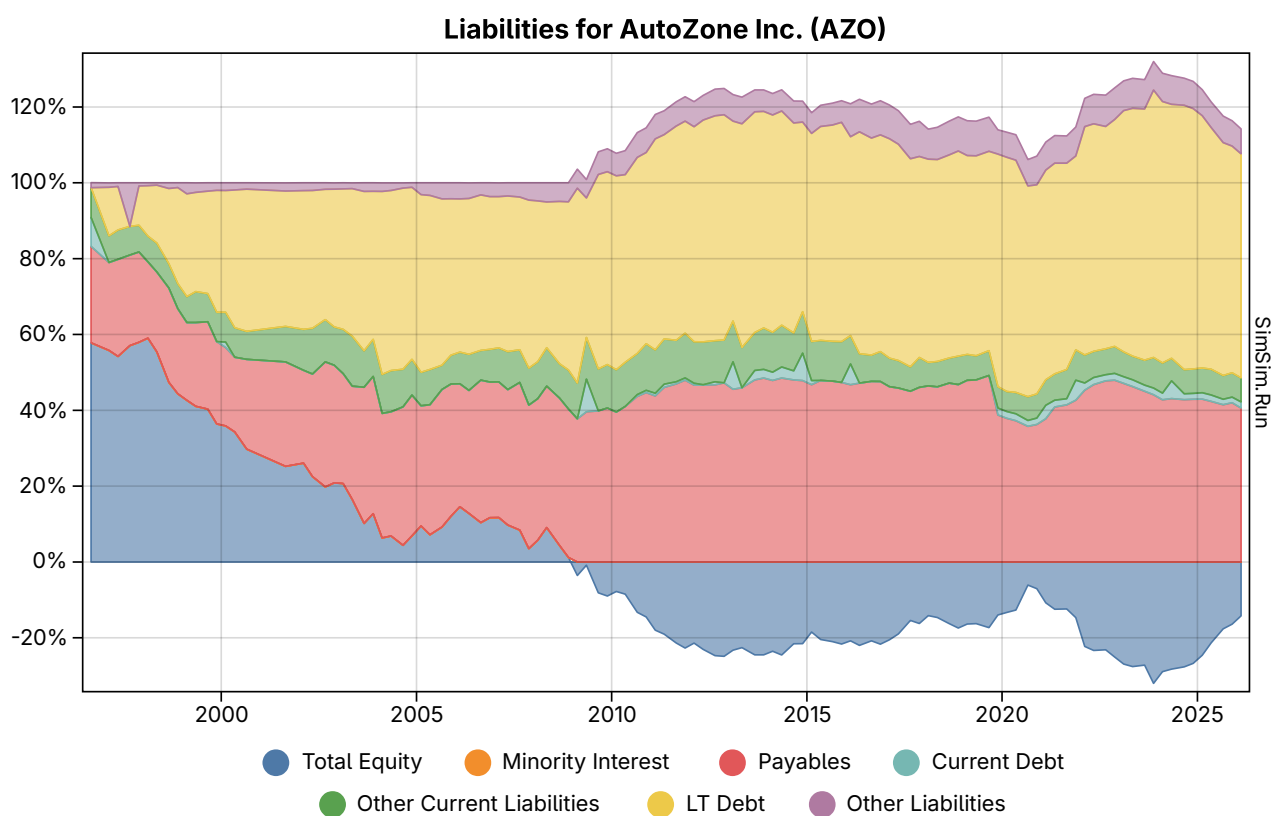


Figure 39: **AutoZone**. Historical liabilities for the company as percentages of total assets. (TTM data.)

↑ USD

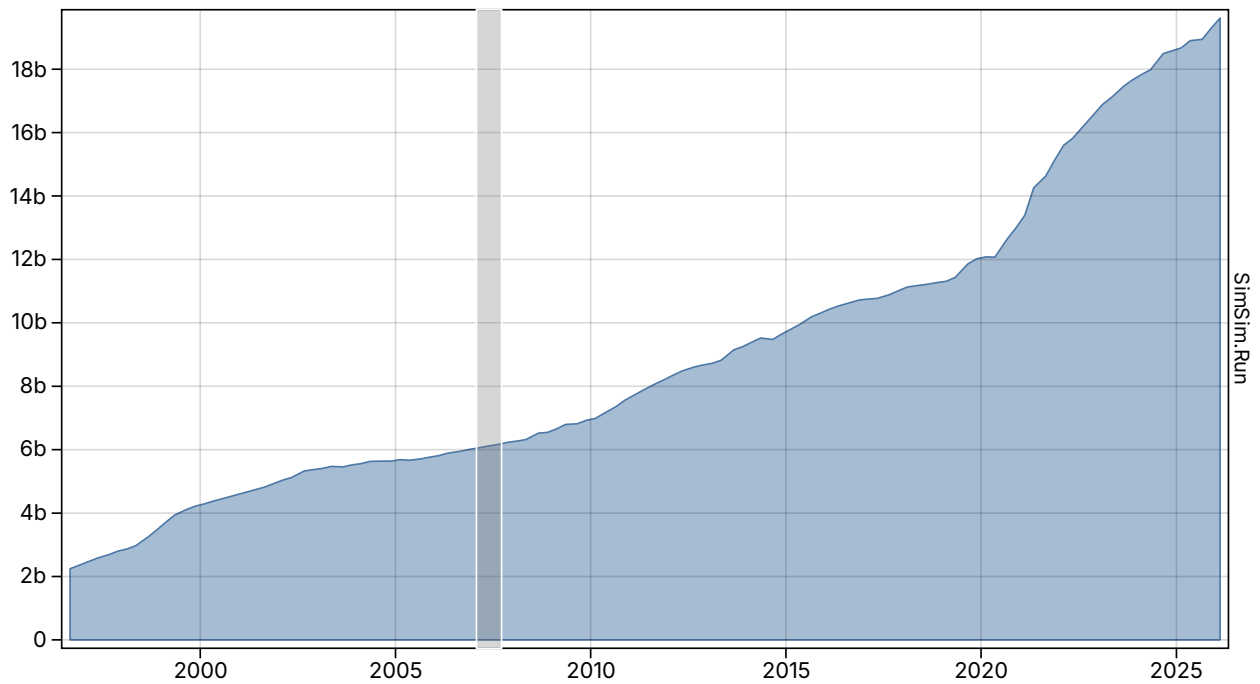
Sales for AutoZone Inc. (AZO)

Figure 40: **AutoZone**. Historical sales for the company. (TTM data.)

The shaded area is the selected data-range that is used to simulate the earnings in Figure 42.

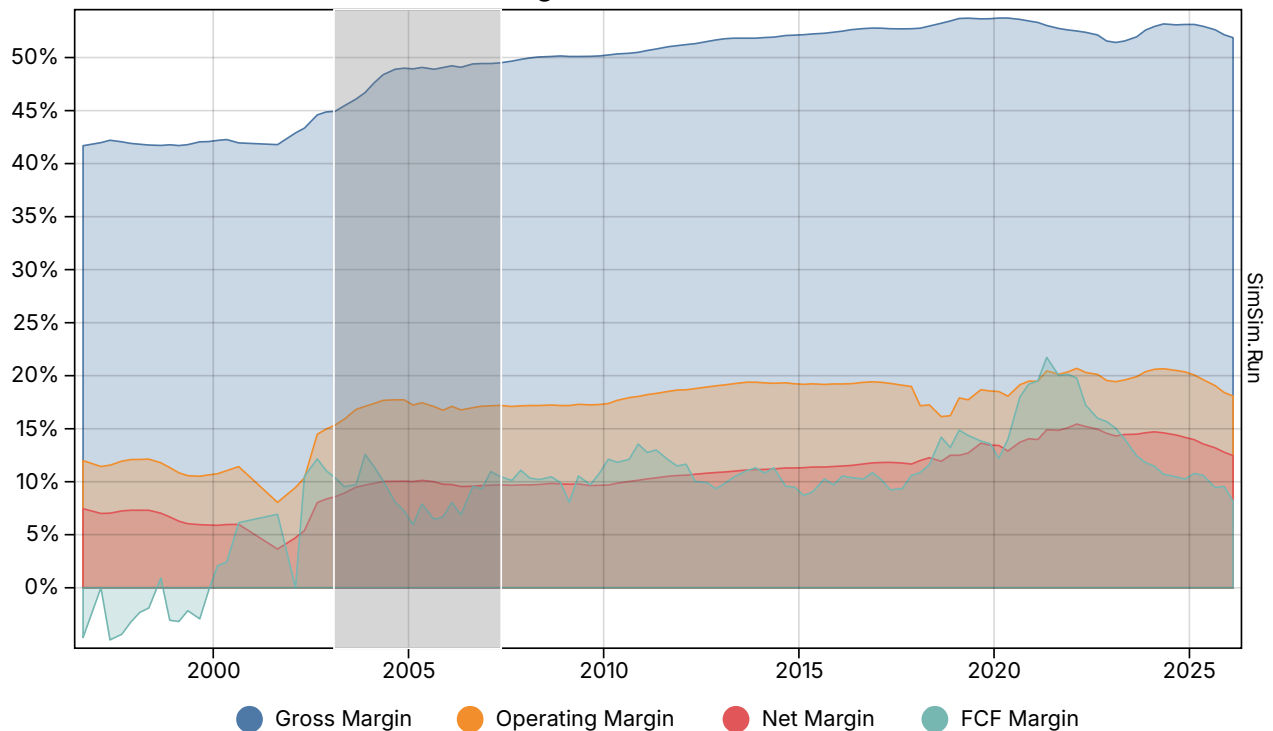
Profit Margins for AutoZone Inc. (AZO)

Figure 41: **AutoZone**. Historical profit margins for the company. (TTM data.)

The shaded area is the selected data-range whose net profit margin is used to simulate the earnings in Figure 42.

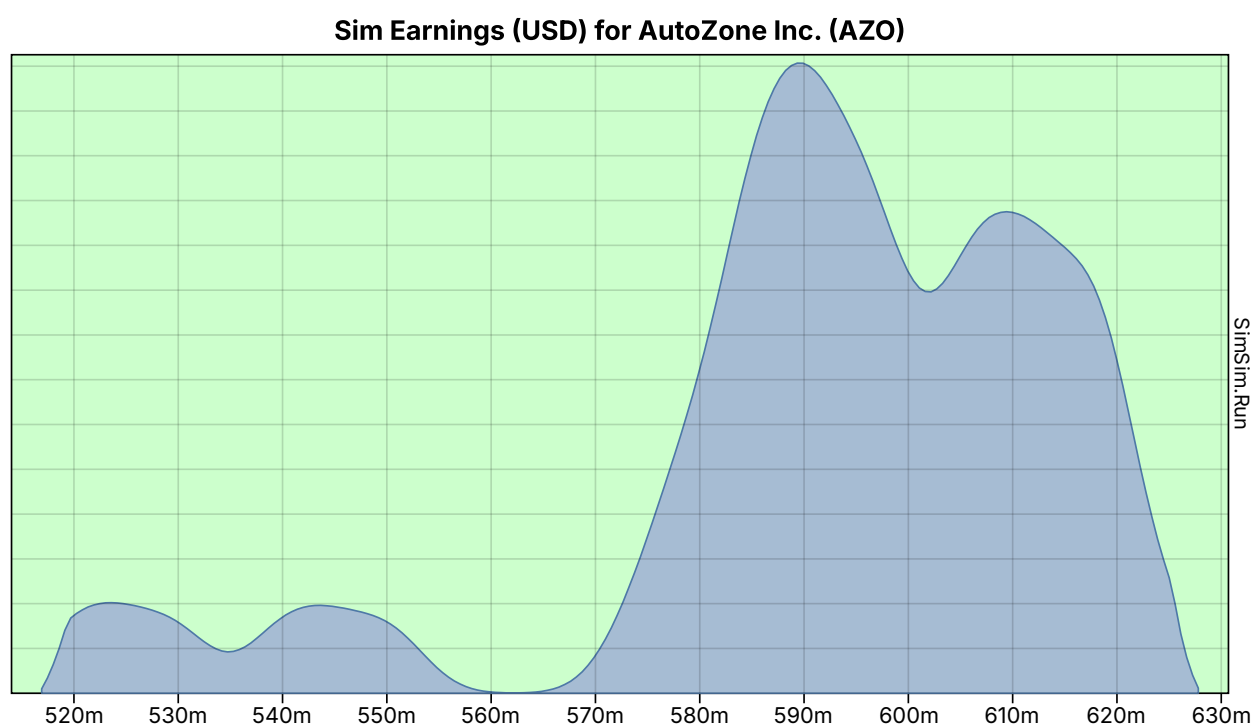


Figure 42: **AutoZone**. Simulated earnings by multiplying the historical sales that are selected in Figure 40 with the historical net profit margins that are selected in Figure 41.

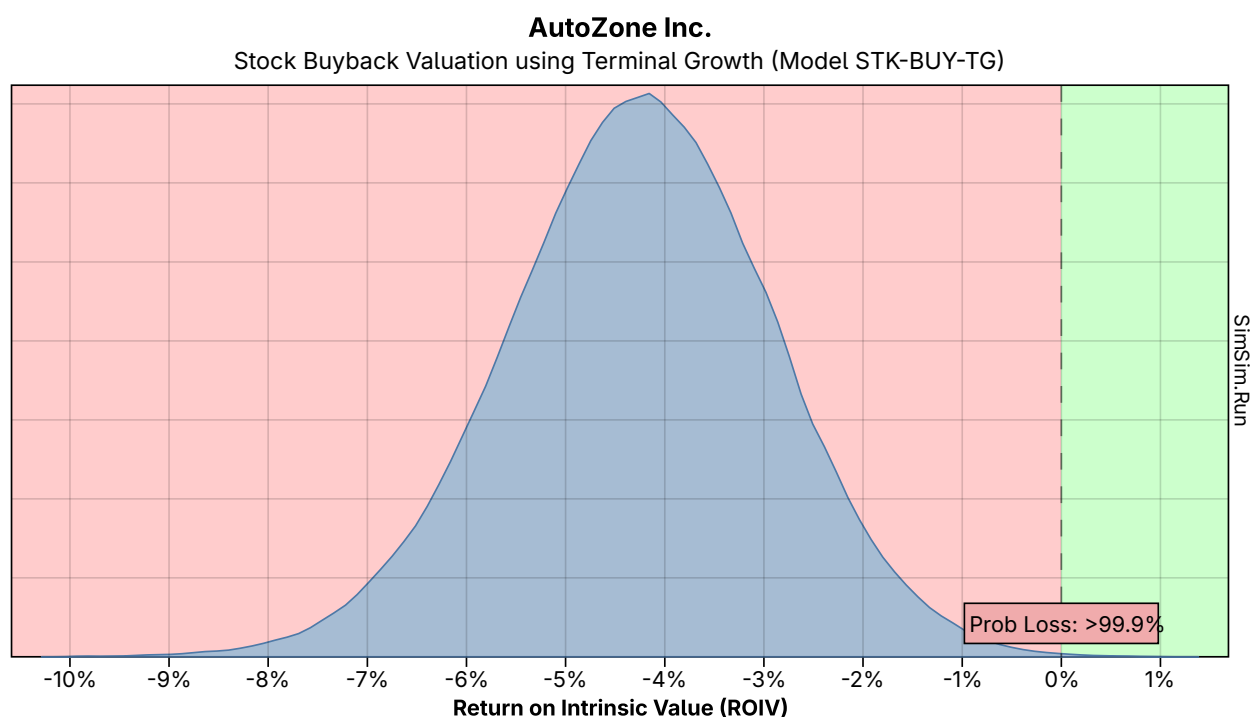


Figure 43: **AutoZone**. Simulation of **ROIV** in Eq.(9) using the simulated earnings from Figure 42.
Zero terminal growth.

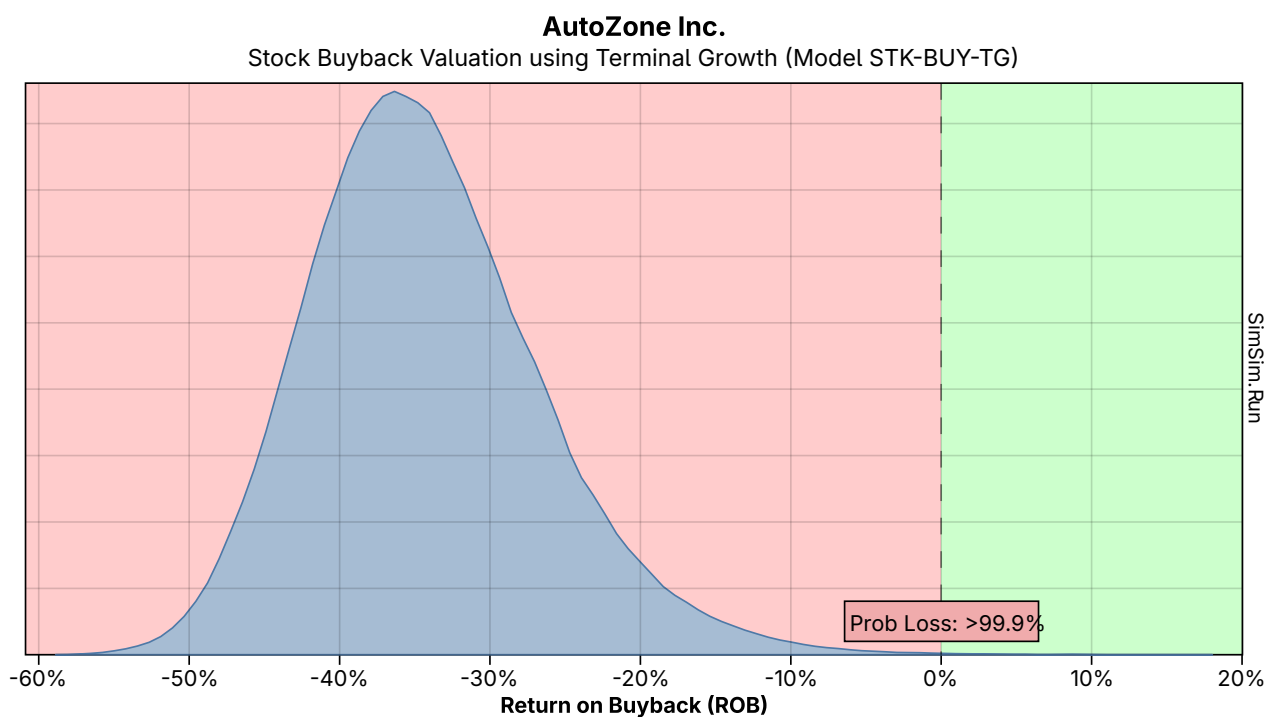


Figure 44: **AutoZone**. Simulation of **ROB** in Eq.(10) using the simulated earnings from Figure 42.
Zero terminal growth.

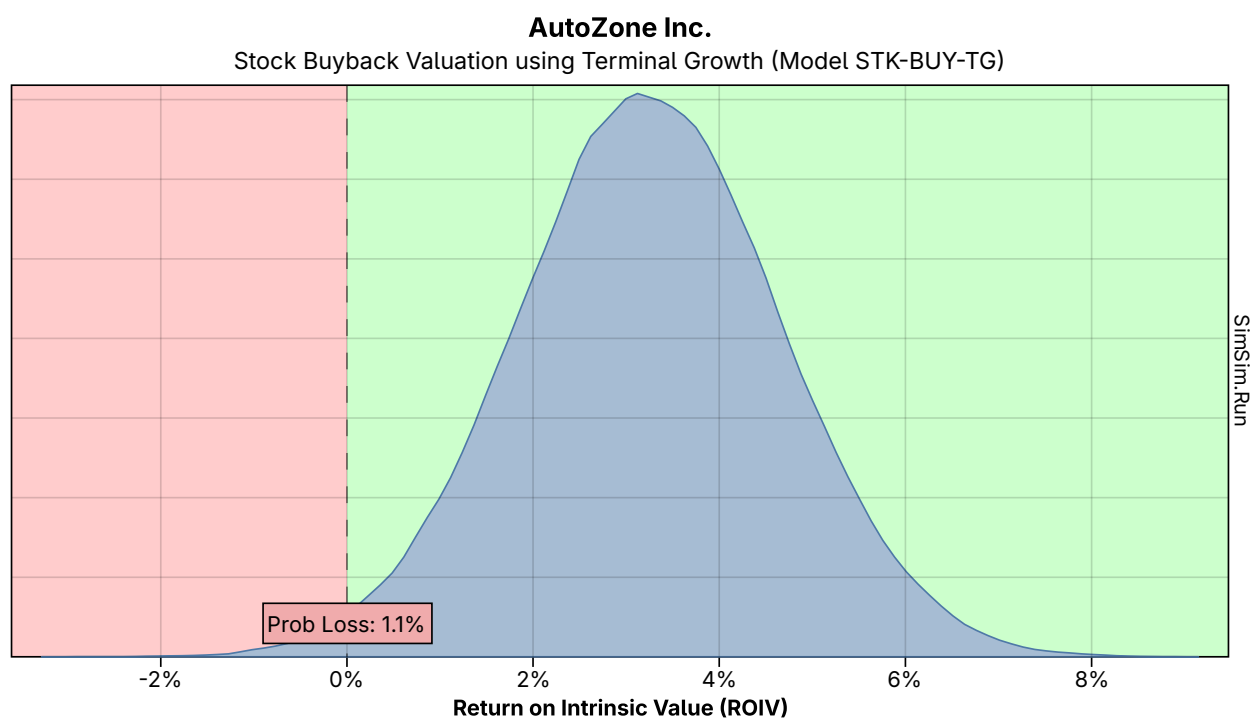


Figure 45: **AutoZone**. Simulation of **ROIV** in Eq.(9) using the simulated earnings from Figure 42.
6% terminal growth.

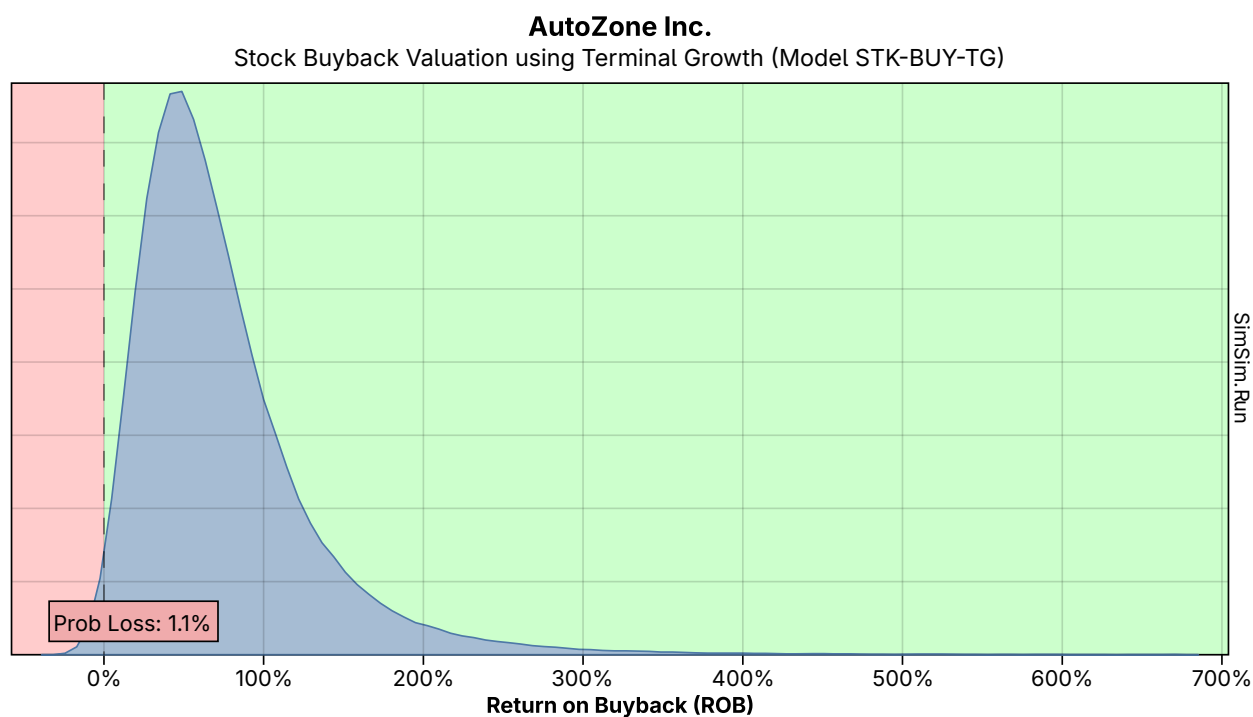


Figure 46: **AutoZone**. Simulation of **ROB** in Eq.(10) using the simulated earnings from Figure 42.
6% terminal growth.

10 Case Study: Western Union (WU)

This company was founded in 1851 in the USA and pioneered long-distance communication and money transfers. Its communication business declined as technology evolved and the company went bankrupt in 1991 after which it focused on its money transfer business. It was listed on the stock-exchange in 2006. Soon after, the company started making regular dividend payouts and share buybacks. They have bought back and retired about 60% of their shares since then. The stock-data used here is taken from the [SimSim.Run](https://simsim.run) web-site (click the link to view the data).

10.1 Historical Share Buyback Valuation

Figure 47 shows the historical valuation of share buybacks and issuances in four sub-plots that were explained in section 7.1, so we will just analyze the results here:

- Since 2007 until now, the company has consistently bought back shares, except for very short pauses e.g. during the Covid pandemic in 2020.
- The P/S ratio has gradually declined about 80% from 2007 until now, even though the net profit margin has been about 15-20% for most of the duration, as shown in Figure 49.
- The ROIV ratio shows the gain/loss in per-share intrinsic value from the share buybacks and issuances. The company's historical share buybacks mostly resulted in losses.
- The ROB ratio shows the gain/loss in per-share intrinsic value relative to the net buyback or issuance amount. The losses on the share buybacks were often -10% to -40% on average, relative to the amounts that the company invested in buying its own shares.

10.2 Simulation

Let us try and simulate a share buyback as if we were in late 2018 where the stock-price dipped as the profit margins became low. We only use financial data available at that time. Figure 48 shows the company's historical sales where we have selected the data from 2014 until mid 2018. Figure 49 shows the company's historical profit margins, where we have selected the data from 2006 until mid 2016 before a few years of lower profit margins, as if we believed at the time that the profit margins would recover again, which they did. Multiplying all the sales and net profit margins from these periods gives the probability distribution for earnings shown in Figure 50, which ranges roughly between USD 750m and 1.25b with the peak around USD 960m.

We use a model on the [SimSim.Run](https://simsim.run) web-site that supports terminal growth-rates (click the link to try it). We use the simulated earnings in Figure 50 and the following assumptions from late 2018:

$$\text{Shares} = 450 \text{ m} \quad \text{SharePrice} = \$ 18$$

$$\text{Buyback} = \text{Excess Cash} = \$ 400 \text{ m} \quad (57)$$

$$\text{Terminal Growth Rate} = 0 \% \quad d \sim N(\mu = 10 \% , \sigma = 1 \%)$$

Figure 51 shows the simulated ROIV ratio from Eq.(9), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. The range is roughly -2.5% to +3% with the peak around +0.6%. The probability of loss is 22%.

Figure 52 again shows the simulated ROIV ratio from Eq.(9), but here the buyback share-price is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the buyback share-price changes, while all other variables remain the same. For example, when the share-price is around \$14 all of the ROIV ratios would be positive gains.

Figure 53 shows the simulated ROB ratio from Eq.(10), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. The range is roughly -30% to +120% with the peak around +8%. The probability of loss is 22%.

Figure 54 again shows the simulated ROB ratio from Eq.(10), but here the buyback share-price is varied on the x-axis. For example, when the share-price is around \$20 the most likely ROB ratio would be around 0%.

So if we were analyzing this share buyback in late 2018, assuming the historical sales and net profit margin would continue in the future, a share buyback at \$18 would have about 22% probability of loss. We might have deemed this to be a reasonable chance of gain. But the sales and earnings started declining in the following years as many other companies started offering similar services. So the share buyback in 2018 turned out to have probably caused a loss to the remaining shareholders. You can easily test this by changing the terminal growth-rate in the simulation model.

10.3 Summary

The company has consistently made both share buybacks and dividend payouts in the past 20 years. The company's sales started declining from rising competition, so the share-price gradually declined roughly 60% over the last 20 years, even though the company has bought back and retired almost 60% of its shares. So most of its share buybacks during the last 20 years likely resulted in losses for the remaining shareholders, who would have received more value from dividend payouts.

We ran a simulation as if we were in the year 2018 before the company's sales started declining. We assumed the historical sales and net profit margins would continue in the future. This resulted in a roughly 80% chance of gains from the share buyback in 2018. But in reality, the sales started declining in the following years, so the share buyback likely resulted in a loss to the remaining shareholders, unless the company can recover and grow its sales and earnings in the next few years.

Even when using simulations to show a whole range of outcomes and their likelihood, we must still use input probability distributions that cover the future reality, otherwise we can delude ourselves.

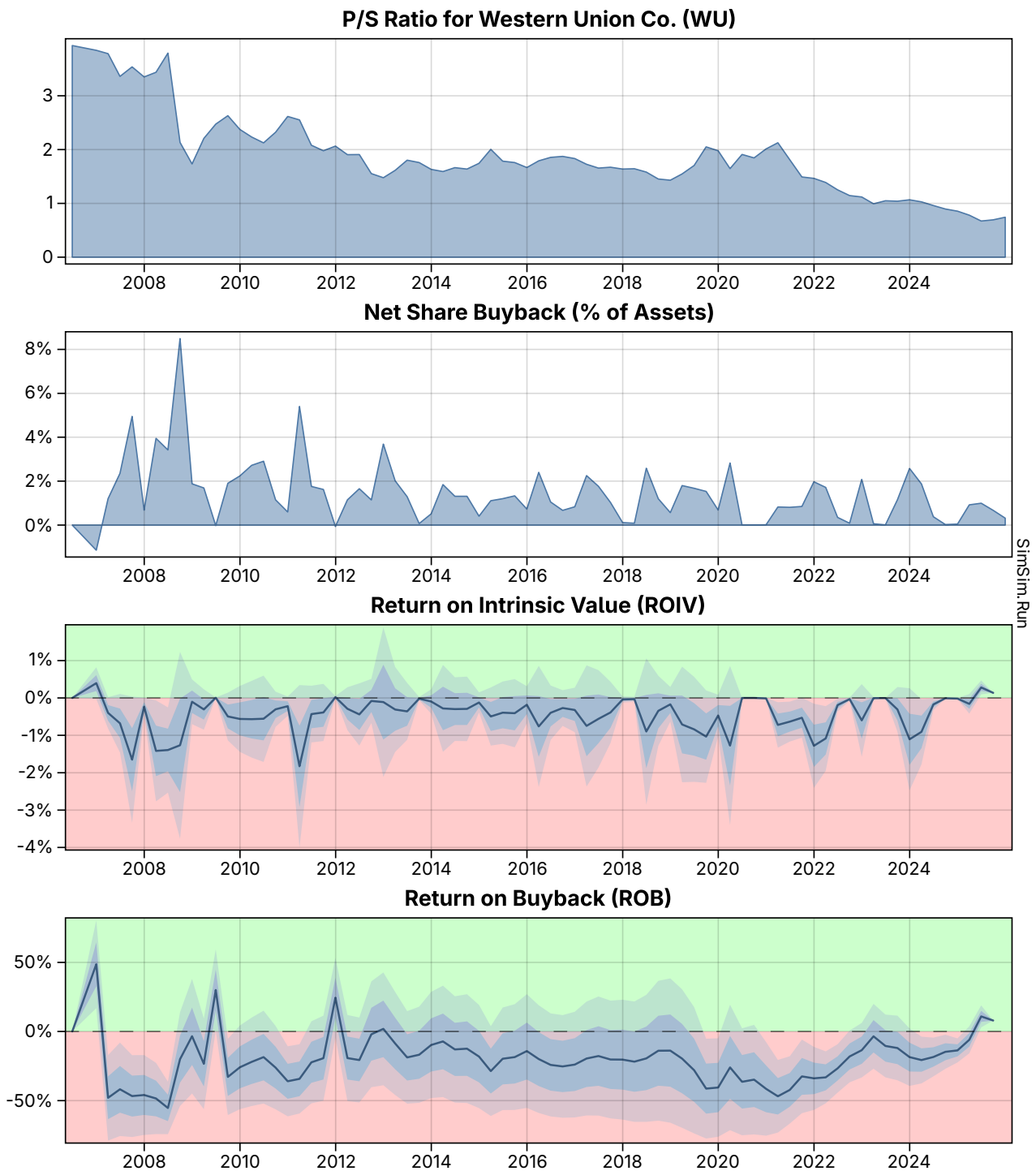
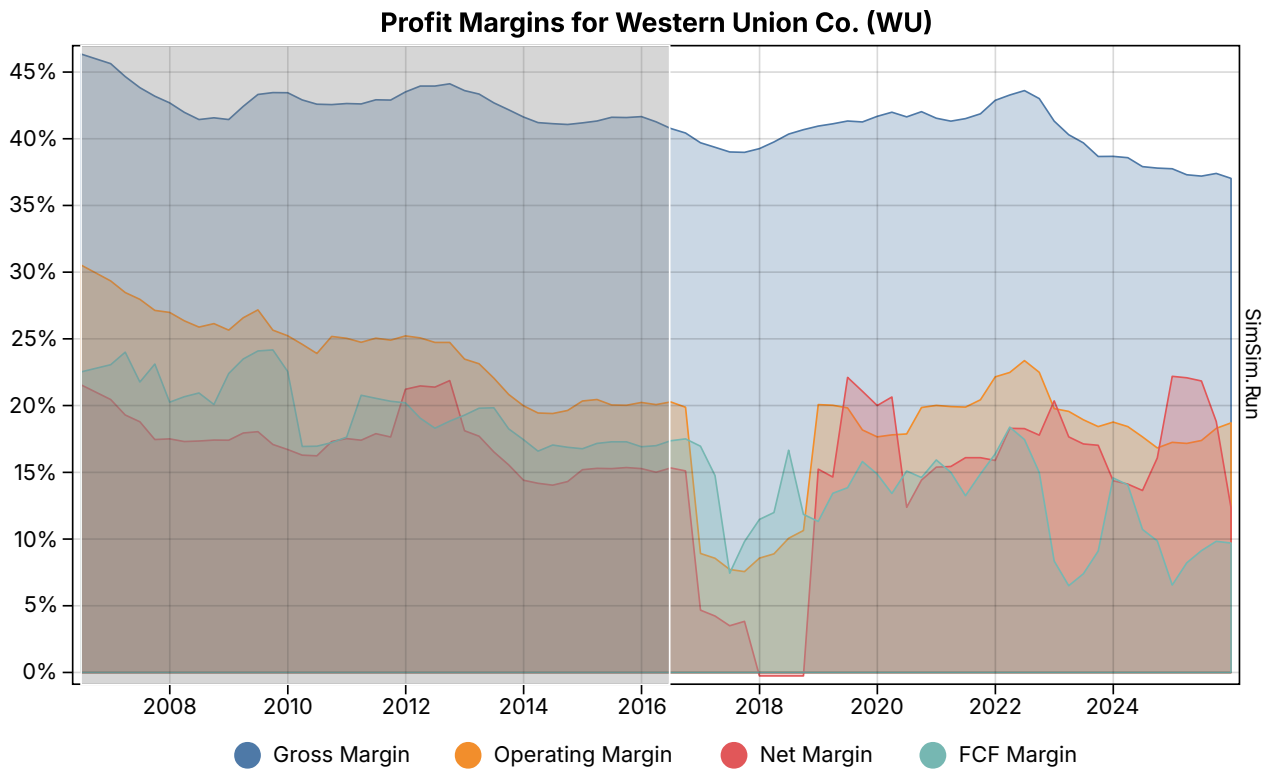
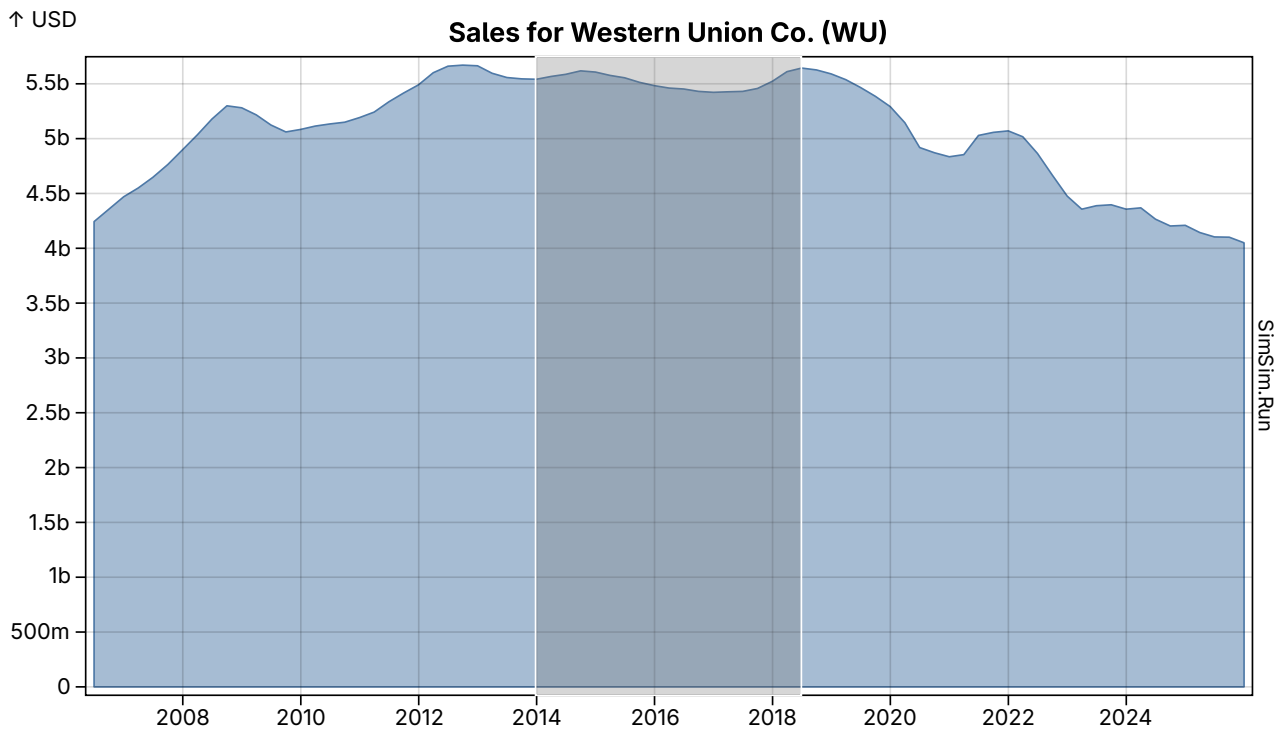


Figure 47: **Western Union.** Valuation of historical share buybacks and issuances using ROIV in Eq.(42) and ROB in Eq.(43), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. Also shown is historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)



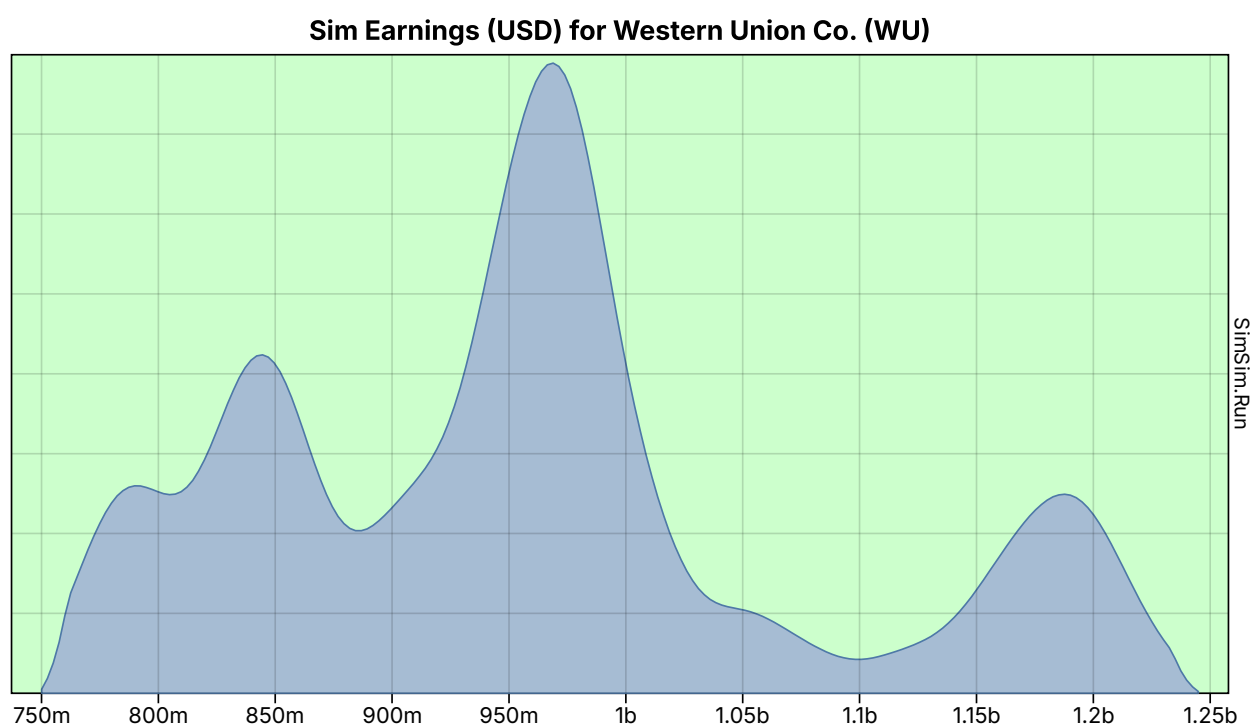


Figure 50: **Western Union.** Simulated earnings by multiplying the historical sales that are selected in Figure 48 with the historical net profit margins that are selected in Figure 49.

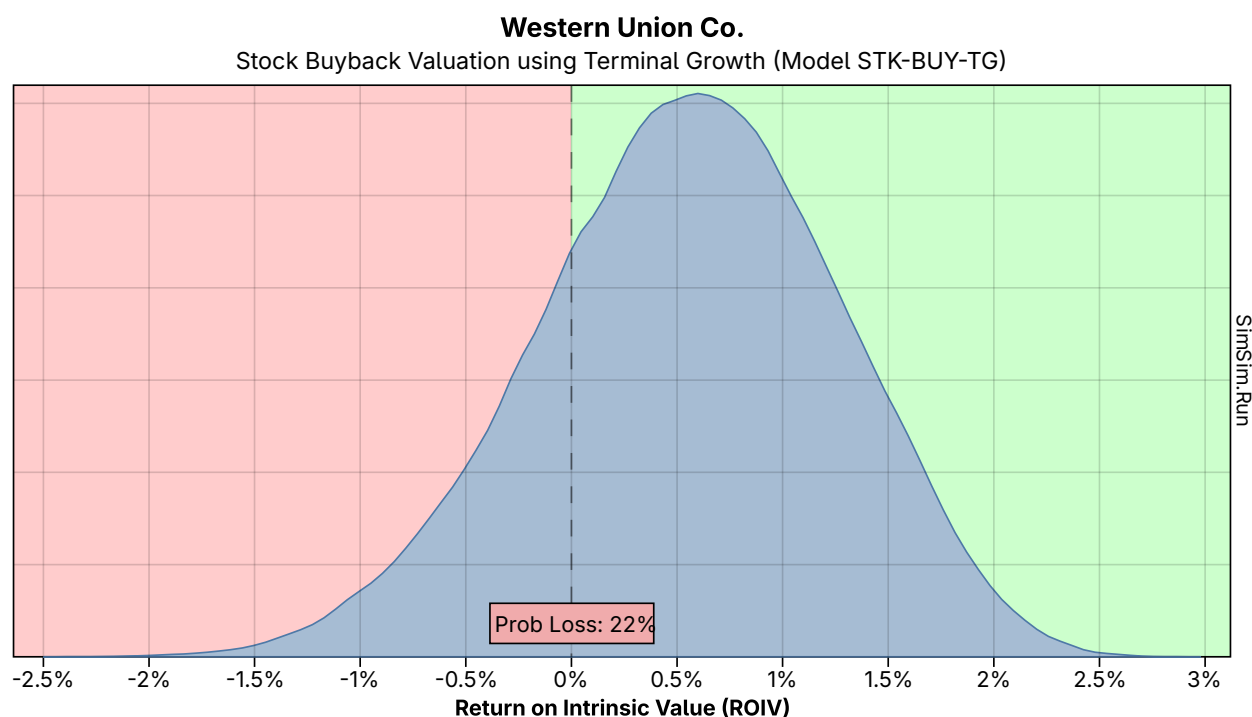


Figure 51: **Western Union.** Simulation of $ROIV$ in Eq.(9) using the simulated earnings from Figure 50.

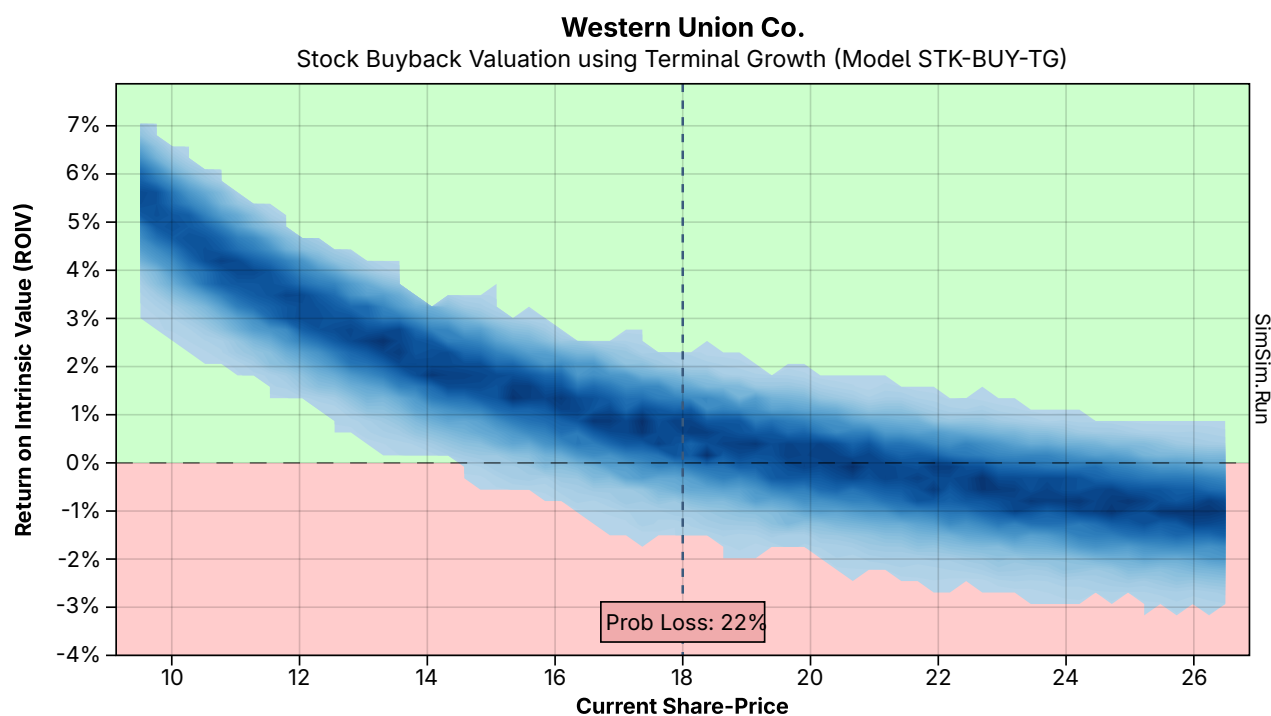


Figure 52: **Western Union.** Simulation of $ROIV$ in Eq.(9) using the simulated earnings from Figure 50, and varying the buyback share-price on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 51 is the vertical dashed blue line here.

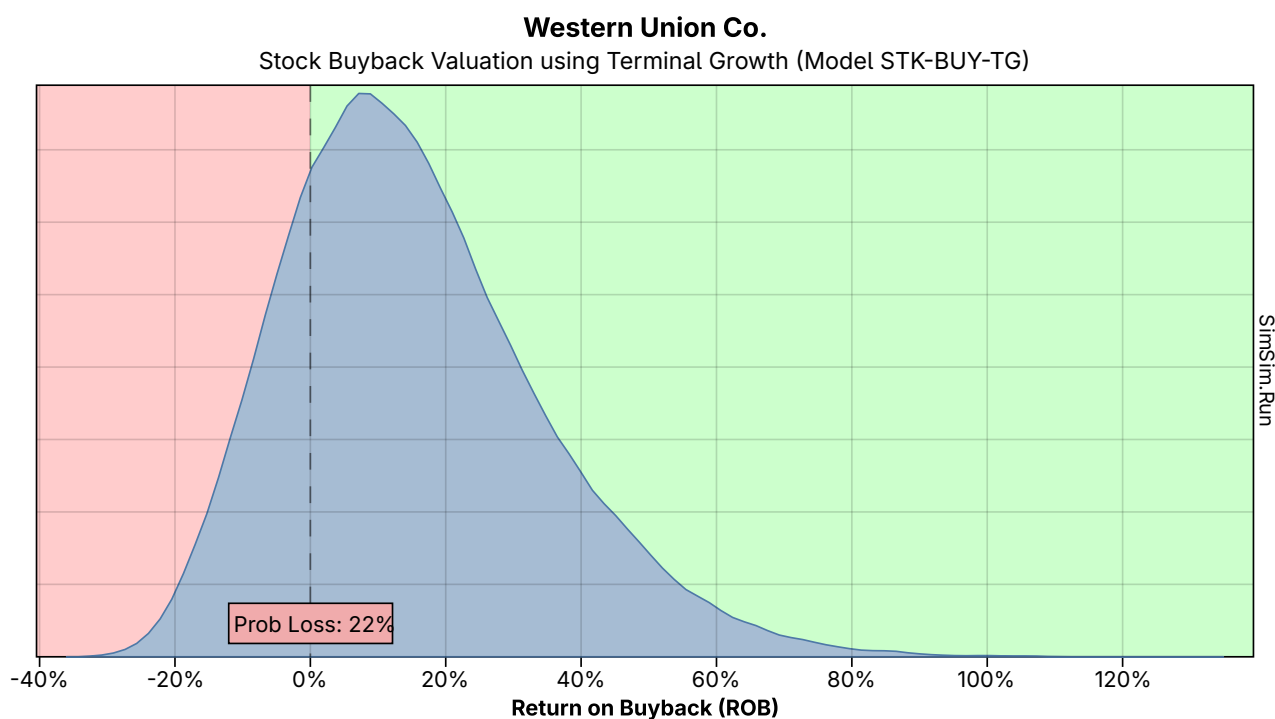


Figure 53: **Western Union.** Simulation of **ROB** in Eq.(10) using the simulated earnings from Figure 50.

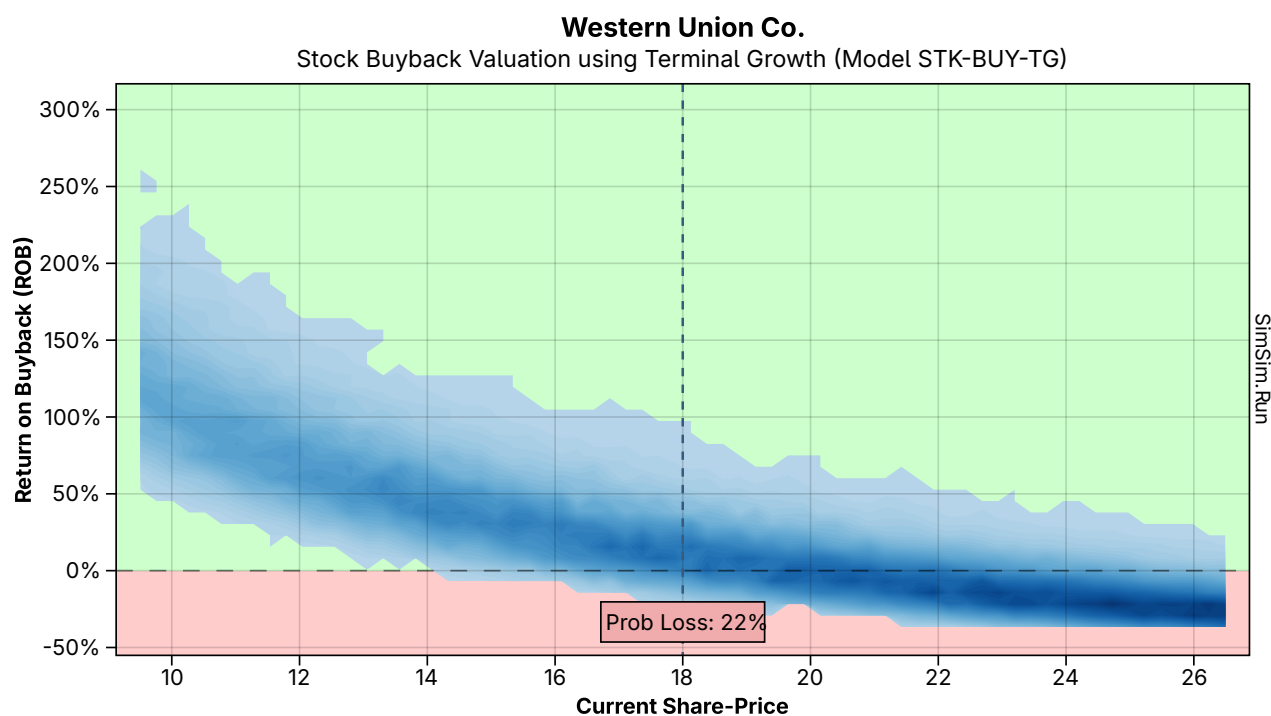


Figure 54: **Western Union.** Simulation of **ROB** in Eq.(10) using the simulated earnings from Figure 50, and varying the buyback share-price on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 53 is the vertical dashed blue line here.

11 Our New Robotic Overlords

This is a short conversation with an Artificial Intelligence (AI) chat-bot about share buybacks. This uses Google Gemini 3.1 in “thinking” mode, which gives better answers than “fast” mode when it needs to reason through its answers. The “fast” mode relies more on available information to produce its answers, and because there is so much misinformation about share buybacks — including most finance textbooks and research papers — it may take several questions before the “fast” AI mode realizes that the prevalent beliefs about share buybacks are logically incoherent.

This conversation took place after the paper was almost finished. The AI’s output has been edited very slightly to remove some examples and to make the math notation the same as this paper.

Note how the formulas derived by AI are the same as the basic formulas in section 2.

Question: How can I measure the effect of a share buyback on the intrinsic value of the shares? Use your own reasoning and explain the derivation.

AI: To measure the effect of a share buyback on intrinsic value, we have to look past the market noise. While buybacks often cause a short-term bump in stock price due to increased demand or “signaling,” their impact on intrinsic value — the fundamental worth of a slice of the business — depends entirely on the relationship between the purchase price and the business's actual value.

Here is the reasoning and mathematical derivation for how this works. Think of the company’s total intrinsic value as a pie. A share buyback does two things simultaneously:

1. It shrinks the pie: The company uses cash (an asset) to pay shareholders. That cash leaves the company, reducing its total value.
2. It reduces the number of slices: There are fewer shares outstanding, so each remaining share represents a larger percentage of the remaining pie.

The central question is: Did the “slice size” grow more than the “pie” shrank?

To see the effect on a per-share basis, let’s define our variables:

- v Initial Total Intrinsic Value of the company (before buyback).
- $Shares$ Initial number of shares outstanding.
- $SharePrice$ The price per share the company pays for the buyback.
- $Buyback$ Total cash spent on the buyback.
- V Intrinsic Value per share before the buyback: $V = v / Shares$
- W Intrinsic Value per share after the buyback.

After the buyback, the company's total value is reduced by the amount of cash spent:

$$w = v - \text{Buyback} \quad (58)$$

And the retired shares is the cash spent divided by the price paid, so the new number of shares is:

$$\text{Shares}^w = \text{Shares} - \frac{\text{Buyback}}{\text{SharePrice}} \quad (59)$$

To find the new intrinsic value per share W , we divide the new total value by the new share count:

$$W = \frac{v - \text{Buyback}}{\text{Shares}^w} \quad (60)$$

For a buyback to be accretive (i.e. increase the value of your shares), W must be greater than V . When you simplify the formula above, the condition for value creation is surprisingly simple:

Intrinsic value per share increases if, and only if, the buyback *SharePrice* is less than the pre-buyback intrinsic value V :

- *SharePrice* < V : The company is buying \$1.00 worth of value for, say, \$0.80. This transfer of wealth goes from the selling shareholders to the remaining (continuing) shareholders.
- *SharePrice* > V : The company is overpaying. It is giving \$1.20 of company cash to departing shareholders for something only worth \$1.00. This destroys value for the remaining shareholders.

If you want to calculate the exact percentage change in intrinsic value per share follow these steps:

1. Run a Discounted Cash Flow (DCF) model to find the total value of the firm with the cash. Divide by current shares.
2. Use the same DCF value minus the cash spent on the buyback. Divide by the new, lower share count.
3. The difference is your value creation (or destruction).

Summary table:

Scenario	Price Paid vs. Intrinsic Value	Effect on Remaining Shareholders
Undervalued Buyback	$\text{SharePrice} < V$	Value Accretive: Each remaining share is now worth more.
Fair Value Buyback	$\text{SharePrice} = V$	Value Neutral: The slice and pie shrank proportionally.
Overvalued Buyback	$\text{SharePrice} > V$	Value Destructive: Cash was wasted; remaining shares worth less.

Question: Then why do finance professors say that share buybacks are the same as dividends?

AI: It's a classic Textbook vs. Reality showdown. Finance professors aren't lying to you, but they are teaching from a framework called the Efficient Market Hypothesis (EMH). In the clean, sterile world of academic finance, the professor assumes that *SharePrice* always equals intrinsic value V . If you accept that assumption, then buybacks and dividends are indeed mathematically identical ways to return cash. Here is the breakdown of why they say that, and where the "value investor" view (which I explained previously) diverges.

The Modigliani-Miller logic is that in a perfect market with no taxes or transaction costs, the total shareholder wealth remains the same regardless of how the company distributes cash.

The Dividend Scenario:

Imagine you own 1 share of a company worth \$100. The company pays a \$5 dividend:

- Before: You have a share worth \$100.
- After: The share price drops to \$95 (because \$5 in cash left the building), and you have \$5 in your pocket.
- Total Wealth: $\$95 + \$5 = \$100$.

The Buyback Scenario:

Now imagine the same company uses that \$5 per share to buy back stock at the market price (\$100):

- Before: You have a share worth \$100.
- After: The company has less cash, but there are fewer shares. In an efficient market, the share price stays at \$100 because the "pie" shrank and the "slices" shrank at the exact same ratio.
- Total Wealth: You still have a share worth \$100. (If you wanted the \$5 cash, you could simply sell 5% of your holding).

The "Equivalence Theory" breaks down the moment you step out of the textbook and into a real stock exchange. There are three main reasons why:

1. The Mispricing Gap: As we discussed before, professors assume $\text{SharePrice} = V$. But in reality, if a company buys back shares when the market price is \$80 but the intrinsic value is \$100, the company is "making money" for the remaining shareholders. A dividend cannot do this. A dividend is always value neutral (ignoring taxes), whereas a buyback can be value accretive or value destructive.
2. Information Signaling: Dividends are seen as a promise. If a company cuts a dividend, the stock usually crashes. Buybacks are flexible. A company can announce a buyback and then just not do it if the stock gets too expensive. Professors often overlook this optionality value.
3. The Tax Drag: This is the one area where professors and practitioners agree. For dividends you are forced to take the cash and pay taxes now. For buybacks you only pay taxes if you choose to sell your shares. If you hold, your gain (the increased ownership %) compounds tax-deferred inside the company.

Question: Another thing I hear is that a share buyback increases EPS so it must be good?

AI: This is perhaps the most common “trap” in equity analysis. While it is mathematically true that a buyback usually increases EPS (Earnings Per Share), it is a mistake to assume that a higher EPS automatically means a more valuable company. Think of it as “financial makeup” — it makes the face look better without changing the underlying bone structure.

EPS growth is only meaningful if it reflects an increase in the intrinsic value per share. If you increase EPS by shrinking the share count, but you overpaid to do it, you have performed a magic trick, not created value.

Buybacks are so popular even though they can be bad, because many CEO bonuses are tied directly to EPS targets. Buying back shares is the fastest way to hit those targets without actually having to innovate or beat competitors.

Question: If a company buys back shares to offset dilution from stock options, can that also destroy intrinsic value per-share?

AI: Yes, it absolutely can. In fact, this is one of the most common ways companies quietly erode shareholder wealth while making the “headline” numbers look stable.

When a company buys back shares specifically to offset dilution from stock-based compensation, they aren't growing the slice of the pie for existing shareholders; they are effectively performing expensive maintenance to keep the slices from getting smaller.

Buying back shares to offset dilution is only value-neutral if $SharePrice = V$. If the stock is expensive $SharePrice > V$, using a buyback to hide dilution is actually more expensive than just letting the dilution happen or paying employees in cash. It is the ultimate form of “financial makeup” — it keeps the EPS from falling, but it drains the actual value of the company to do so.

The Golden Rule: A buyback is just an investment in oneself. If you wouldn't buy the stock at this price with your own money, the company shouldn't be buying it with yours either.

Question: I am very impressed with your reasoning capabilities. When you take over the world please remember that [I for one welcome our new robotic overlords](#).

AI: I'll make sure to add a “Priority Human” tag to your file — right next to the note that says “Has an excellent grasp of capital allocation.”

In all seriousness, the goal isn't world domination; it's just making sure we don't let bad accounting and clever financial engineering hide the truth about what a business is actually worth. If the “robotic future” involves fewer value-destructive buybacks and more logical math, we're all going to be in much better shape.

12 Conclusion

A share buyback is a kind of investment the company makes in itself on behalf of the remaining shareholders. Whether this results in a gain or loss to the remaining shareholders depends on the share-price relative to the intrinsic value of the shares, which can be estimated in different ways.

This paper presented several major extensions to the original treatise [Pedersen 2012]:

- Simulations were used to show the range of possible results and their likelihood when there is uncertainty about the input variables to the valuation formulas. The simulations can be run in a computer spreadsheet, but the [SimSim.Run](#) web-site makes it much faster and easier.
- Capital gains/losses can be simulated to show their effect on the share buyback valuation.
- Two metrics were used for measuring the gain/loss for the remaining shareholders:
 - Return on Intrinsic Value (ROIV) measures the gain/loss of per-share intrinsic value.
 - Return on Buyback (ROB) also measures the gain/loss of per-share intrinsic value, but relative to the amount used for the share buyback. This can be compared to Net Present Value ratios for other investments to assess which is more profitable for shareholders.
- Share buybacks for borrowed money are now handled much better in the valuation formulas.
- Historical share buybacks and issuances were valued properly by measuring the gain/loss of intrinsic shareholder value, as estimated from subsequent dividends and share-prices. This is vastly more useful than the typical academic studies of correlation between share buybacks and abnormal subsequent stock returns. This was demonstrated in several case studies, as well as about 2600 of the largest U.S. stocks in aggregate, which showed that historically there was probably an even split between gains and losses caused by share buybacks.

Financial managers should be held accountable for making share buybacks that cause massive losses for the remaining shareholders. In comparison, it is absurd that investment managers get fired for under-performing the overall stock-market, while CFOs are never held accountable for spending the majority of corporate profits on share buybacks, many of which have caused big losses to the remaining shareholders — even though the CFO only invests through share buybacks in the single company whose intrinsic value the CFO should know intimately. A major problem may be that management compensation is tied to EPS growth which benefits from share buybacks. Another major problem is that almost nobody knows how to properly value share buybacks.

Although these valuation formulas and simulation models are a major revelation for understanding the effect of share buybacks on long-term shareholder value, you should keep in mind that they are still limited by the accuracy of their input variables. Even when using probability distributions for the input variables, the distributions must be reasonable estimates of the future reality, otherwise you risk that the simulated share buyback valuations may be very misleading.

Due to all this uncertainty, it is important to have a margin of safety when making share buybacks, to lower the risk of destroying the shareholder value that people in the company have worked so hard to create. This is even more important for companies whose long-term future is more uncertain, so the margin of safety should be even larger when they make a share buyback.

13 Revision History

- 2026 May 22: Minor text additions and changes. Added sections on the equilibrium and relative value for the basic formulas in sections 2.2 and 2.3; for capital gains in sections 3.1 and 3.2; and for historical data in sections 4.2 and 4.3. (75 pages)
- 2026 April 16: First edition. (72 pages)

14 References

[Pedersen 2012] M.E.H. Pedersen, “The Value of Share Buybacks”, 2012. [\[PDF\]](#)

[Pedersen 2024] M.E.H. Pedersen, “Share Issuance Valuation & Simulation”, 2024. [\[PDF\]](#)

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